

Optimal Carbon Tax with Dynamic Skill Heterogeneity

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August 17, 2026[†]
Work in Progress: [Latest version](#)

Abstract

How should a government price carbon when it also cares about redistribution? I extend the dynamic Mirrlees model to two goods, one polluting, over a life cycle of privately observed skill shocks. Under weak separability, the dirty good should be taxed at the social cost of carbon (SCC) while labor and savings distortions keep their standard forms. Only non-separability makes it nonlinear, raising it above the SCC when the dirty good is a weaker labor complement. In a US calibration with energy standing in for the dirty good however, uniform taxation captures 98% of the welfare gain of an optimal nonlinear carbon tax. When labor and capital distortions are frozen at the levels implied by the US tax system, this sufficiency result is overturned as the best uniform tax only captures 1.5% of the welfare gain of the best nonlinear carbon tax. Here, a flat tax at an SCC of \$190/tCO₂ would be welfare-reducing and the best flat tax would deflate to an equivalent of \$47/tCO₂. This happens to be in the ballpark of where many jurisdictions price carbon, consistent with a government optimizing the carbon margin alone.

Keywords: *Carbon tax, dynamic Mirrlees taxation, second-best environmental policy*

JEL Classification: *D62; D82; H21; H23*

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[†]I thank my advisor Guillaume Sublet for his invaluable guidance and support, as well as the members of his workshop and the participants of the CIREQ Weekly PhD Student Research Workshop, the McGill University microbreakfast, the University of Montréal macro brownbag, the CIREQ-CIRANO-RRECQ Environmental and Natural Resource Economics Workshop, the 19th CIREQ PhD Students' Conference, the 59th CEA Annual Conference, the 35th CREEA Annual Conference, the 65th SCSE Conference, the 60th CEA Annual Conference, the 25th LAGV conference, and the 7th WCERE conference for their comments and feedback. I am grateful for the support and funding from the Fonds de Recherche du Québec Société et Culture (FRQSC) through the Doctoral Training Scholarships program. DOI: <https://doi.org/10.6977/329014>.

1 Introduction

Carbon pricing is the economist’s favorite climate policy, and the voter’s least. France abandoned its carbon tax plans after the *gilets jaunes* protests and Canada repealed its consumer carbon tax in 2025 after it became an electoral wedge issue on the grounds that the tax unfairly burdens low-income households. This disconnect might in part be driven by the fact that most carbon policy is informed by integrated assessment models (Environment and Climate Change Canada (2016), Interagency Working Group on Social Cost of Greenhouse Gases (2021)), which are representative agent models that are not designed to address this concern.

Whether carbon taxes are regressive is a topic of intense study, with one aspect of the debate relating to the choice of measure. When households are ranked by annual income, taxes on gasoline and other energy appear to fall more heavily on the poor. However, when households are ranked by lifetime income, the burden appears much flatter because households smooth consumption and today’s low earners include the young and temporarily poor (Poterba 1989; Hassett et al. 2009). The welfare-relevant burden of a carbon tax is therefore a lifetime object rather than an annual one and it’s important to consider the timing of an optimal carbon tax.

However, most research focuses on the incidence of existing carbon taxes rather than asking how they should be designed in the first place. Barrage (2020) and Douenne et al. (2026) are the exceptions, but the first ties the tax system to an *ad hoc* revenue target rather than optimal redistribution and the second restricts attention to linear instruments even though income taxes are in practice nonlinear. How then should a government price carbon when it cares about redistribution, and does the presence of pollution change how income should be taxed?

The answer turns on how the income tax is modeled. The Ramsey tradition treats it as a schedule of linear instruments and usually finds that carbon should be priced below the Pigouvian level because the carbon tax compounds the distortions the rest of the tax system already imposes. The Mirrleesian tradition allows the government to levy nonlinear income taxes and usually finds that the carbon tax should be set at the Pigouvian level, barring exceptions in the structure of preferences.

I take the Mirrleesian approach into a dynamic setting with two goods, one clean and one dirty, in which agents draw privately observed skill shocks over their life cycle and their consumption of the dirty good feeds a stock externality. To my knowledge, this is the first dynamic-Mirrleesian derivation of the optimal consumer carbon tax. Compared to the static irrelevance theorems of Kaplow (2012) and Jacobs and Mooij (2015), dynamics introduce savings and history-dependence.

I find that the static irrelevance result holds in a dynamic environment. Under weak separability, the carbon wedge equals the social cost of carbon, the labor wedge keeps the Golosov et al.

(2016) ABCD form, and the savings wedge keeps its inverse-Euler form. Redistributing through the optimal nonlinear income tax restores the split between corrective and redistributive taxation. That is, the carbon price should not be adjusted to perform redistribution.

I then quantify the separation by solving the optimal mechanism, treating energy as a stand-in for the dirty good and calibrating the model to US data. I find that the optimal wedge on energy is the social cost of carbon scaled up by a history-dependent Corlett-Hague premium, starting at the Pigouvian level, rising with age with the labor wedge, and ending the working life about 9% above the Pigouvian level. That premium is however small in welfare terms. When the income tax is set optimally alongside it, a flat carbon tax at the social cost of carbon captures 98% of the welfare gain of the full optimum.

This result however assumes that every margin is set optimally. I therefore also consider the case where the carbon wedge is set when labor and savings wedges are fixed to the levels implied by the US tax system. I find that the carbon wedge is pulled down when labor is overdistorted and up when it is underdistorted. The frozen savings margin instead rotates the carbon wedge across the life cycle, pushing it above the social cost of carbon early in life and below it late.

Solving this constrained optimum upends the previous analysis. A flat tax at the EPA's \$190/tCO₂ social cost of carbon now *costs* 0.115% of consumption relative to no carbon price at all. In a jointly optimized tax system even an unoptimized flat tax, set at the social cost of carbon, captures 98% of the welfare gain of a nonlinear tax; in this constrained setting even the best flat tax captures only 1.5% of it. Moreover, this tax is equivalent to \$47/tCO₂, about a quarter of the social cost of carbon. The lesson: mitigation is constrained by a suboptimal income tax system rather than by suboptimal carbon pricing design.

Section 2 reviews the literature. Section 3 presents the modeling framework. Section 4 builds the two-good dynamic Mirrlees economy and derives the optimal distortions. Section 5 derives optimal goods distortions when labor and savings distortions are held fixed. Section 6 solves both cases numerically with a model calibrated to US data. Section 7 concludes.

2 Literature review

This paper sits between two literatures that reach opposite verdicts on how distortionary taxation should bend the carbon price and draws on a third that has the tools to adjudicate between them, yet has never been aimed at the environment. Should a government facing a distortionary tax system price carbon below marginal damages because the environmental tax compounds distortions elsewhere, above marginal damages because the environmental tax can be used to ease distortions elsewhere, or exactly at marginal damages?

The Ramsey tradition typically finds that the optimal carbon tax should lie below marginal

damages, but that the structure of the optimal income tax is unchanged (Sandmo 1975; Bovenberg and Mooij 1994; Bovenberg and Smulders 1996). Barrage (2020) extends their static results with a dynamic climate-economy Ramsey model and finds that the optimal carbon tax should be 8-24% below the first-best Pigouvian level in the presence of distortionary income taxation. However, this analysis assumes a representative agent, focusing on efficiency and ignoring any distributional concerns.

Adding heterogeneous agents to this model overturns this result. Douenne et al. (2026) extend Barrage’s dynamic Ramsey model to heterogeneous agents, taking inspiration from the model developed in Werning (2007), and find that the optimal carbon tax is nearly Pigouvian. Douenne et al. (2025) show that this is robust to the inclusion of uninsurable income risk. Whether such a flat carbon tax is regressive and whether revenue recycling is progressive is still an open debate (Klenert et al. 2018; Goulder et al. 2019) and a parallel policy and ethics literature presses for differentiated carbon taxation (Chancel 2022; Dietsch 2024).

Without personalized carbon taxes, the exact restrictions on a uniform carbon tax reshape both its level and incidence. Using a heterogeneous agent microsimulation model calibrated to German data and focusing on revenue-neutral carbon tax reforms, Ploeg et al. (2025) find that the carbon tax should exceed marginal damages when agents have high inequality aversion. In a model calibrated to German data where households differ in income and energy efficiency, Hänsel et al. (2022) show that the first-best keeps the carbon tax uniform at the Pigouvian level and carries all redistribution through lump-sum transfers conditioned on the household’s income–energy type while leaving the income tax undistorted.

Separation can also fail for a reason orthogonal to instrument constraints. When climate damages fall unequally across households and compensation is incomplete, the social cost of carbon itself becomes distribution-dependent (Dennig et al. 2015; Kornek et al. 2021). I hold damages homogeneous to isolate the redistributive-instrument channel, so the separation I study is the Atkinson–Stiglitz one.

A static Mirrleesian literature establishes that separation, and overturns the Ramsey verdict in doing so: the below-damages result is not intrinsic to distortionary taxation but to holding the tax system away from the optimum. Once the income tax is the optimal nonlinear schedule of Mirrlees (1971), the logic of Atkinson and Stiglitz (1976) applies and differential commodity taxes are redundant under weak separability. Cremer et al. (1998), Cremer et al. (2001), and Cremer et al. (2010) develop the environmental version of the problem where a nonlinear income tax alongside a strictly Pigouvian levy is efficient, so the public-funds correction of the Ramsey tradition disappears. Kaplow (2012) shows that any environmental reform can be made distribution-neutral by a type-by-type adjustment of the income tax, so a move to Pigouvian pricing is Pareto-improving whatever the schedule. Jacobs and Mooij (2015) show that, when goods and environmental quality are weakly separable from labor, the optimal pollution tax equals marginal damages exactly and the marginal cost of public funds is one.

When separability fails, the tax follows the logic established in Corlett and Hague (1953), which Mirrlees (1976) formalized in his model. A good is taxed above the Pigouvian level for a leisure complement, the case West and Williams (2007) substantiate for gasoline, and below it for a leisure substitute. Saez (2002) extends the benchmark to preferences that vary with ability. In a linear-tax setting, Jacobs and Ploeg (2019) show that the pollution tax stays Pigouvian when Engel curves are linear and slips below only when the poor spend disproportionately on the dirty good under nonlinear Engel curves. Every result in this literature is static, with no savings and no life-cycle dynamics. Therefore, it cannot speak to the age profile or the time path of a carbon tax.

The tools for this dynamic treatment come from the New Dynamic Public Finance literature. Golosov et al. (2003) derive a rationale for positive capital taxation in a dynamic Mirrlees economy, since it is harder to elicit future work from households that save privately. This overturns the classic zero-capital-tax benchmark common in the Ramsey tradition (Judd 1985; Chamley 1986). Moreover, the authors find that the Atkinson-Stiglitz uniform-commodity result carries over into the dynamic setting. Kocherlakota (2005) shows how to implement the capital wedge into a wealth-tax and Albanesi and Sleet (2006) derive a tractable recursive form of the problem. Kapička (2013) extends the first-order approach to persistent shocks, Pavan et al. (2014) give the general envelope characterization, Farhi and Werning (2013) characterize the labor wedge over the life cycle, and Golosov et al. (2016) extend the ABC form of the labor wedge derived in the static frameworks of Mirrlees (1971) and Saez (2001) into a dynamic ABCD form. The optimal wedges the dynamic Mirrlees model delivers vary with age, connecting to the age-dependent-tax literature (Weinzierl 2011; Bastani et al. 2013; Erosa and Gervais 2002), which prices age-varying labor rather than a commodity. None of this machinery has been aimed at commodity taxation under non-separable preferences or corrective taxation.

I extend the dynamic Mirrleesian apparatus from one consumption good to two, letting the second pollute, and characterize the optimal carbon tax alongside the labor and savings wedges over the life cycle. To my knowledge this is the first dynamic Mirrleesian derivation of the optimal carbon tax. The heterogeneous-agent Ramsey results and this paper share the Jacobs and Mooij (2015) and Kaplow (2012) irrelevance result, which both approaches carry into a climate economy by different routes. Douenne et al. (2026) carry it with an optimally-set linear transfer, showing analytically that the marginal cost of public funds averages to one across a balanced-growth path and equals one period by period when preferences are logarithmic. Douenne et al. (2025) find that this result still exists when agents face uninsurable labor risk. I consider uninsurable labor risk with a fully nonlinear income tax and obtain a carbon wedge equal to the social cost of carbon at every history under weak separability and any preference satisfying mild conditions, alongside a Mirrleesian labor wedge and a non-zero savings wedge following the inverse Euler condition. The two approaches converge in their policy conclusions on the condition that the tax system is jointly optimized. What matters is therefore not so much linearity of instruments, but whether other instruments are misset.

3 Environment

The economy is a two-good extension of the dynamic Mirrlees setting of Golosov et al. (2003), Farhi and Werning (2013), and Golosov et al. (2016). The two goods are a clean numéraire and a polluting good. To isolate how household income inequality shapes the optimal carbon tax, I abstract from international and intergenerational equity, and consider a single country as well as a single generation living for $T + 1$ periods.

Each household consumes a clean good c_t and a polluting good q_t , supplies labor l_t , and is exposed to the aggregate pollution stock S_t . Expected lifetime utility is

$$\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(c_t, q_t, l_t, S_t) \right]$$

$\beta \in (0, 1)$ is the discount factor and utility is non-separable between its arguments. The clean good c is the numéraire, with prices normalized to 1. A unit of output transforms one-for-one into either good, so the producer price of the dirty good is also 1. The clean good is emission-free, whereas each unit of the dirty good emits a unit of carbon, so that date- t emissions equal aggregate dirty-good consumption Q_t . Emissions accumulate into a persistent pollution stock S_t , a function of current and past aggregate emissions.

$$S_t = S(Q_0, Q_1, \dots, Q_t)$$

where $Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta] dF_0(\theta)$ is aggregate consumption of the dirty good at date t . I assume that S is differentiable and non-decreasing in each argument, but otherwise leave it unrestricted. The linear special case $S_t = \sum_{j=0}^t (1 - d_j) Q_{t-j}$ in which each past emission flow leaves an additive trace $(1 - d_j)$ in the current stock, is the carbon-cycle representation of Golosov et al. (2014b). The general form above nests it and allows nonlinear accumulation. A convenient leading example keeps a fraction δ of the stock from one period to the next.

$$S_t = \sum_{j=0}^t \delta^j Q_{t-j}$$

This is the recursion $S_t = \delta S_{t-1} + Q_t$ from an initial pollution stock. To keep the state finite, I assume emissions stop affecting the pollution stock after s periods.

$$S_t = S(Q_{t-s}, \dots, Q_t)$$

Aside from the polluting good, the model follows the standard dynamic Mirrlees setup. At $t = 0$, agents draw an initial skill (type) θ_0 from a distribution $F_0(\theta)$. Later skills follow a Markov process $F_t(\theta|\theta_{t-1})$. At period t , each agent knows their skill draws up to time t , which I denote by a skill history vector $\theta^t = (\theta_0, \dots, \theta_t)$. Θ^t is the set of all possible skill histories.

Agents retire at an exogenous date $T_R \leq T$, common to all and not a choice. Periods $t < T_R$ are working periods and periods $t \geq T_R$ are retirement. From T_R on the agent supplies no labor and draws no further skills, so period utility is the same U evaluated at $l = 0$. Nothing is left to screen: the incentive constraints below hold trivially, the costate vanishes (Appendix C.12), and the labor wedge has no content. The savings and commodity margins survive, so a retiree still splits consumption between the two goods and still faces a carbon price. This is the life-cycle structure of Farhi and Werning (2013), with one difference that matters later: there retirement is a pure consumption-smoothing problem, while here the retiree's bundle still contains the dirty good.

An agent of type θ_t who supplies l_t units of labor produces $y_t = \theta_t l_t$ units of output. Everyone observes total output, but skill θ_t and labor supply l_t are private. Consumption of both goods is also publicly observable¹. Let $c_t(\theta^t), q_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ and $y_t(\theta^t) : \Theta^t \rightarrow \mathbb{R}_+$ be the agent's allocation of the clean good, the polluting good, and output, respectively. Let $\sigma^t(\theta^t) : \Theta^t \rightarrow \Theta^t$ describe how an agent with skill history θ^t reports it, and let Σ^t be the set of all possible reporting strategies. Capital transfers resources across periods through a linear technology with exogenous constant gross return $R > 0$. The social welfare function is

$$\int_0^\infty \alpha(\theta) \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U(c_t, q_t, l_t, S_t) \right] dF_0(\theta)$$

$\alpha(\theta)$ is a Pareto weight on the welfare of an agent with initial skill θ , normalized so that $\int_0^\infty \alpha(\theta) dF_0(\theta) = 1$. A utilitarian planner sets $\alpha(\theta) = 1$ for all θ . Let U_c and U_q denote the partial derivatives with respect to the clean and polluting goods, and U_l with respect to labor. Since $y = \theta l$, I can write $U(c, q, l, S)$ as $U(c, q, \frac{y}{\theta}, S)$. Let U_y and U_θ be the partial derivatives of utility with respect to y and θ .

Assumption 3.1 (Preferences).

U is twice continuously differentiable with $U_c, U_q > 0$, $U_l, U_S < 0$, $U_{cc}, U_{qq}, U_{ll}, U_{SS} \leq 0$, strictly quasi-concave in (c, q) with $2U_c U_q U_{cq} - U_c^2 U_{qq} - U_q^2 U_{cc} > 0$ (regular strict quasi-concavity: a

¹This is a plausible assumption for some goods such as electricity and air travel. For goods whose consumption cannot be observed, the planner is further constrained and the history-dependent commodity wedges of Sections 4 and 5.1 no longer apply. The anonymous price of Section 5.2 survives: an anonymous linear price requires observing total expenditure, not its composition. The assumption binds hardest on Section 5.1, where the commodity margin is the only instrument the planner still sets freely and Section 6 finds its history dependence to be where the welfare gains concentrate.

strictly convex, non-degenerate indifference map), and satisfies Inada conditions in (c, q) . Single crossing holds at every S .

$$\frac{\partial}{\partial \theta} \frac{U_l(c, q, y/\theta)}{U_c(c, q, y/\theta, S)} \geq 0 \quad \text{for all } (c, q, y, \theta)$$

At the optimal allocation, $1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) > 0$, where $\epsilon_t, \gamma_{c,t}$ are defined in (1) below. The conditions involving labor are imposed on working periods, where the labor margin has content; from T_R on labor is zero, U is evaluated at $l = 0$, and the inverse Frisch elasticity ϵ_t of (1) is not defined.

Assumption 3.2 (Shock process).

For all $1 \leq t < T_R$, f_t is differentiable in both arguments with $f'_t = \partial f_t / \partial \theta$ and $f_{2,t} = \partial f_t / \partial \theta_{t-1}$. For all θ_{t-1} , the ratio

$$\frac{\theta_{t-1} \int_{\theta}^{\infty} f_{2,t}(x|\theta_{t-1}) dx}{\theta f_t(\theta|\theta_{t-1})}$$

is bounded, and $\lim_{\theta \rightarrow \infty} \frac{1 - F_t(\theta|\theta_{t-1})}{\theta f_t(\theta|\theta_{t-1})}$ is finite.

Any admissible allocation must be incentive compatible: the agent must weakly prefer truthful reporting to any other strategy. The pollution stock S_t is an aggregate, so no single agent's deviation moves it and it enters both sides of the constraint identically. For every initial type θ and every reporting strategy,

$$\begin{aligned} \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \\ \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t \right) \middle| \theta \right] \end{aligned}$$

Over a finite horizon, global incentive compatibility is equivalent to the absence of a profitable one-shot deviation after every history (Fernandes and Phelan 2000; Kapička 2013). I follow the first-order approach of Kapička (2013), Farhi and Werning (2013), and Golosov et al. (2016), replacing these constraints with their local counterpart. Let $u_t(\theta^t)$ be the lifetime utility an agent obtains from reporting truthfully at every date. Truth-telling is locally optimal when the envelope condition holds.

$$\dot{u}_t(\theta^t) = U_{\theta} \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) + \beta w_{2,t}(\theta^t)$$

$w_{2,t}(\theta^t) = \int_0^\infty u_{t+1}(\theta^t, \theta_{t+1}) \partial_{\theta_t} f_{t+1}(\theta_{t+1} | \theta_t) d\theta_{t+1}$ is the marginal continuation value of type θ^t , and the dot denotes a derivative with respect to the current draw θ_t . The continuation term is the future information rent of the current draw: with i.i.d. shocks, $\partial_{\theta_t} f_{t+1} = 0$ and it vanishes, since today's skill then carries no information about tomorrow's. Under Assumptions 3.1 and 3.2, this local condition is necessary for incentive compatibility (Kapička 2013). The monotonicity requirements on the allocation belong to the sufficiency direction, not to necessity. The local conditions define a relaxed problem and are necessary but not sufficient for global incentive compatibility. Indeed, no general primitive restriction guarantees sufficiency in dynamic models with multiple goods (Goloso et al. 2016). I solve the relaxed problem throughout and verify global incentive compatibility numerically in Section 6.

The aggregate resource constraint must hold in every period,

$$\int_0^\infty \mathbb{E}_0[c_t(\theta^t) + q_t(\theta^t) | \theta] dF_0(\theta) + K_{t+1} \leq \int_0^\infty \mathbb{E}_0[y_t(\theta^t) | \theta] dF_0(\theta) + RK_t$$

where K_t is the aggregate capital stock, with K_0 given and $K_{T+1} = 0$. Dividing each period's constraint by R^t , summing over t , and using these boundary conditions to telescope the capital terms consolidates the sequence into a single intertemporal constraint,

$$\int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{c_t(\theta^t) + q_t(\theta^t)}{R^t} \middle| \theta \right] dF_0(\theta) \leq \int_0^\infty \mathbb{E}_0 \left[\sum_{t=0}^T \frac{y_t(\theta^t)}{R^t} \middle| \theta \right] dF_0(\theta) + RK_0$$

The planner chooses allocations to maximize welfare subject to the local incentive constraint, the intertemporal resource constraint, and the pollutant law of motion.

$$\max_{\{c_t(\theta^t), q_t(\theta^t), y_t(\theta^t), S_t\}_{\theta^t \in \Theta^t, t=0, \dots, T}} \int_0^\infty \alpha(\theta) \left(\mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \right) dF_0(\theta)$$

Three wedges, the implicit marginal taxes on the labor, savings, and commodity margins, describe the distortions the optimal allocation imposes.

$$\begin{aligned}
1 - \tau_t^y(\theta^t) &\equiv -\frac{U_l\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\theta_t U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)} \\
1 - \tau_t^s(\theta^t) &\equiv \frac{1}{\beta R} \frac{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{\mathbb{E}_t\left[U_c\left(c_{t+1}(\theta^{t+1}), q_{t+1}(\theta^{t+1}), \frac{y_{t+1}(\theta^{t+1})}{\theta_{t+1}}, S_{t+1}\right)\right]} \\
1 + \tau_t^q(\theta^t) &\equiv \frac{U_q\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}{U_c\left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t\right)}
\end{aligned}$$

I define six preference statistics used throughout. Two are the standard own-curvature elasticities, namely the inverse Frisch elasticity and the inverse intertemporal elasticity of substitution.

$$\epsilon_t \equiv \frac{U_{ll,t} l_t}{U_{l,t}} \quad \sigma_t \equiv -\frac{U_{cc,t} c_t}{U_{c,t}} \quad (1)$$

Four are cross-partial derivatives that measure how the marginal utilities of the two goods and of the pollutant move with labor as well as with each other. In order, they represent the labor complementarity with the clean good, the labor complementarity with the dirty good, the labor complementarity of environmental quality, and the cross-elasticity between the two goods.

$$\gamma_{c,t} \equiv \frac{U_{cl,t} l_t}{U_{c,t}} \quad \gamma_{q,t} \equiv \frac{U_{ql,t} l_t}{U_{q,t}} \quad \xi_t \equiv \frac{U_{Sl,t} l_t}{U_{S,t}} \quad \sigma_{cq,t} \equiv -\frac{U_{cq,t} q_t}{U_{c,t}} \quad (2)$$

Note that complementarity is measured with respect to labor rather than leisure here, so a positive γ signals a good whose marginal utility rises as the agent works more. To get the leisure complementarity interpretation, it suffices to flip the sign. Most of the commodity-wedge results hinge on the gap $\gamma_{c,t} - \gamma_{q,t}$ between the clean and dirty goods' complementarities with labor. If the gap is positive, the clean good is the relative labor complement, meaning an agent who works more wants more of it. Likewise, the results pertaining to the social cost of carbon hinge on the gap $\xi_t - \gamma_{c,t}$, which measures how the marginal damage of pollution varies with labor supply. If the gap is positive, an agent who works more suffers more from pollution and a cleaner environment is a relative labor complement. Both complementarities are proportional to l_t , so both vanish in retirement: from T_R on $\gamma_{c,t} = \gamma_{q,t} = 0$ and the gap closes mechanically.

One combination of these statistics appears whenever the commodity margin is constrained:

$$\mathcal{I}_t \equiv \frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t(1 + \tau_t^q)} \quad (3)$$

This is the income-effect gap evaluated at the prevailing consumer price of the dirty good. It is positive when the dirty good is normal since the marginal propensity to spend on it is proportional to $\mathcal{I}_t$ (Appendix C.3).

4 Optimal distortions

4.1 Planner's problem

First, following Kapička (2013)², we can write the planner's problem recursively. Appendix A.2 shows that the presence of externalities does not invalidate the approach. No individual report moves the aggregate pollution stock, so the agent's incentive problem is the one above with the pollution stock entering utility as a date-specific parameter. Aggregates are deterministic, so internalizing them appends a single new state, the emission window $\mathbf{S}_- = (Q_{t-s}, \dots, Q_{t-1})$ of lagged aggregate emissions feeding current and future pollution. The continuation value receives the updated window, with the current aggregate Q_t appended.

$$V_t(\hat{w}, \hat{w}_2, \theta_-, \mathbf{S}_-) = \min_{c, q, y, u, w, w_2, S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), \theta, \mathbf{S}) \right] f_t(\theta | \theta_-) d\theta$$

s.t.

²Kapicka's consumption set is convex and compact (his Assumption 1), so consumption can be a vector rather than a scalar. This paper uses only the necessity direction, his Theorem 4, which is dimension-free and does not invoke his Assumption 5. That assumption, that U is affine in the consumption vector, carries the sufficiency direction, and he gives two routes to it: utility separable across the components and affine in the shock, $U(x, \theta) = U_1(x_1) + \theta U_2(x_2)$, which a non-separable $U(c, q, l, S)$ does not satisfy, or convexification by lotteries over consumption, which would replace the deterministic allocations solved here. Sufficiency is accordingly not assumed but verified numerically in Section 6.

$$\begin{aligned}
\dot{u}(\theta) &= U_\theta \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w_2(\theta) \\
\hat{w} &= \int_0^\infty u(\theta) \bar{\alpha}_t(\theta) f_t(\theta | \theta_-) d\theta \\
\hat{w}_2 &= \int_0^\infty u(\theta) f_{2,t}(\theta | \theta_-) d\theta \\
u(\theta) &= U \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) + \beta w(\theta) \\
S &= S \left(\dots, \int q(\theta) f(\theta) d\theta \right)
\end{aligned}$$

The planner minimizes resource cost while offering an agent expected lifetime utility $\hat{w}$ with marginal continuation value $\hat{w}_2$, now internalizing what the assigned dirty good does to the pollution stock. $\mathbf{S}$ is the updated emission state, $\mathbf{S}_-$ with Q appended. $\bar{\alpha}_t(\theta)$ is the continuation Pareto weight attached to delivering utility to type θ . It equals the primitive Pareto weight in the initial period, $\bar{\alpha}_0(\theta) = \alpha(\theta)$, and $\bar{\alpha}_t(\theta) = 1$ for $t > 0$. As in Golosov et al. (2016), the lifetime Pareto weights enter only at date 0. From date 1 on, the weight collapses to one.

Two multipliers from this problem carry through the rest of the paper. $\lambda_{1,t}$ prices promise keeping, and the costate $\psi(\theta)$ prices the envelope condition, tied down at both ends of the type space by $\psi(0) = \psi(\infty) = 0$. Every wedge below is built on its normalized form $\psi U_{c,t} / (\theta f_t)$, the resources the planner gains by relaxing the local incentive constraint at a node. That object is zero wherever screening has no value, as in retirement (Appendix C.12).

4.2 Optimal distortions

Proposition 4.1 below describes the planner's optimality conditions in terms of labor supply, savings, and the split between the two goods.

Proposition 4.1 (Optimal wedges).

The labor wedge is

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta)$$

$$\begin{aligned}
A_t(\theta) &= 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) \\
B_t(\theta) &= \frac{1 - F_t(\theta)}{\theta f_t(\theta)} \\
C_t(\theta) &= \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x}) \frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
D_t(\theta) &= \frac{A_t(\theta) U_{c,t}(\theta)}{A_{t-1} U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}
\end{aligned}$$

Let $\widehat{U}_{c,t} \equiv U_{c,t}/(1 - \chi_t)$, with $\chi_t \equiv \frac{\gamma_{c,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y}$ as the incentive credit. Savings satisfy the generalized inverse Euler equation

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right]$$

The carbon wedge departs from the social cost of carbon by

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{D_t(\theta)}{A_t(\theta)} \right] = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y} \quad (4)$$

with SCC_t characterized in Proposition 4.2.

Proof. See Appendices A.3, A.4, and A.5. □

The first formula is the standard ABCD labor wedge formula, as in Golosov et al. (2016) and Farhi and Werning (2013). None of the terms change, except for term C_t which represents the social value of the resources extracted from types above θ , where the exponential term captures the income effect induced on those types induced by the redistribution. This term now includes the cross-curvature between the two goods, such that the wedge rises all things equal when the goods are Edgeworth substitutes ($\sigma_{cq,t} > 0$) and falls when they are complements ($\sigma_{cq,t} < 0$). Intuitively, when the goods are substitutes, taking away the numéraire from these types renders the dirty good more pleasurable and allows the planner to extract more resources from them.

The second formula is the generalized inverse Euler condition, expressed as a reweighted inverse Euler condition similar to Hellwig (2021). When preferences are non-separable between labor and consumption, the cost of delivering utility is weighted by $1 - \chi_t$, where χ_t is the incentive credit. When the numéraire complements labor, a unit of consumption makes work less painful and shrinks the information rent that truth-telling requires, rebating χ_t from the delivery cost. Neither the presence of the second good nor of externalities changes the structure of the savings wedge.

The third formula describes how the planner allocates each good across types. It is the dynamic analogue to the optimal commodity wedge derived in Jacobs and Mooij (2015). The left-hand side is the commodity distortion net of the social cost of carbon, described in the next section, and the right-hand side is the labor distortion scaled by the relative labor complementarities of the two goods. When preferences are weakly separable, $\gamma_{c,t} = \gamma_{q,t}$ and the deviation vanishes, so the carbon wedge is exactly the social cost of carbon; with no externality this is the zero commodity wedge of Golosov et al. (2003). Otherwise, when one of the goods is relatively more complementary to labor than the other, the planner would like to distort the weaker complement to labor more heavily. Because the labor wedge is history dependent, so is the goods wedge, which then inherits both the shape and the time path of the labor wedge, bent by the ratio of relative labor complementarities and labor response elasticities. This is the dynamic version of the Corlett-Hague rule that states that the weaker complement of labor should be distorted more heavily since this acts as an indirect tax on leisure.

4.3 Social cost of carbon

The social cost of carbon in (4) is not the textbook Pigouvian price, but rather the discounted stream of marginal damages where each damage term is reweighted by how pollution interacts with labor distortions.

Proposition 4.2 (The social cost of carbon).

The shadow price of current emissions (14) is

$$\text{SCC}_t = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \left[1 + \left(\frac{\tau_{t+j}^y}{1 - \tau_{t+j}^y} \right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} \right] \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta \quad (5)$$

with $\Phi_j \equiv \partial S_{t+j} / \partial Q_t$ the effect of current aggregate emissions on the date- $(t+j)$ pollution stock ($\Phi_j = 0$ for $j > s$ by construction) and the expectation taken over future skill histories. The Pigouvian price of the same emission is the same stream with every bracket set to one,

$$\text{SCC}_t^P = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta$$

and $\text{SCC}_t = \text{SCC}_t^P$ when $\xi_{t+j} = \gamma_{c,t+j}$ at every future node or when labor is undistorted.

Proof. See Appendix A.6. □

The bracket is a fiscal-adjustment factor, which is active whenever labor is distorted and pollution interacts with labor supply. Suppose that $\xi_{t+j}, \gamma_{c,t+j} > 0$. When $\xi_{t+j} > \gamma_{c,t+j}$,

pollution harm discourages labor supply more so than consuming more of the numéraire encourages it. Pollution then causes direct harm to utility and indirect harm through labor discouragement, both of which the planner would like to correct. Hence, the social cost of carbon rises above the Pigouvian level. When $\xi_{t+j} < \gamma_{c,t+j}$, pollution discourages labor less than consuming more of the numéraire encourages it, so the planner would like to underdistort the dirty good, shrinking the social cost of carbon. This correction operates on every term of the discounted stream of damages and is the dynamic extension of Jacobs and Mooij (2015), under different notation. The two statistics enter only through their difference, and that difference has a compact reading: $\xi_t - \gamma_{c,t}$ is the elasticity of the marginal rate of substitution U_S/U_c with respect to hours, so wherever labor is distorted the bracket exceeds one exactly when working more raises the household's valuation of a cleaner environment, measured in units of the numéraire.

4.4 Separation

Proposition 4.3 (Three-instrument separation).

Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, the corrective and redistributive margins separate.

$$\tau_t^q = \text{SCC}_t \quad \frac{\tau_t^y}{1 - \tau_t^y} = A_t B_t C_t + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t \quad \frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right]$$

The carbon price carries no redistributive term: the income tax and the savings wedge do all the redistribution and insurance whereas the goods wedge is purely corrective.

Proof. Immediate from (4) at $\gamma_{c,t} = \gamma_{q,t}$. □

Corollary 4.1 (Dynamic Atkinson-Stiglitz).

With no externality, weak separability gives $\tau_t^q = 0$ and the goods wedge vanishes.

Proof. Proposition 4.3 at $\text{SCC}_t = 0$. □

Redistributing through an optimal nonlinear income tax separates externality correction from redistribution. The carbon price is set at the social cost of carbon, unlike a dynamic Ramsey planner with representative agents like that in Barrage (2020) who prices carbon below it. Douenne et al. (2026) recover an average near-Pigouvian price with linear instruments, whereas the separation here is exact at every history. However, the social cost of carbon may be below or above the Pigouvian level depending on whether a clean environment complements labor. Corollary 4.1 is the dynamic Atkinson-Stiglitz result of Golosov et al. (2003) and Proposition 4.3 generalizes the Mirrlees irrelevance result of Jacobs and Mooij (2015) to a dynamic setting.

5 Constrained optimum

These results assume the government can implement any incentive-compatible allocation. Real carbon taxes are set alongside a tax system whose implemented wedges differ from the planner's preferred ones. I therefore consider a problem where the planner adjusts allocations to optimize one wedge, holding the other two constant at an exogenous schedule. This restriction applies history by history and can take an arbitrary form, whether linear or nonlinear.

5.1 Constrained-optimal commodity wedges

When setting a carbon tax, a relevant question is how to account for preexisting distortions. In this framework, I represent this by a planner who distorts both goods under the constraint that the labor and savings distortions must stay fixed at an exogenous level. At every type, the labor wedge is held at a schedule $\bar{\tau}_t^y(\theta) < 1$, with a Lagrange multiplier of $\eta(\theta)f_t(\theta|\theta_-)$.

$$\Gamma(\theta) \equiv -\frac{U_l\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)}{\theta U_c\left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S\right)} - (1 - \bar{\tau}_t^y(\theta)) = 0 \quad (6)$$

Γ is the distance between the margin the agent requires to supply the assigned hours and the net-of-wedge rate the schedule allows, so that $\Gamma = \bar{\tau}_t^y - \tau_t^y$: the constraint holds the implemented labor wedge at the frozen schedule, not at the wedge the planner would choose. The distance to that one is the labor gap of Lemma 5.2 below. Moreover, the savings wedge is held at $\bar{\tau}_t^s(\theta) < 1$, with Lagrange multiplier $\vartheta_{t+1}(\theta^t)f_t(\theta|\theta_-)$.

$$U_{c,t}(\theta^t) = (1 - \bar{\tau}_t^s(\theta_t)) \beta R \mathbb{E}_t[U_{c,t+1}(\theta^{t+1})] \quad (7)$$

The labor constraint is static, whereas the savings constraint links marginal utilities across dates and needs an additional state, the marginal-utility requirement $\hat{M} = \int_0^\infty U_{c,t}(\theta) f_t(\theta|\theta_-) d\theta$ that each component problem owes its predecessor. The planner's problem is

$$V_t(\hat{w}, \hat{w}_2, \hat{M}, \theta_-, \mathbf{S}_-) = \min_{c,q,y,u,w,w_2,S} \int_0^\infty \left[c(\theta) + q(\theta) - y(\theta) + \frac{1}{R} V_{t+1}(w(\theta), w_2(\theta), M(\theta), \theta, \mathbf{S}) \right] f_t(\theta|\theta_-) d\theta$$

with the two constraints added to the planner's problem of Section 4.1.

Lemma 5.1 (The freeze leaves the screening problem intact).

Constraints (6) and (7) restrict allocations rather than reporting strategies, so an allocation

incentive compatible under them would also be without them. Three consequences follow. (i) The envelope condition and the costate's boundary conditions $\psi(0) = \psi(\infty) = 0$ are those of Section 4.1 and the costate equation is unchanged in form, with ϕ carrying the two multipliers. (ii) The labor constraint is static: it enters the date- t problem pointwise with multiplier $\eta(\theta)f_t$ and adds no state. (iii) The savings constraint adds one state, the requirement $\hat{M}$ above, with shadow value $\partial V_t/\partial \hat{M} = -\vartheta_t$. That state is unconstrained at the two ends of life. The date-0 problem is the continuation of no history and inherits no requirement such that $\vartheta_0 = 0$, while the date- T problem has no successor and therefore passes no requirement.

Proof. See Appendix B.1. □

The first-order approach therefore applies as before, and the entire cost of the restriction enters as two multipliers, one per constraint, added to the first-order conditions for c , q , y , and S . Three statistics organize the results.

$$\Gamma_{c,t} \equiv (1 - \bar{\tau}_t^y) \frac{\sigma_t}{c_t} - \frac{\gamma_{c,t}}{y_t} \quad \Lambda_t \equiv \frac{\Gamma_{c,t}}{1 - \bar{\tau}_t^y} \quad \Delta_t \equiv (1 - \bar{\tau}_t^y) \frac{\sigma_t}{c_t} + \frac{\epsilon_t - 2\gamma_{c,t}}{y_t} \quad (8)$$

Derived in Appendix B.3, these statistics measure how the labor wedge responds to a perturbation of the allocation. $\Gamma_{c,t} = \partial \Gamma / \partial c$ moves consumption alone at fixed labor supply and is positive when leisure is a normal good since extra consumption then raises the margin the agent requires to stay put. With the wedge frozen, labor supply falls to restore (6). Λ_t is $\Gamma_{c,t}$ rescaled into the units of the wedge ratio $\bar{\tau}_t^y / (1 - \bar{\tau}_t^y)$ so that the savings multiplier enters Lemma 5.2 in the same units as the labor wedge gap. Δ_t moves consumption and labor supply together along a utility-compensated path and carries the opposite sign to the second derivative of the agent's objective in labor supply, so $\Delta_t > 0$ is the second-order condition of her labor choice at the frozen wedge. I assume this throughout³.

The two constraints enter the conditions for c , q , y , and S additively. The savings constraint appears in each of these conditions twice, so the two occurrences collect into a single shadow price of relaxing the intertemporal restriction at a node:

$$\eta_t^s(\theta^t) \equiv \left[\vartheta_t - \frac{\vartheta_{t+1}(\theta^t)}{(1 - \bar{\tau}_t^s) \beta R^2} \right] U_{c,t}(\theta^t) \quad (9)$$

³ $\Delta_t > 0$ is equivalent to a positive compensated earnings response, $s_{yy,t} = 1 / ((1 - \bar{\tau}_t^y) \Delta_t) > 0$. It fails only when the numéraire complements labor strongly enough that $2\gamma_{c,t}$ exceeds ϵ_t by more than $(1 - \bar{\tau}_t^y) \sigma_t y_t / c_t$, and the calibration of Section 6 clears it with room, at $\epsilon_t \approx 2$ against a $\gamma_{c,t}$ smaller by more than an order of magnitude. The progressive earnings-indexed schedule of Appendix B.12 weakens the requirement further, since Δ_t then gains p_t / y_t .

η_t prices the labor schedule and η_t^s prices the savings schedule. The two ϑ 's are dated by the problem that inherits the requirement the savings constraint imposes: ϑ_t prices what the date- t problem owes its predecessor and $\vartheta_{t+1}(\theta^t)$ what it passes on, which is why the date- t constraint carries the later index. ϑ_t and $\vartheta_{t+1}(\theta^t)$ affect the carbon price through the net shadow price $\zeta_t \equiv \eta_t(1 - \bar{\tau}_t^y) - \eta_t^s$. The labor wedge multiplier affects it through a second channel as well.

Proposition 5.1 (Constrained-optimal carbon price).

Let $\mathcal{I}_t$ be the income-effect gap (3), positive when the dirty good is normal. At every history before retirement,

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left(\frac{\psi U_{c,t}}{\theta f_t} + \frac{\eta_t}{y_t} \right) - \zeta_t \mathcal{I}_t$$

The social cost of carbon keeps the discounted-damage form of (5), with the bracket replaced at each future node by

$$1 + (\xi_{t+j} - \gamma_{c,t+j}) \left(\frac{\psi_{t+j} U_{c,t+j}}{\theta f_{t+j}} + \frac{\eta_{t+j}}{y_{t+j}} \right) + \zeta_{t+j} \frac{\sigma_{t+j} + \xi_{c,t+j}}{c_{t+j}}$$

$\xi_{c,t} \equiv U_{Sc,t} c_t / U_{S,t}$ is the complementarity of the pollutant with the clean good, such that $(\sigma_t + \xi_{c,t}) / c_t = \partial_c \ln(U_S / U_c)$. From retirement onward, the labor constraint is not imposed, so the terms it generates are absent and $\zeta_t = -\eta_t^s$. Corollary 5.3 states the retiree's carbon price.

Proof. See Appendices B.9 and B.10, with Appendix C.12 for $\psi \equiv 0$ at the retirement nodes. $\square$

Corollary 5.1 (Constrained-optimal carbon price with separable preferences).

Under weak separability or $\gamma_{c,t} = \gamma_{q,t}$,

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = -\zeta_t \mathcal{I}_t$$

where SCC_t is the constrained social cost of carbon of Proposition 5.1. When the dirty good is normal the carbon price lies below it where $\zeta_t > 0$ and above it where $\zeta_t < 0$.

Proof. Proposition 5.1 at $\gamma_{c,t} = \gamma_{q,t}$. $\square$

Corollary 5.2 (Accidental separation).

Under weak separability, $\tau_t^q = \text{SCC}_t$ at every node where $\eta_t(1 - \bar{\tau}_t^y) = \eta_t^s$.

Proof. Corollary 5.1 at $\zeta_t = 0$. $\square$

Two things change in Proposition 5.1 relative to Proposition 4.1. First, the screening weight $\psi U_{c,t}/(\theta f_t)$ gains a term η_t/y_t , so a binding labor constraint scales the nonseparable motive up or down before any income effect enters. Second, there is a new income-effect channel $-\zeta_t \mathcal{I}_t$.

The fiscal-adjustment factor $\xi_{t+j} - \gamma_{c,t+j}$ for the social cost of carbon picks up the same companion η_{t+j}/y_{t+j} and it gains a term in ζ_{t+j} , scaled by the rate at which consumption moves the valuation of marginal damages.

Corollary 5.1 carries into a Mirrleesian environment a verdict the rest of the public-finance literature reached long ago, and sharpens it. Whether a fixed margin should be priced at social marginal cost when another margin carries a distortion the planner cannot remove is an old question. Guesnerie (1979) labels the configuration the case of irreducible distortions, and the answer outside the Mirrleesian literature is generally no: Boadway and Harris (1977) show that piecemeal policy in a control sector requires, for any set of fixed distortions, the uncompensated demand derivatives of every other sector with respect to control-sector prices to vanish, a condition of which weak separability is only a necessary consequence.⁴ In the static analogue of this subsection's partition only one margin is fixed, and Boadway and Harris (1977) note that a single fixed distortion is trivial with a fixed-supply numéraire and lump-sum transfers, since proportional pricing in the control sector then attains the first best. Private information removes the triviality, because the frozen schedules bind type by type and redistribution across types is limited to what the envelope condition permits. The frozen savings margin adds a restriction the static taxonomy has no slot for: an intertemporal one.

The two shadow prices affect the carbon price through the same channel and enter as a difference such that they can cancel out. Shifting the assigned bundle between the two goods moves $U_{c,t}$ and $\mathcal{I}_t$ measures how much it moves per unit shifted. The labor constraint prices that movement at $\eta_t(1 - \bar{\tau}_t^y)$ because it displaces the labor supply margin. The savings wedge constraint prices the same movement at $-\eta_t^s$ because it displaces the marginal-utility requirement the frozen Euler equation carries across dates. The planner tilts the bundle toward whichever good displaces $U_{c,t}$ less and the two can therefore cancel without either constraint being slack, per Corollary 5.2. The social cost of carbon there still carries the modified damage weight of Proposition 5.1 wherever the multipliers are nonzero at future nodes.

A node where the two prices cancel is observationally identical to one at the unconstrained

⁴They argue that the condition holds universally across partitions only for the linear logarithmic class, and even where it holds the proportionality factor is not one, so control-sector prices are proportional to marginal cost rather than equal to it. Jewitt (1981) settles the single-consumer case at implicit separability, pseudo-separability in his terms, and Blackorby et al. (1991) show that with many consumers the requirement tightens to a two-sector representation of community preferences, for which implicit separability of every consumer's preferences is not sufficient, even when all consumers have identical preferences; single-consumer results give a misleadingly optimistic view of when piecemeal policy is valid. Golosov et al. (2014a) use variational arguments to characterize optimal systems within classes such as age-independent, linear, or separable.

optimum and has nothing to do with it. A carbon price sitting at the social cost of carbon therefore does not certify that the income tax is set right. It certifies only that the two frozen margins are mispriced by offsetting amounts at that node, and the offset cannot hold at every age. In retirement there is no labor multiplier to do the offsetting, so $\zeta_t = -\eta_t^s$ by Corollary 5.3 below, and a savings multiplier vanishing at every retirement date would reduce the law of motion of Proposition 5.2, read at the retiree's $\tau_t^{s,*} = 0$, to $\bar{\tau}_t^s = 0$. Any nonzero capital income tax therefore breaks the cancellation somewhere.

Neither of Proposition 5.1's displays is a formula in primitives. The carbon price carries $1 + \tau_t^q$ on the left and again inside $\mathcal{I}_t$, and the damage weight carries ζ_{t+j} , hence through Lemma 5.4 the whole future allocation. Both characterize a fixed point, and the loss formula of Proposition 5.4 is the section's one sufficient-statistics object. Both corollaries turn on ζ_t , which so far is only a pair of Lagrange multipliers and not something a tax authority would recognize. The rest of the subsection gives each of them an economic reading and then puts them back together: Lemma 5.2 takes the labor schedule, Proposition 5.2 and Lemma 5.3 the savings schedule, and Lemma 5.4 the pair.

Lemma 5.2 (The labor multiplier).

Define the desired labor wedge $\tau_t^{y,*}$ by $\tau_t^{y,*}/(1 - \tau_t^{y,*}) \equiv A_t \psi U_{c,t}/(\theta f_t)$, with $A_t = 1 + \epsilon_t - \gamma_{c,t}$. By (16) it is the labor wedge of Proposition 4.1, evaluated here at the constrained allocation, so it is what the planner would set on this margin were it free. Let $E_t \equiv \bar{\tau}_t^y/(1 - \bar{\tau}_t^y) - \tau_t^{y,*}/(1 - \tau_t^{y,*})$ be the labor gap, the distance at a node between the frozen wedge and the desired one. At every history with $t < T_R$,

$$\eta_t \Delta_t = E_t + \eta_t^s \Lambda_t$$

with Δ_t and Λ_t as in (8). Given η_t^s , the labor multiplier is pinned node by node. Since $\Delta_t > 0$, η_t is positive where the frozen schedule holds the labor wedge above the desired wedge corrected for the savings multiplier, negative where it holds it below, and zero where the two cross.

Proof. See Appendix B.4. □

The gap E_t is what the two multipliers have to account for, each scaled by how strongly its constraint bites on the hours margin. For the labor multiplier that scale is Δ_t , the movement in the margin along a compensated change in consumption and labor together. For the savings multiplier it is Λ_t : the frozen Euler equation binds on consumption alone, and Λ_t carries the rate at which consumption alone moves the same margin, $\Gamma_{c,t}$ rescaled into the wedge-ratio units the condition is written in, positive when leisure is normal.

Reading η_t as a gap between two wedges is what lets the rest of the section speak about tax policy rather than about multipliers, and it carries three loads. The carbon price of Proposition 5.1 contains η_t/y_t on its own, outside ζ_t , so the labor multiplier needs a reading of its own; Lemma 5.4 is obtained by solving the condition of Lemma 5.2 for η_t and substituting, which is

what leaves the labor gap standing alone; and the best flat labor rate of Proposition 5.5 below, in Section 5.3, is the schedule that averages η_t to zero, which is a condition on the gap.

Setting $\eta_t \equiv \eta_t^s \equiv 0$, which happens when the frozen schedules coincide with the ones the unconstrained optimum would deliver, gives the standard labor wedge formula. Zeroing one multiplier rather than both gives the single-freeze problems: at $\eta_t^s \equiv 0$ the conditions describe a frozen labor schedule alone, and at $\eta_t \equiv 0$ a frozen savings schedule alone, with the labor wedge solved rather than imposed. The freeze ladder of Section 6.4 climbs through the first of these: its lower rung is this problem at $\eta_t^s \equiv 0$ and its upper rung the joint problem, with both multipliers alive. The intertemporal margin is pinned by the frozen schedule rather than by the planner, so the inverse Euler condition survives in a different role.

Proposition 5.2 (Intertemporal condition).

Define the incentive credit and the savings-free delivery cost

$$\chi_t \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t \Gamma_{c,t} + \eta_t^s \frac{\sigma_t}{c_t} \quad \mathcal{G}_t \equiv 1 - \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} + \eta_t \Gamma_{c,t}$$

so that $1 - \chi_t = \mathcal{G}_t - \eta_t^s \sigma_t / c_t$, and define the desired savings wedge by $1 - \tau_t^{s,*} \equiv \mathcal{G}_t / \mathbb{E}_t[\mathcal{G}_{t+1}]$. At $\eta_t = \eta_t^s = 0$, χ_t collapses to the incentive credit of Proposition 4.1, which it generalizes. The generalized inverse Euler equation holds in the adjusted marginal utility $U_{c,t}/(1 - \chi_t)$,

$$\frac{1 - \chi_t}{U_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}}{U_{c,t+1}} \right]$$

and, read together with the frozen Euler equation, it pins the savings multiplier by a law of motion: for $t < T$, when the date- $(t + 1)$ draw is degenerate,

$$\eta_t^s \frac{\sigma_t}{c_t} - (1 - \bar{\tau}_t^s) \eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} = \mathcal{G}_{t+1} (\bar{\tau}_t^s - \tau_t^{s,*})$$

The level is not tied down here at all: it is tied down at the two ends of life by the endpoint conditions of Lemma 5.1, which (9) converts into conditions on η^s where they are used, in Appendix B.6. Away from a degenerate draw the date- $(t + 1)$ terms enter as conditional expectations and the right-hand side carries an additional covariance term (Appendix B.5). I maintain $\mathcal{G}_t > 0$ at every date, which the desired wedge and the sign of the law of motion both need and which nothing signs in general.

Proof. See Appendix B.5. □

The intertemporal allocation is pinned by (7), so the inverse Euler equation no longer characterizes savings. It prices the restriction instead. Delivering utility through the numéraire costs $1/U_{c,t}$ in resources, discounted by $\gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t}$ when the numéraire complements labor. The labor schedule

adds a second term: the same unit of consumption raises the margin $-U_l/(\theta U_c)$ by $\Gamma_{c,t}$, and with the wedge frozen, labor supply falls to restore (6). The planner values the induced fall at η_t , so where the schedule holds labor above the desired wedge and leisure is normal, consumption pushes hours further below where the planner wants them and delivery costs more. The savings schedule adds the third term by a different route: a unit of the numéraire moves $U_{c,t}$ itself, by σ_t/c_t in proportional terms, and $U_{c,t}$ is the object (7) constrains. Delivery is cheaper where $\eta_t^s > 0$ and dearer where $\eta_t^s < 0$, and Lemma 5.3 below forces both signs over the life cycle.

The law of motion differs from Lemma 5.2 in what the wedge gap determines, and the difference is the section's pivot. The gap is measured against a wedge the planner would actually choose: setting $\eta_t^s \equiv 0$ leaves $1 - \chi_t = \mathcal{G}_t$, so at a degenerate draw $\mathcal{G}_t/\mathcal{G}_{t+1}$ is the intertemporal margin the mechanism delivers when only the labor schedule is frozen. A frozen labor wedge is a static restriction, so its gap sets η_t node by node. A frozen savings wedge restricts a ratio across dates, so its gap sets only the weighted *difference* of adjacent multipliers, while the level stays pinned at birth and death alone. Where the schedule sits at its desired counterpart at every date that difference vanishes, the recursion is homogeneous, and $\eta^s \equiv 0$ solves it. Anywhere else the recursion is inhomogeneous between two fixed endpoints: a local gap can point the same way at every age, an accumulated one cannot.

Lemma 5.3 (Sign of the savings multiplier).

Fix a history. If $\eta_t^s \geq 0$ at every date along it, or $\eta_t^s \leq 0$ at every date along it, then $\vartheta_t = 0$ and $\eta_t^s = 0$ at every date along it. A schedule that leaves the multiplier alive anywhere therefore produces one that takes both signs over the life cycle. The argument uses only (9) with the two endpoint conditions of Lemma 5.1, and no first-order condition, so it holds whatever the labor schedule and whether or not the date- $(t + 1)$ draw is degenerate.

Proof. See Appendix B.6. □

The two results close on each other. By the law of motion at a degenerate draw, $\eta^s \equiv 0$ forces $\bar{\tau}_t^s = \tau_t^{s,*}$ at every date, so the multiplier is alive at exactly the schedules that misprice the intertemporal margin. Away from that draw the covariance term can absorb a wedge gap instead, but wherever the multiplier is alive it changes sign, whatever the draw.

Both multipliers now have a reading, and ζ_t is their difference. Substituting one into the other separates them.

Lemma 5.4 (Decomposition of the net shadow price).

With E_t the labor gap of Lemma 5.2, for $t < T_R$,

$$\zeta_t = \frac{1 - \bar{\tau}_t^y}{\Delta_t} E_t - \frac{A_t - 1}{y_t \Delta_t} \eta_t^s \quad (10)$$

Since $\Delta_t > 0$ and $1 - \bar{\tau}_t^y > 0$, the coefficient on E_t is positive. Suppose $A_t > 1$ at every working age. The coefficient on η_t^s , taken with its sign, is then negative at every date of the life cycle and not only at those the display covers: $-(A_t - 1)/(y_t \Delta_t)$ during working life, and -1 in retirement, where $\zeta_t = -\eta_t^s$ (Corollary 5.3 below). Three consequences follow. (i) Fix a history. If $\eta^s \not\equiv 0$ along it, the savings contribution to ζ_t is strictly positive at some dates along it and strictly negative at others: the multiplier takes both signs by Lemma 5.3, and a coefficient that never changes sign carries both through. (ii) No such restriction disciplines the labor contribution. Its sign profile is that of E_t , whatever profile the schedule delivers. (iii) Where the schedule overdistorts labor at every date, so that $E_t > 0$ throughout, the labor contribution is positive at every date, including at nodes where η_t itself is negative, and raising the labor level can only shrink the set of dates at which $\zeta_t < 0$.

Proof. See Appendix B.8. □

The substitution is not a relabeling. In the raw form $\zeta_t = \eta_t(1 - \bar{\tau}_t^y) - \eta_t^s$ the savings constraint appears twice, once explicitly and once inside η_t by Lemma 5.2, so no part of the carbon-price deviation can be attributed to either margin. Equation (10) purges the implicit occurrence, and only in this form does each constraint enter through a single object. The freeze ladder of Section 6.4 attributes the two restrictions the same way, but it is a different operation and neither licenses the other: the identity splits ζ_t at the joint solution, both terms evaluated at the allocation the two constraints produce together, while the ladder differences separate solves and is order-dependent. That the two agree on what belongs to each margin is a check on both. What makes the identity collapse is that $\Gamma_{c,t}$ and Δ_t share the term $(1 - \bar{\tau}_t^y)\sigma_t/c_t$ exactly. The income effect on labor supply therefore cancels out of their difference and only labor-side curvature survives, as $A_t - 1 = \epsilon_t - \gamma_{c,t}$.

The two terms are not parallel objects, and the language has to keep them apart. The labor channel enters through E_t , a gap between two wedges; the savings channel enters through η_t^s , a multiplier. Only the first is an “error” in the sense of a schedule sitting away from the desired wedge the constrained problem prices it against, and the distinction is not pedantic: at the calibration of Section 6 the savings *gap* is one-signed, since the frozen wedge exceeds $\tau_t^{s,*}$, the wedge the mechanism would set with only the labor schedule frozen, at every age, while the savings *multiplier* is not. Reading the second as though it were the first replaces a reversal with a level shift, and gets the late-life deviation backwards. The two are asymmetric in what pins them, too. The labor schedule is free to move date by date, so E_t takes whatever sign its gap takes at a node and may keep it for the whole working life; there is no analogue of Lemma 5.3 on the labor side, because nothing ties E_t down at the two ends of life. The savings term inherits the reversal that lemma forces, and inherits it only because its own coefficient does not change sign: a sign-changing multiplier scaled by a sign-changing coefficient need not

contribute both signs. $A_t > 1$ at working ages is the sufficient condition.⁵

Only one of the two channels earns a name. A frozen savings margin cannot move the carbon price the same way at every age: whatever the schedule, it raises ζ_t at some ages and lowers it at others. A frozen labor schedule is under no such discipline. It can hold one sign, if the schedule sits on one side of the desired wedge at every age, and it can change sign, which is what it does at the calibration of Section 6, where the labor gap crosses at mid-career. Naming it for either behavior would claim more than the lemma delivers. The asymmetry is the result: one channel is forced to reverse, the other is free.

A schedule that holds the labor wedge too high throughout, paired with one that overdistorts saving, therefore gives a carbon price held below the social cost of carbon by an amount set by the labor gap and moved around that position by the savings multiplier, with the reversal running from above to below as the cohort ages. Appendix B.6 proves the reversal, Appendix B.7 signs it from the law of motion, and Appendix B.8 adds the labor level to it. That the multiplier reverses is draw-free; the order in which it does is not, since the ordering is read off the law of motion at a degenerate date- $(t + 1)$ draw. Retirement delivers that draw whatever the skill process, but the random walk of Section 6 does not, so over working life the ordering is a solved property of the calibration rather than a consequence of the argument. The savings channel is a force rather than a guaranteed crossing: by (10) the deviation vanishes where the savings term *offsets* the labor term, not where η_t^s changes sign, so whether ζ_t itself reverses depends on the size of the savings swing against the labor level. Which force dominates at a given age is a quantitative question, and Section 6 answers it at US parameters.⁶

The reversal is the commodity margin doing part of the savings margin's work. The savings schedule constrains the marginal-utility requirement across dates, and the one margin the constrained planner still sets freely can move that requirement: because the dirty good is normal, a change in its relative price at date t shifts the marginal utility of the numéraire at

⁵Assumption 3.1 delivers $A_t > 0$, which is weaker, and it is imposed at the optimal rather than the constrained allocation. $A_t > 1$ is $\epsilon_t > \gamma_{c,t}$, the inverse Frisch elasticity above the numéraire's labor complementarity, and it holds with room in the calibration of Section 6: $\alpha = 3$ gives $\epsilon_t \approx 2$ against a $\gamma_{c,t}$ smaller by more than an order of magnitude, so $A_t \approx 3$. The direction of the condition is not arbitrary. Retirement fixes the coefficient at -1 whatever the preferences, so $A_t > 1$, which signs the working-life coefficient the same way, is what keeps one sign over the whole life cycle. $A_t < 1$ throughout would flip the working-life coefficient against the retirement one, and the savings contribution could then keep one sign despite the multiplier's reversal. Under the earnings-indexed schedule of Appendix B.12, the one Section 6 freezes, the requirement weakens to $A_t - 1 + p_t > 0$, since p_t then survives the difference $\Gamma_{c,t} - \Delta_t$ instead of being absorbed into Δ_t alone.

⁶Two readings are excluded. Equation (10) is an identity at the solved allocation rather than a comparative static, since E_t contains ψ and is evaluated at the constrained allocation: what the substitution removes is the explicit η_t^s , not the labor gap's dependence on the savings constraint through the allocation. And it is a working-life statement, since ϵ_t is undefined once labor is zero and $\Gamma_{c,t}$, Λ_t and Δ_t all carry $1/y_t$; the retiree needs no decomposition, because $\eta_t \equiv 0$ and $\zeta_t = -\eta_t^s$ by Corollary 5.3 below, where the labor term is gone and the savings multiplier is the whole deviation. The two regimes still meet: (40) shows the working-life coefficient tending to -1 as hours go to zero, so retirement's -1 is its limit rather than a separate case.

that date, and the marginal propensity to spend on the dirty good, $\mathcal{I}_t$ up to a positive factor, sets the strength of the channel. The path of the carbon price is therefore an intertemporal price. A schedule that overdistorts saving relative to its desired counterpart front-loads the consumption path the planner would otherwise assign, and the allocation answers by pricing the dirty good above the social cost of carbon early in life and below it late, leaning against the front-loading the frozen Euler equation enforces. That is why the deviation of Corollary 5.1 runs from above to below as the cohort ages rather than settling at one sign, and it is the sense in which the commodity margin repairs part of what the frozen schedule breaks; Section 6.5 prices the repair.

What the savings freeze does to the allocation has an exact aggregate measure. When the savings schedule does not vary with the type, taking unconditional expectations in (7) and iterating forward gives

$$\mathbb{E}_0[U_{c,t}] = \prod_{j=0}^{t-1} \frac{1}{(1 - \bar{\tau}_j^s) \beta R} \mathbb{E}_0[U_{c,0}] \quad (11)$$

so the schedule pins the growth rate of the population mean of marginal utility at every date, whatever else the planner does, and leaves its level free. This is the allocation-side counterpart of the multiplier's level being tied down only at the two ends of life. At the calibration of Appendix D.1, $\beta R = 1$ and $\bar{\tau}^s \approx 0.009$: mean marginal utility then compounds at $\bar{\tau}^s/(1 - \bar{\tau}^s) \approx 0.9\%$ per period and rises by about seventy percent over the sixty-year life of the quantitative model. A frozen savings wedge does not so much distort the intertemporal margin date by date as bend the whole path of marginal utility. The carbon-price deviation of Corollary 5.1 prices that bending date by date.⁷

Retirement separates the two forces cleanly.

Corollary 5.3 (Retirement).

From the retirement date T_R of Section 3 on there is no labor to report, so $\psi \equiv 0$ and (6) has no content, giving $\eta_t \equiv 0$ and $\zeta_t = -\eta_t^s$. The retiree's carbon price satisfies

$$\frac{\tau_t^q - \text{SCC}_t}{1 + \tau_t^q} = \eta_t^s \mathcal{I}_t$$

⁷Lemma 5.3 disciplines the sign of η_t^s , not the size of ϑ_t . The chain (34) compounds ϑ_t at $(1 - \bar{\tau}_t^s) \beta R^2$ per period, so it grows with the accumulated bending in (11) and can end the life cycle orders of magnitude above the per-period gap $\bar{\tau}_t^s - \tau_t^{s,*}$, even when the schedule sits close to its desired counterpart at every date. The formulas price η_t^s , a difference of adjacent multipliers that nets out the accumulated level, and by Proposition 5.2 its drift is of the order of that gap. A large ϑ_t and a small carbon-price deviation therefore coexist: ϑ_t measures how far the constraint has bent the path of marginal utility, η_t^s how much bending the current date adds.

whether or not preferences are weakly separable, since $\gamma_{c,t} = \gamma_{q,t} = 0$ mechanically at $l = 0$: the deviation is the savings multiplier alone. Retirees face a purely corrective price when the carbon price is chosen anonymously, Corollary 5.5 below; that result survives a miset labor schedule and fails against a miset savings schedule.

Proof. See Appendix C.12 for $\psi \equiv 0$, Appendix B.9 for the condition, and Appendix B.10 for SCC_t . Both are rederived at $l_t = 0$ rather than taken as a limit of Proposition 5.1. $\square$

The asymmetry is informative. A miset labor schedule distorts the allocation of the numéraire across types within a date, so its damage dies with the labor margin. A miset savings schedule distorts the allocation of the numéraire across dates within a life, so it operates through retirement, where the desired savings wedge is zero and any nonzero frozen wedge keeps injecting error. Retirees' carbon price deviates from marginal damages for reasons that have nothing to do with screening their labor.

5.2 Constrained carbon wedge

Real carbon prices are legislated rather than derived, anonymous rather than personalized, and set without reference to the income tax. The binding restriction is anonymity: the instrument is one number where Section 5.1 priced a schedule in type. This subsection prices that restriction, and the answer is the one result in this section written entirely in demand-side and tax-side statistics. It turns on the cross-sectional dispersion of the node-specific wedge the planner forgoes, which is small when the income tax is set optimally alongside the price and large when it is frozen, because the forgone wedge then carries the deviation $-\zeta_t \mathcal{I}_t$ of Corollary 5.1, dispersed across histories rather than nearly common.

The machinery mirrors the previous subsection. The planner holds the carbon wedge to an exogenous schedule $\bar{\tau}_t^q(\theta)$ and lets the income-side wedges respond. A common rate, the form every carbon tax in force actually takes, is the member of that family flat in type; the problem in which the planner still chooses that rate rather than inheriting it is a second member. Both are read off the same multiplier.

The planner solves the problem of Section 4.1 subject to

$$\frac{U_q \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right)}{U_c \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right)} = 1 + \bar{\tau}_t^q(\theta) \equiv \bar{p}_t(\theta), \quad (12)$$

with multiplier $\eta^q(\theta) f_t(\theta | \theta_-)$. The constraint is static, so the multiplier is pinned by a static condition, and the regularity it needs is the second-order condition $\mathcal{D}_t < 0$ below, guaranteed by Assumption 3.1.

Three demand statistics organize the results, alongside the income-effect gap $\mathcal{I}_t$ and the normalized shadow price $\psi U_{c,t}/(\theta f_t)$ of the local incentive constraint. Let $\mathcal{D} \equiv U_{qq} - 2\bar{p}U_{cq} + \bar{p}^2U_{cc} < 0$ be the curvature of utility along the budget line. It is negative at whatever price the household faces, since the split sets $U_q = \bar{p}U_c$ and there $\mathcal{D} = -(2U_cU_qU_{cq} - U_c^2U_{qq} - U_q^2U_{cc})/U_c^2$, which Assumption 3.1 signs. Define

$$m_q \equiv \bar{p} \frac{\partial q^d}{\partial x} = \bar{p}^2 |s_{qq}| \mathcal{I}_t \quad s_{qq} \equiv \left. \frac{\partial q^d}{\partial \bar{p}} \right|_u = \frac{U_c}{\mathcal{D}} < 0$$

the marginal budget share of the dirty good and its compensated own-price response (Appendix C.3). Two more objects carry the price gap itself:

$$\varpi_t \equiv \frac{\bar{\tau}_t^q - \text{SCC}_t}{\bar{p}_t} \quad \text{and} \quad \varpi_t^* \equiv (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t}$$

ϖ_t is the screening component of the given price, the ad-valorem wedge net of the part that pays for damages. ϖ_t^* is the wedge the planner would set at that node if the instrument were free, which is the commodity wedge of Proposition 4.1. The whole subsection turns on the gap between them.

When the schedule is flat in type, the restricted economy collapses to a one-good economy in expenditure.

Lemma 5.5 (Anonymity and the one-good reduction).

Suppose $\bar{p}_t(\theta) = \bar{p}_t$ for every type. An allocation satisfies (12) if and only if it assigns each type the private split of some pair (x, y) of expenditure and output at the common price. With indirect utility

$$\tilde{U}(x, l; \bar{p}) \equiv \max_{c, q \geq 0} U(c, q, l) \quad \text{s.t.} \quad c + \bar{p}q \leq x,$$

the split is unique since $\mathcal{D} < 0$, and incentive compatibility in the restricted economy is that of a one-good economy in (x, y) with period utility $\tilde{U}(x, y/\theta; \bar{p}_t)$. The envelope condition is unchanged,

$$\dot{u}(\theta) = -\frac{\tilde{U}_l l}{\theta} + \beta w_2(\theta) = -\frac{U_l l}{\theta} + \beta w_2(\theta)$$

Proof. See Appendix C.2. □

The reduction is what makes an anonymous carbon price tractable: the planner assigns expenditure and the agent splits it, so the two-good screening problem is a one-good screening problem in a reshaped utility. It fails for a schedule that varies with type, since then no single budget line

rationalizes the assigned bundles. Section 6 solves the restricted mechanism through exactly this reduction.

Lemma 5.6 (The price of a given carbon wedge).

At every history,

$$\eta_t^q = \bar{p}_t |s_{qq,t}| (\varpi_t^* - \varpi_t)$$

The multiplier is the price gap converted into resources by the compensated substitution response: $\eta_t^q > 0$ where the schedule underprices the dirty good relative to the node's Corlett-Hague level, $\eta_t^q < 0$ where it overprices it, and $\eta_t^q = 0$ where the two cross.

Proof. See Appendix C.4. □

Nodes at which the consumer substitutes away from the dirty good easily convert a given price gap into a large multiplier, which is the deadweight-loss weighting of the Corlett-Hague tradition. Every result below is this multiplier read at a different schedule.

Proposition 5.3 (Wedges under a given carbon price).

At every history the labor wedge and the savings adjustment are

$$\frac{\tau_t^y}{1 - \tau_t^y} = \frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} + \eta_t^q \bar{p}_t \left[\mathcal{I}_t - \frac{\gamma_{c,t} - \gamma_{q,t}}{(1 - \tau_t^y) y_t} \right] \quad \chi_t^q \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q \bar{p}_t \mathcal{I}_t$$

with the generalized inverse Euler equation holding in $U_{c,t}/(1 - \chi_t^q)$ and, to second order,

$$\tau_t^s \approx \text{Var}_t(\ln U_{c,t+1}) + (\chi_t^q - \mathbb{E}_t \chi_{t+1}^q) + \text{Cov}_t(\chi_{t+1}^q, \ln U_{c,t+1})$$

The social cost of carbon keeps its discounted-damage form with the damage weight

$$1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta^q \left[\bar{p} \frac{\sigma + \xi_c}{c} - \frac{\sigma_{cq} + \xi_q}{q} \right]$$

at each future node, where $\xi_{q,t} \equiv U_{Sq,t} q_t / U_{S,t}$ is the complementarity of the pollutant with the dirty good.

Proof. See Appendices C.5, C.6, and C.11. □

The labor wedge the proposition reports is the earnings-margin wedge alone, and it is not what the screening condition pins down. When an agent earns a marginal unit of output on a compensated path, the planner collects τ_t^y through the earnings margin and $\bar{\tau}_t^q dq^d/dy|_u$ through the commodity margin, where $dq^d/dy|_u \equiv (1 - \tau^y) \partial_x q^d + \partial_l q^d / \theta$ is the utility-compensated response of dirty-good demand to earnings; of the commodity part, $\text{SCC}_t dq^d/dy|_u$ only pays for the marginal damages the extra dirty consumption causes. Their sum net of that corrective piece

$$\Omega_t \equiv \frac{\tau_t^y}{1 - \tau_t^y} + \frac{\bar{\tau}_t^q - \text{SCC}_t}{1 - \tau_t^y} \frac{dq_t^d}{dy} \Big|_u = \tilde{A}_t \frac{\psi U_{c,t}}{\theta f_t} \quad (13)$$

is the comprehensive wedge the mechanism sets, the dynamic counterpart of the effective labor wedge of Jacobs and Boadway (2014), and $\tilde{A}_t$ is built on composite statistics that average the two goods' curvatures by their marginal expenditure shares (Appendix C.7). One property of those composites is worth stating in the text: the effective inverse Frisch elasticity satisfies $\tilde{\epsilon} \leq \epsilon$, a Le Chatelier property, because the agent's freedom to re-optimize the split cushions labor shocks, so delivering expenditure is a weakly better hedge than delivering a rigid bundle. The comprehensive wedge inherits the full *ABCD* structure of Proposition 4.1 in those composites (Appendix C.8), and Proposition 5.3 then dictates how Ω_t splits between the income-tax wedge and the commodity-revenue term.

The correction to the income-tax wedge has two parts of opposite sign. The expenditure part $(1 - \tau^y)\partial_x q^d$ is positive for a normal good. The labor part $\partial_l q^d/\theta$ is the statistic of Christiansen (1984), negative when the dirty good is a labor substitute, since working more at fixed expenditure shifts spending away from it. When the labor-substitute part dominates, discouraging labor is fiscally cheaper than it looks, because the agent who works less spends more on the taxed good and part of the forgone output returns as commodity revenue.

Under weak separability the whole apparatus collapses to a single statistic.

Corollary 5.4 (The income tax collects the carbon the carbon price misses).

Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$, we have $\varpi_t^* \equiv 0$, so $\eta_t^q \bar{p}_t = -\bar{p}_t^2 |s_{qq,t}| \varpi_t$ and the corrections collapse to the marginal budget share times the uncollected carbon price:

$$\frac{\tau_t^y}{1 - \tau_t^y} = \frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} + m_{q,t} \frac{\text{SCC}_t - \bar{\tau}_t^q}{\bar{p}_t} \quad \chi_t^q = \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - m_{q,t} \frac{\text{SCC}_t - \bar{\tau}_t^q}{\bar{p}_t}$$

A carbon price given at the social cost of carbon is harmless: $\bar{\tau}_t^q = \text{SCC}_t$ gives $\varpi_t = 0$, hence $\eta_t^q = 0$, and returns every wedge to its Section 4 form. A price given below it raises the labor wedge above the desired level; one given above it lowers the labor wedge.

Proof. See Appendix C.5. □

The mechanism is embedded carbon. A marginal compensated dollar of earnings is spent, a fraction $m_{q,t}$ of it on the dirty good, and each of those units carries uncollected marginal damage $(\text{SCC}_t - \bar{\tau}_t^q)/\bar{p}_t$. The income tax adds that charge, taxing the carbon content of marginal spending that the carbon price misses. The savings side applies the same charge intertemporally: delivering consumption at date $t + 1$ rather than t carries the uncollected damage of the extra dirty consumption it drags along, so the savings wedge rises where

underpricing worsens over the life cycle and falls where pricing improves. A legislated price that stays flat while the social cost of carbon grows is the worsening case. Both corrections persist into retirement, since the price gap is a within-period object that never expires. This is the Mirrleesian counterpart of Hänsel et al. (2022), who find that a mandated uniform carbon tax reshapes the optimal income tax; here the reshaping has a closed form, and it is the marginal budget share times the price gap.

5.2.1 When the planner chooses the common rate

Suppose now that the price is anonymous but not inherited: the schedule is flat, $\bar{p}_t(\theta) = \bar{p}_t$, and the planner chooses its level at every date. This is the problem a government solves when it may legislate the carbon tax but not personalize it, and it is the configuration every carbon tax in force approximates.

Proposition 5.4 (The optimal anonymous price).

The optimal level sets the population average of the multiplier to zero at every date, $\mathbb{E}_0[\eta_t^q] = 0$. Equivalently, the screening component is the substitution-weighted average of the desired node wedges,

$$\varpi_t = \frac{\mathbb{E}_0[|s_{qq,t}| \varpi_t^*]}{\mathbb{E}_0[|s_{qq,t}|]}$$

and the optimal price splits additively into a corrective and a screening part,

$$1 + \tau_t^q = 1 + \text{SCC}_t + \bar{\tau}_t^{\text{screen}} \quad \frac{\bar{\tau}_t^{\text{screen}}}{1 + \tau_t^q} = \frac{\mathbb{E}_0[|s_{qq,t}| (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi_{U_{c,t}}}{\theta_{f_t}}]}{\mathbb{E}_0[|s_{qq,t}|]}$$

To second order in the dispersion of ϖ_t^* , the date- t resource-equivalent loss from charging this common rate rather than the node-specific schedule of Proposition 4.1 is

$$\text{Loss}_t \approx \frac{1}{2} \frac{\bar{p}_t^3}{1 + \text{SCC}_t} \mathbb{E}_0[|s_{qq,t}|] \text{Var}_t^{|s|}(\varpi_t^*)$$

with $\text{Var}_t^{|s|}$ the $|s_{qq,t}|$ -weighted cross-sectional variance over date- t histories, minimized at the level above. The $1 + \text{SCC}_t$ in the denominator converts the ad-valorem wedge into its screening component, $d\bar{\tau}^q = \bar{p}^2 d\varpi / (1 + \text{SCC}_t)$, and drops out in the no-externality economy.

Proof. See Appendices C.9, C.11, and C.10. □

A single instrument averages the schedule the planner would set node by node, weighting each history by how easily the consumer can substitute away from the dirty good there.⁸ The condition is implicit: the weights and the node wedges are themselves evaluated at the restricted allocation, so the level is characterized as a fixed point rather than in closed form. The cost of not personalizing is the substitution-weighted cross-sectional variance of the wedge the planner forgoes. Because that wedge tracks the labor wedge by Proposition 4.1, its dispersion is first-order in the dispersion of τ^y , and what the tracking is worth is what anonymity gives up. Every object in the formula is a demand-side or tax-side statistic, so this is a sufficient-statistics answer to the cost-of-mispricing question. When the income tax is set optimally alongside the price, that variance is small, and the near-sufficiency of a flat carbon price documented in Section 6 is precisely this. The same reasoning predicts its reversal. Against a frozen income tax the node-specific wedge the planner forgoes carries the deviation $-\zeta_t \mathcal{I}_t$ of Corollary 5.1 on top of the Corlett-Hague term, and that deviation is dispersed across histories rather than nearly common, so anonymity gives up far more.

Corollary 5.5 (When the screening component vanishes).

The screening component of the optimal anonymous price is zero, and the price is exactly Pigouvian, whenever $\varpi_t^* \equiv 0$ across the date- t cross-section. Two cases deliver this. Under weak separability, $\gamma_{c,t} = \gamma_{q,t}$ pointwise, so $\varpi_t^* \equiv 0$ at every history and the restricted optimum coincides with the unconstrained optimum of Corollary 4.1. From the retirement date T_R on there is no labor to report, so $\psi \equiv 0$ and again $\varpi_t^* \equiv 0$: the screening component of the carbon price dies with the labor margin and retirees face pure Pigouvian pricing. More generally the restriction is costless at date t if and only if ϖ_t^* is constant across the date- t cross-section, since only its dispersion enters the loss.

Proof. See Appendix C.12 for $\psi \equiv 0$; the rest is immediate from Proposition 5.4. $\square$

The additive split is the section's main reading, and it is a statement about which margin the restriction binds. The corrective part of the optimal dirty-good price was uniform already at the unconstrained optimum, since a ton of emissions carries the same social cost whoever emits it and the emissions margin cannot screen. What anonymity constrains is only the Corlett-Hague screening part, forced from a history-dependent schedule down to a substitution-weighted scalar. A legislated uniform carbon price therefore sacrifices nothing on the corrective margin: the whole cost of the restriction falls on screening, which the loss formula measures. The level of the social cost of carbon is nonetheless co-determined with the constrained allocation, its damage weight carrying the multiplier node by node through Proposition 5.3; this reprices

⁸The construction parallels the many-person Ramsey rule of Diamond (1975) in collapsing a heterogeneous population into a single linear-tax condition: there the optimal excise rule, here $\mathbb{E}_0[\eta_t^q] = 0$. The weighting differs in kind. Diamond aggregates through the distributional characteristic, the covariance of demand with the net social marginal utility of income; here the weights are pure compensated substitution responses, with the redistributive content carried inside the node wedges being averaged rather than in the weights.

carbon in the constrained second best but is not a loss on the corrective margin, which has no cross-sectional dispersion to forgo. This is the separation of Section 4.4 restated for the instrument governments actually use.

Between the two readings of this subsection lies the whole family. A price given at the social cost of carbon leaves every wedge at its Section 4 value. A price given anywhere else routes the uncollected damage into the income tax at the rate Corollary 5.4 sets. And a price chosen anonymously but optimally sits at the zero-average point, $\mathbb{E}_0[\eta_t^q] = 0$, where the errors cancel in the cross-section without vanishing node by node.

5.3 A Mirrlees-Ramsey bridge

The section's exercises assemble into a bridge between the two traditions the literature review set against each other. Each margin now admits three readings. Left free, it carries its Mirrleesian wedge. Given flat with the level chosen optimally, it is a linear instrument of the Ramsey kind. Given anywhere else, it is at the third best, and the multiplier's profile prices the error. Only the middle reading is still missing on the income side: Proposition 5.4 supplied it for the carbon price, and the following is its counterpart for the labor and savings rates.

Proposition 5.5 (Optimal uniform schedules).

Let both frozen schedules of Section 5.1 be uniform, $\bar{\tau}_t^y(\theta) = \bar{\tau}_t^y$ and $\bar{\tau}_t^s(\theta) = \bar{\tau}_t^s$. The best uniform schedules satisfy

$$\mathbb{E}_0[\eta_t] = 0 \quad \text{and} \quad \mathbb{E}_0[\vartheta_{t+1} U_{c,t}] = 0$$

Equivalently, for the labor wedge,

$$\frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} = \frac{\mathbb{E}_0 \left[s_{yy,t} \left(\frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} - \eta_t^s \Lambda_t \right) \right]}{\mathbb{E}_0[s_{yy,t}]} \quad s_{yy,t} \equiv \frac{\partial y}{\partial (1 - \bar{\tau}_t^y)} \Big|_u = \frac{1}{(1 - \bar{\tau}_t^y) \Delta_t}$$

with $s_{yy,t}$ the compensated response of earnings to the net-of-wedge rate, holding the commodity split fixed (Appendix B.3), and $\mathbb{E}_0[\eta_t^s] = \mathbb{E}_0[\vartheta_t U_{c,t}]$ for the savings rate, which is guaranteed to vanish only at $t = 0$.

Proof. See Appendix B.11. □

The uniform labor wedge is the compensated-earnings-response-weighted average of the desired Mirrleesian wedges, net of the savings multiplier at each node. It mirrors the optimal anonymous carbon price of Proposition 5.4, with s_{yy} in place of $|s_{qq}|$: a single rate averages the schedule the

planner would set node by node, weighting each node by how elastically its earnings respond. Even at the optimal uniform pair, ζ_t is generically nonzero node by node, so the carbon-price deviations of Corollary 5.1 do not vanish.

Optimality on all three margins therefore takes the same form, a zero-average condition,

$$\mathbb{E}_0[\eta_t^q] = 0 \quad \mathbb{E}_0[\eta_t] = 0 \quad \mathbb{E}_0[\vartheta_{t+1} U_{c,t}] = 0$$

The parallel is exact on the two static margins, where the object averaging to zero is the multiplier on the wedge constraint itself. The savings condition is the same idea one step removed: the rate controls only the requirement passed forward, so what averages to zero is the passed-forward multiplier weighted by current marginal utility, while the per-node price η_t^s averages to $\mathbb{E}_0[\vartheta_t U_{c,t}]$, zero at birth and drifting thereafter. Margin by margin, the optimal linear instrument is the zero-average point of the third-best family, and the second best of Section 4 is the point where every multiplier vanishes identically.

6 Quantitative analysis

How large is the Mirrleesian correction to the carbon price, how much welfare does it deliver over a flat carbon tax, and how much of the answer survives an income tax the planner did not choose? I answer all three by solving the model's optimal mechanism globally, calibrated to US data, rather than evaluating the wedge formulas at borrowed sufficient statistics.

The exercise runs in two worlds that differ in one respect. In the first the labor and savings wedges are set optimally alongside the carbon price, which is Section 4. In the second they are frozen at the schedules the US tax system implies, which is Section 5.1. Everything else is held fixed: the same preferences, the same calibration, the same solver family, and the same simulated skill histories. Every result below is a comparison across those two worlds, and the contrast is sharper than the level of either.

The solver extends the dual-costate recursion of Farhi and Werning (2013) and Golosov et al. (2016) to two consumption goods, a clean numéraire and a polluting energy good, and prices the externality as a constant marginal damage e entering the planner's resource constraint, so the social producer price of energy is $1 + e$. Two assumptions take the social cost of carbon of Section 4.3 to that constant. First, the cohort is small relative to the world whose damages e measures: its emissions carry the full global social cost but do not move the world pollution stock, so nothing feeds back into its own utility and no pollution-stock state is carried. Second, damages are denominated in consumption units rather than utils, so a unit of energy costs $1 + e$ out of the resource pool that finances consumption. This is exactly the restriction $\xi_t = \gamma_{c,t}$ of Proposition 4.2: the fiscal-adjustment bracket is one at every node, the social cost of carbon equals its Pigouvian counterpart, and the benchmark is $\tau^q = \text{SCC}^P = e$. The same specification makes the extra damage weight of Proposition 5.1 inert in the frozen world, which is what lets

the third best be benchmarked against the same e . Writing damages as $U(c - D(S), q, l)$ gives $U_{cS} = -U_{cc}D'$ alongside $U_S = -U_cD'$, so $\xi_{c,t} = U_{cS}c/U_S = U_{cc}c/U_c = -\sigma_t$ and the term $\zeta_t(\sigma_t + \xi_{c,t})/c_t$ vanishes identically, whatever the net shadow price does.

Both are scope restrictions rather than results. The social cost of carbon that Proposition 4.2 derives is a within-cohort object, pricing the damage this cohort's emissions do to this cohort over this cohort's horizon. The number policy uses is not that object: the EPA's \$190 prices damages to people outside the cohort and beyond its life, which a single-cohort model cannot generate. Rather than derive a smaller and less relevant number, I import the policy one and hold it fixed. This splits the carbon price cleanly in two. Its level comes from the integrated-assessment literature and is exogenous here; its structure over the life cycle and across skill histories is endogenous, and is what this section measures. Every result below is a ratio to e or a deviation from it, so rescaling the social cost of carbon rescales the levels and leaves the structure intact. The pollution-stock dynamics of Section 4.3 are a separate exercise.

6.1 Model, solvers, and calibration

The life-cycle block is Farhi and Werning (2013)'s, replicated and validated before the two-good extension. Agents live sixty years, work for forty and retire for twenty, so $T = 59$ and $T_R = 40$, and period t is age $25+t$: labor-market entry is at 25, the last working year is 64, and retirement runs from 65 to 84. I report ages throughout; log-skills follow a geometric random walk with innovation variance $\hat{\sigma}^2 = 0.0095$; $\beta = 1/R = 0.95$, so $\beta R = 1$; skills are private information and the incentive constraints are handled by the first-order approach. The planner is utilitarian. Following Farhi and Werning (2013), newborns are ex ante identical, so the initial cross-section is the first innovation of the random walk: the initial skill dispersion is already the $\hat{\sigma}^2 = 0.0095$ above, there is no separate initial-skill variance to set, and the common initial promised value is pinned by the economy's resource constraint. Each period the agent consumes a clean numéraire and an energy good (motor fuel, electricity, gas, fuel oil), both with producer price one. Period utility is

$$U(c, q, l) = \frac{1}{\rho} \log[(1 - b_q) c^\rho + b_q e^{-\kappa l^\alpha} (q - \bar{q})^\rho] - \frac{l^\alpha}{\alpha} \quad \rho \equiv 1 - \frac{1}{\varsigma}$$

a CES aggregate over the clean good and energy net of subsistence $\bar{q}$, inside the log shell that preserves Farhi and Werning (2013)'s balanced-growth homogeneity, with isoelastic labor disutility of curvature $\alpha = 3$ (Frisch elasticity one half). ς is the elasticity of substitution between the goods, b_q the energy weight, and $\bar{q} = 0$ in the homothetic benchmark; $\bar{q} > 0$ is the Stone-Geary necessity variant. The twist $e^{-\kappa l^\alpha}$ makes energy a relative leisure complement: its utility weight falls with labor effort, so $U_{ql} \neq 0$ with the sign, in the notation of Section 3, that makes the complementarity gap $\gamma_c - \gamma_q$ positive. This is the empirically relevant case (West and Williams (2007); Christiansen (1984)), and the one that pushes the optimal

carbon tax above Pigouvian. The pollution stock does not appear in U because damages are denominated in consumption units: writing them as $U(c - D(S), q, l)$ gives $U_S/U_c = -D'$ and $U_{Sl}/U_S = U_{cl}/U_c$, so $\xi_t = \gamma_{c,t}$ holds identically, the fiscal-adjustment bracket of Proposition 4.2 is one at every node, and SCC_t coincides with its Pigouvian counterpart throughout this section; every derivative above is unchanged.

The two-good calibration is disciplined by four US moments (Table 1), two of which bind in the homothetic benchmark. The energy expenditure share is 7%, the round benchmark spanning the Consumer Expenditure Survey's 7.6% in 2022 and 6.5% in 2023; motor fuel is half the composite and its price fell sharply between the two years. The own-price elasticity of energy is 0.4, which at a 7% share implies a substitution elasticity $\varsigma \approx 0.35$; I use $\varsigma = 0.4$, so the model's own-price elasticity at the calibrated share is 0.44. This is a long-run value, the horizon a forty-year model asks for, and a conservative one: Labandeira et al. (2017) put the long-run elasticity of aggregate energy demand at about 0.6, against a short-run value roughly one third as large, and the welfare numbers scale with it. The income elasticity of energy is 0.45, below one because energy is a necessity. This is a cross-sectional Engel gradient rather than a long-run elasticity, and it sits inside but well below the top of the range in the residential-energy meta-analyses, which report income elasticities from about 0.15 in the short run to near unity in the long run (Espey and Espey 2004; Zhu et al. 2018). Nothing in the paper's headline turns on it: the homothetic benchmark sets it to one, and only the Stone-Geary variant imposes 0.45. The share moment pins the small CES energy weight b_q , and under Stone-Geary the income-elasticity moment implies a subsistence level $\bar{q} \approx 0.04$, about 57% of mean energy spending, so more than half of energy consumption is price-inelastic at the mean. The leisure complementarity $\varkappa$ is set so that, with the externality off, the second-best energy wedge at mid-career is about 10% ad valorem. West and Williams (2007) put the zero-externality gasoline tax at 33% of the pre-tax price and attribute 13 points of it to nonseparability, which is the part $\varkappa$ parameterizes; 10% is that figure tempered for the household-energy composite, whose labor cross-effects are weaker than motor fuel's.⁹ The marginal damage is $e = 0.494$: the EPA's central social cost of carbon of \$190 per ton of CO₂ (U.S. Environmental Protection Agency 2023), in 2020 dollars at a 2% near-term Ramsey discount rate for emissions in 2020, times the carbon intensity of household energy spending, 0.0026 tons per producer dollar.¹⁰ Because energy's producer price is one, the Pigouvian wedge $\tau^q = e$ is a 49% ad valorem tax on energy.

⁹They estimate a four-good demand system in gasoline, male leisure, female leisure, and other goods, so there is no home-energy counterpart to extrapolate from, and they caution against extrapolating to goods used differently from gasoline. The tempering is therefore an assumption, defensible only in that it errs toward understatement. Their sign is not unconditional either: dropping state fixed effects reverses gasoline from a leisure complement to a substitute, which would flip $\varkappa$ and with it the sign of the premium.

¹⁰The intensity aggregates 2023 emission factors and pre-tax prices for motor fuel, electricity, and natural gas; the unrounded spending-weighted average is nearer 0.0027, which would put e at about 0.51. I round down, so the benchmark is conservative, and e scales one-for-one with this coefficient. Pairing a social cost in 2020 dollars with an intensity whose denominator is a 2023 dollar understates e by a further margin, in the same direction.

Table 1: Calibration

parameter	value	moment or source
<i>life-cycle block (Farhi and Werning 2013)</i>		
discount factor $\beta = 1/R$	0.95	FW13 ($\beta R = 1$)
labor-disutility curvature α	3	Frisch elasticity 0.5
log-skill innovation variance $\hat{\sigma}^2$	0.0095	US male wages, via FW13
working / retirement periods	40/20	FW13
<i>two-good block</i>		
energy expenditure share	0.07	Consumer Expenditure Survey, 2022–23
own-price elasticity of energy	0.4	long run, energy-demand meta-analyses
income elasticity of energy	0.45	Engel gradient; unity in the homothetic benchmark
leisure complementarity $\varkappa$	0.25	mid-career zero-externality wedge $\approx 10\%$ (West and Williams 2007)
marginal damage e	0.494	\$190/tCO ₂ , 2020\$ (U.S. Environmental Protection Agency 2023) $\times 0.0026$ tCO ₂ /\$
<i>frozen schedules, Appendix D.1</i>		
labor progressivity p	0.181	Heathcote et al. (2017)
capital income tax τ^k	0.18	effective marginal rate, Burnham and Carloni (2022)
implied savings wedge $\bar{\tau}^s$	0.9% per year	(58) at $\beta R = 1$

The frozen schedules of the second world are the ones calibrated in Appendix D.1: the Heathcote et al. (2017) tax function at $p = 0.181$ for the labor margin, and a capital income tax at the US effective marginal rate $\tau^k = 0.18$ for the savings margin, which by (58) implies $\bar{\tau}^s = 0.009$ at $\beta R = 1$. Because the carbon price turns out to be far more sensitive to the savings schedule than to the labor schedule, Section 6.4 solves the model across a grid of τ^k running up to the 0.36 of the aggregate effective-rate measures; everything else in this section is at the benchmark.

Three solvers deliver the results, and each is anchored on the one before it. The *second best* assigns full consumption bundles, so the commodity wedge is unrestricted; this solver requires homothetic preferences. *Linear commodity taxation* layers an anonymous rate τ^q on the optimal nonlinear income and savings mechanism through the exact one-good reduction of Lemma 5.5, which is valid with subsistence and hence usable for the necessity calibration. The *third best* adds the two frozen schedules, with the marginal-utility requirement $\hat{M}$ of Lemma 5.1 carried as

a third state, and runs either with the commodity margin free or with it restricted to a flat rate. Two agreements discipline the stack. At $\gamma_c = \gamma_q$ the second best sets $\tau^q = e$ at every history and every age, the dynamic Atkinson-Stiglitz result, and coincides with a flat Pigouvian tax to solver precision. With the frozen schedules set at their desired counterparts the third-best solver returns the second best and its multipliers vanish, which is the $\eta_t \equiv \eta_t^s \equiv 0$ limit of Section 5.1. All welfare comparisons use common random numbers, over 300,000 simulated histories for the second-best ladder and 60,000 for the third-best solvers, so within each comparison the same agents face every regime, and every regime pays for its own emissions in the planner's budget at the social price $1 + e$ whatever the consumer price: no rung is flattered by unpriced damages. Flat rates are optimized by a bracketing search with the solver's discretization and the simulated histories held fixed across rates, confirmed by a solve at the fitted optimum. Under the frozen income tax this gives $\tau^{q*} = 0.121$ at the benchmark capital tax and 0.087 at $\tau^k = 0.36$, each with a standard deviation of about 0.002 across simulation panels and a welfare surface flat to about a thousandth of a percentage point within 0.03 of the optimum; the rate at which the gain returns to zero is bracketed by solves on the same geometry at 0.24 and 0.32. Appendix D.2 documents the implementation and Appendix D.3 the validation battery.

The incentive compatibility of the first-order-approach solution is verified ex post by a one-shot-deviation audit at simulated states, run on the second-best benchmark, on the Stone-Geary and placebo calibrations, and on the third-best stores. The third-best production grid is far coarser than the second best's, so its audit is read against an unconstrained solve at the identical grid, whose menu-interpolation noise bounds what the audit can resolve there: the frozen-wedge solutions show no deviation gains in excess of that control, and the retirement-junction rows, where the continuation value is closed-form and the audit is sharpest, are clean in every store. Appendix D.4 tabulates the audit.

6.2 The carbon wedge over the life cycle

Figure 1 is the paper's central quantitative fact, and the two curves in it are the same object computed under the same calibration.

When the income tax is set optimally, the carbon wedge barely leaves the social cost of carbon. At labor-market entry the labor wedge is near zero, there is no incentive motive yet, and the carbon wedge opens within a point of the Pigouvian level $e = 0.494$. It rises concavely to about 0.63 near retirement, a premium over the Pigouvian gross price that tops out around 9%, and at retirement it falls back to exactly 0.494: with no labor to report the costate vanishes, the screening term dies with it, and retirees face pure Pigouvian pricing (Appendix C.12). The whole path lies within a tenth of the social cost of carbon.

When the income tax is instead the one the US actually levies, the wedge starts in the same place and then rotates away from it. It opens at 51%, less than two points above the Pigouvian level, crosses it at 28, falls to 22% by 45 and -8% at 64, and reaches a 34% energy *subsidy*

Table 2: Cross-sectional mean wedges by age over the working life, income tax optimal. The premium is measured against the consumer price $1 + \tau^q$, the normalization of (4), so that it and the identity below it are the same object; against the Pigouvian gross price $1 + e$ it reaches 0.092 at 64.

age	25	35	45	55	64
labor wedge τ^y	0.021	0.180	0.280	0.341	0.363
carbon wedge τ^q	0.504	0.573	0.609	0.625	0.632
s.d. across histories	0.001	0.005	0.015	0.031	0.055
premium $(\tau^q - e)/(1 + \tau^q)$	0.0065	0.0505	0.0713	0.0806	0.0844
identity $\kappa l^\alpha \tau^y / (1 - \tau^y)$	0.0064	0.0499	0.0705	0.0796	0.0826

at 84. That is a rotation through 85 points of ad-valorem price, against the 14 points the second-best wedge covers over the same sixty years, and its working-life mean is 23%, about half the Pigouvian level. The freeze does not move the level at which the carbon price starts; it bends the whole path.

The premium in the first world is a scaled labor wedge (Table 2). Its first-order value is $\kappa l^\alpha \tau^y / (1 - \tau^y)$, and this identity tracks the solved premium to within 2.1% at every age. The identity sits just below the solved premium throughout, by about 1% through mid-career and by 2.1% at 64, where the labor wedge is flattest and the first-order approximation to it is worst. The non-Pigouvian part of the carbon wedge therefore inherits the labor wedge’s level and its life-cycle rise, which is the proportionality result of Section 4.2 made quantitative. The income-side wedges are in turn essentially undisturbed by the second good: the labor wedge runs from 0.02 to 0.36 over the working life against 0.02 to 0.37 in the one-good economy, and the savings wedge falls from 0.56% to 0.02%, the Farhi and Werning (2013) pattern intact. The commodity margin adds a corrective and screening instrument without reshaping the other two, which is the separation showing up in the solved policies.

History dependence separates the two worlds as sharply as the path does. The second-best wedge’s cross-sectional standard deviation fans out from 0.1 points early in life to 5.5 points near retirement, mirroring the spreading of the labor wedge as skill histories diverge. The third-best wedge’s reaches 18 points in the early sixties, more than three times as wide, and the bands in Figure 1 separate from mid-career on. Against a given income tax, the price the mechanism would charge depends heavily on which history an agent has drawn.

6.3 Who bears the wedge

The optimal carbon wedge rises with skill in both worlds (Figure 2). Under an optimal income tax it is indistinguishable across quintiles early in life, when every history looks alike, and fans

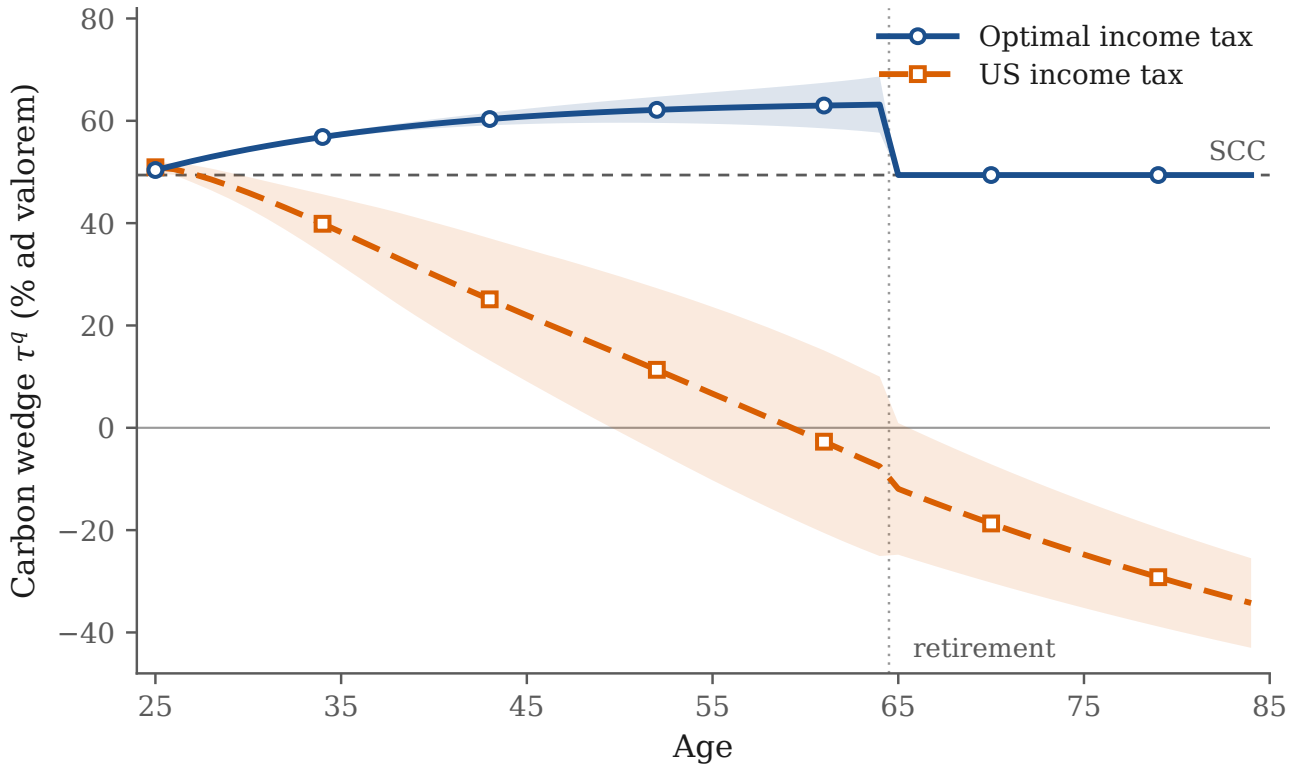


Figure 1: The optimal carbon wedge over the life cycle, benchmark calibration, when the income tax is set optimally alongside it and when it is frozen at the US schedules. Bands are ± 1 cross-sectional standard deviation across skill histories; the dashed horizontal line is the Pigouvian level $\tau^q = e = 0.494$. The retirement segment of the optimal-income-tax path is analytical rather than simulated (Appendix C.12).

out from mid-career: by 64 it reaches about 70% for the top skill quintile against 57% for the bottom, a spread of about 14 points. Under the US income tax the same ranking holds and the spread is three times as wide, 44 points at 64, running from a 29% energy subsidy for the bottom quintile to a 15% tax on the top. High current productivity signals a history whose incentive constraints are expensive, and the commodity margin is the planner's lever on exactly those histories, so the screening value of the instrument concentrates on high-skill agents whether or not the income tax is optimal.

Two readings follow, and the second is the one that survives dropping the optimality assumption. The optimal carbon wedge is not regressive: the planner taxes the energy of high-skill households more, not less, so the redistributive worry about carbon pricing is misplaced at the level of the optimal type-contingent wedge. And this is not an artifact of assuming the rest of the tax system is right. Freezing the income tax at the observed US schedules leaves the progressivity intact and sharply strengthens it. The regressivity of real carbon taxes comes from elsewhere:

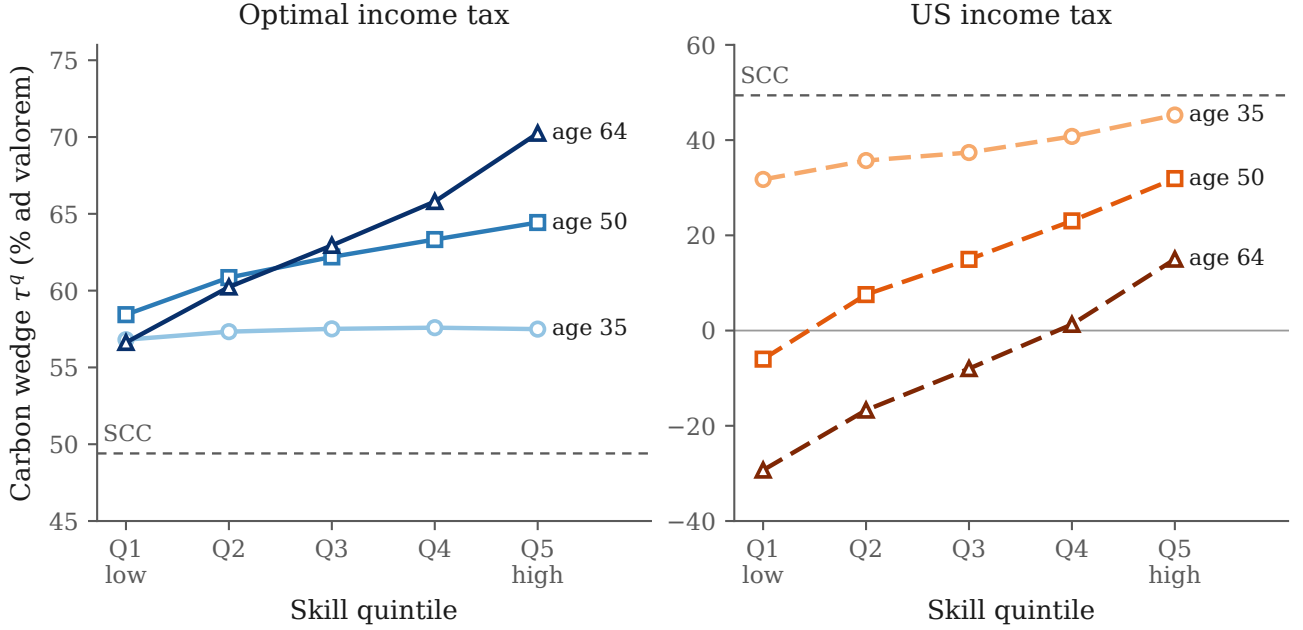


Figure 2: The optimal commodity wedge τ^q by skill quintile at ages 35, 50 and 64, when the income tax is set optimally (left) and when it is frozen at the US schedules (right). Shading runs light to dark in age. The dashed horizontal line is the Pigouvian level. Note the different vertical scales: the comparison the figure is drawn for is the slope in skill, whose magnitudes are given in the text.

from the flat-rate restriction, since an anonymous point-of-sale tax cannot condition on skill, and, in the necessity calibration, from energy's falling budget share.

Figure 2 is a schedule of marginal wedges by skill, not a map of the burden each household bears. That burden is set jointly by the income tax and transfers alongside the wedge, so the schedule's progressivity speaks to the corrective instrument itself, not to the full incidence of carbon pricing.

6.4 Where the rotation comes from

Section 5.1 predicts that the two frozen margins reach the carbon price through one statistic, $\zeta_t = \eta_t(1 - \bar{\tau}_t^y) - \eta_t^s$, and that they enter it asymmetrically (Lemma 5.4). The labor gap is a within-date object whose sign profile follows the schedule's; here it holds each sign for roughly half the working life. The savings contribution can never hold one sign: whatever the schedule, it raises ζ_t at some ages and lowers it at others (Lemma 5.3), so a savings freeze cannot move the whole profile one way. The two enter ζ_t additively, with coefficients that are one-signed under the condition $A_t > 1$ of Lemma 5.4, which the calibration below satisfies with room.

Imposing the two restrictions in sequence separates their contributions. The first solve freezes the labor schedule and leaves the savings margin free, which isolates the labor channel. The second freezes both, which measures what the savings restriction adds. Figure 3 reports the sequence. Throughout, the *pull* of a restriction is the difference between consecutive solves, so the two pulls sum to the total by construction but their split depends on the order in which the restrictions are imposed. Only the total is order-invariant.

The labor gap is a mismatch in shape rather than in level. The frozen schedule is nearly flat in age, falling from 33% at 25 to 30% at 64, and its slope is a composition effect rather than a modeling choice, since the Heathcote et al. (2017) function conditions on earnings and earnings fall over the life cycle here. The Mirrleesian wedge meanwhile climbs from 2% to 36%. The two cross at 49. Labor is overdistorted for the first two decades of working life and underdistorted for the last, so by Proposition 5.1 the carbon wedge is pulled below its second-best counterpart early and above it late. It opens 27 points below that counterpart, sits 8 points above it at 64, and the pull changes sign at 50. The two ages differ by a year because η_t is signed against the desired wedge of Lemma 5.2 evaluated at the constrained allocation, not against the second-best wedge that panel (a) draws. Nothing about this makes the carbon price low on average: with only the labor schedule frozen, the wedge runs from 23% at 25 to about 71% at 64, well above Pigouvian at the end of working life, and returns to the social cost of carbon at retirement.

The savings gap is one-signed and its pull is not. Panel (b) is drawn to keep the two apart. The frozen wedge is 0.9% per year against an unconstrained-optimal wedge that starts at 0.56% and decays to zero, so the US system overdistorts the intertemporal margin at every age. A frozen savings wedge restricts a ratio of marginal utilities across dates rather than a level within a date, so what matters is the distortion accumulated by each age rather than the gap at that age. By (11) mean marginal utility is forced 71% above its date-25 value over the life cycle, where the optimal allocation leaves it essentially flat. The pull on the carbon wedge is *positive* at entry, 28 points, crosses zero at 34, and reaches -84 points at 84. That crossing is not the sign change of η_t^s discussed below, which sits a decade later: a pull is a difference between two solves, and freezing the savings margin moves η_t as well.

The two pulls are equal and opposite at entry, 27 points down against 28 up, so the wedge opens at 51%. They cross at 39, and the savings pull's margin over the labor pull widens for the rest of life. The asymmetry at the other end of life is sharper: the labor pull dies at 65 and the savings pull does not (panel d), which is Corollary 5.3 in the solved policies.

Weighting each age by the emissions it carries converts the two pulls into prices. The labor freeze lowers the effective carbon price, the discounted-emissions-weighted mean of the wedge defined in Section 6.5.2, from \$218 to \$169 per ton. The savings freeze lowers it a further \$62, to \$107. Two fifths of that \$62 comes from the twenty retirement years alone, where the labor multiplier is dead and the savings multiplier runs unopposed, even though those years carry only 9% of discounted emissions.

Less than half of the savings channel is the capital-tax rate. The rest is the restriction itself, which binds even at a zero rate. Table 3 and the left panel of Figure 4 report the grid. At $\tau^k = 0$ the deviation is negative at every age and nearly flat, $\delta = -0.27$ at 25 against -0.02 at 64, and the working-life mean carbon price is already down to 0.36 from a second best near 0.60. A zero capital income tax is not a neutral benchmark on this margin: any constant $\bar{\tau}^s$ freezes the Euler equation, the second best satisfies the inverse Euler, and by Jensen's inequality the two cannot hold together under uncertainty, so the restriction binds at every τ^k including zero. Turning on the capital tax rotates the profile around that level, and the rotation grows monotonically in τ^k : at the benchmark $\tau^k = 0.18$ the deviation runs from $+0.01$ at 25 to -1.27 at 84, and at the $\tau^k = 0.36$ upper bound from $+0.29$ to -2.98 . The sign pattern is the one Appendix B.7 derives analytically at a degenerate draw, with the price above the social cost of carbon early and below it late. Over the random walk of working life that argument delivers the reversal but not its date, so where the crossing falls is a solved property of this calibration rather than a consequence of it.

Two features of the rotation are worth separating from its size. The age at which $\hat{\eta}^s$ changes sign moves by at most a year across the capital-tax grid, from 38–39 at $\tau^k = 0$ to 39–40 at every positive rate: the multiplier's shape is a property of the boundary conditions $\vartheta_0 = 0$ and $\eta_T^s = \vartheta_T U_{c,T}$, not of the size of the wedge gap. Lemma 5.3 forces a reversal but does not locate it; that the date barely moves across the grid is a quantitative finding about the grip of those boundary conditions. That date is not the crossing age of Table 3, and Lemma 5.4 says why the two need not coincide: by (10) the deviation vanishes where the savings term offsets the labor term, not where η_t^s itself does. The crossing age therefore depends on the two terms' relative size and moves with τ^k , from 28 at the benchmark to 36 at the upper bound, while the sign change of the multiplier is fixed by the boundary conditions and does not move at all. The same reading accounts for the dashes in the table. At the three lowest rates the savings term never grows large enough to reverse ζ_t over the working life, and η_t^s holds one sign through retirement, where it is the whole of ζ_t ; the deviation therefore keeps one sign at every reported age even though η_t^s still changes sign just before 40. The savings channel pushes toward a crossing without delivering one. And the deviation survives into retirement while the labor channel does not, which is Corollary 5.3: from T_R on the labor constraint has no content and $\zeta_t = -\eta_t^s$, so the entire retirement deviation is the savings multiplier.

The deviation at labor-market entry is monotone in τ^k and crosses zero at $\tau^k \approx 0.17$, just below the benchmark (Figure 4, right panel). At the US effective marginal rate on capital income the labor and savings multipliers therefore very nearly cancel at entry: the young face a carbon price within a point of the social cost of carbon, and the whole third-best deviation is a rotation with age rather than a shift in level. This is Corollary 5.2 realized at an observed capital-tax rate rather than at a knife-edge.

Nothing in the section rests on that coincidence. The grid is reported in full so that the reading

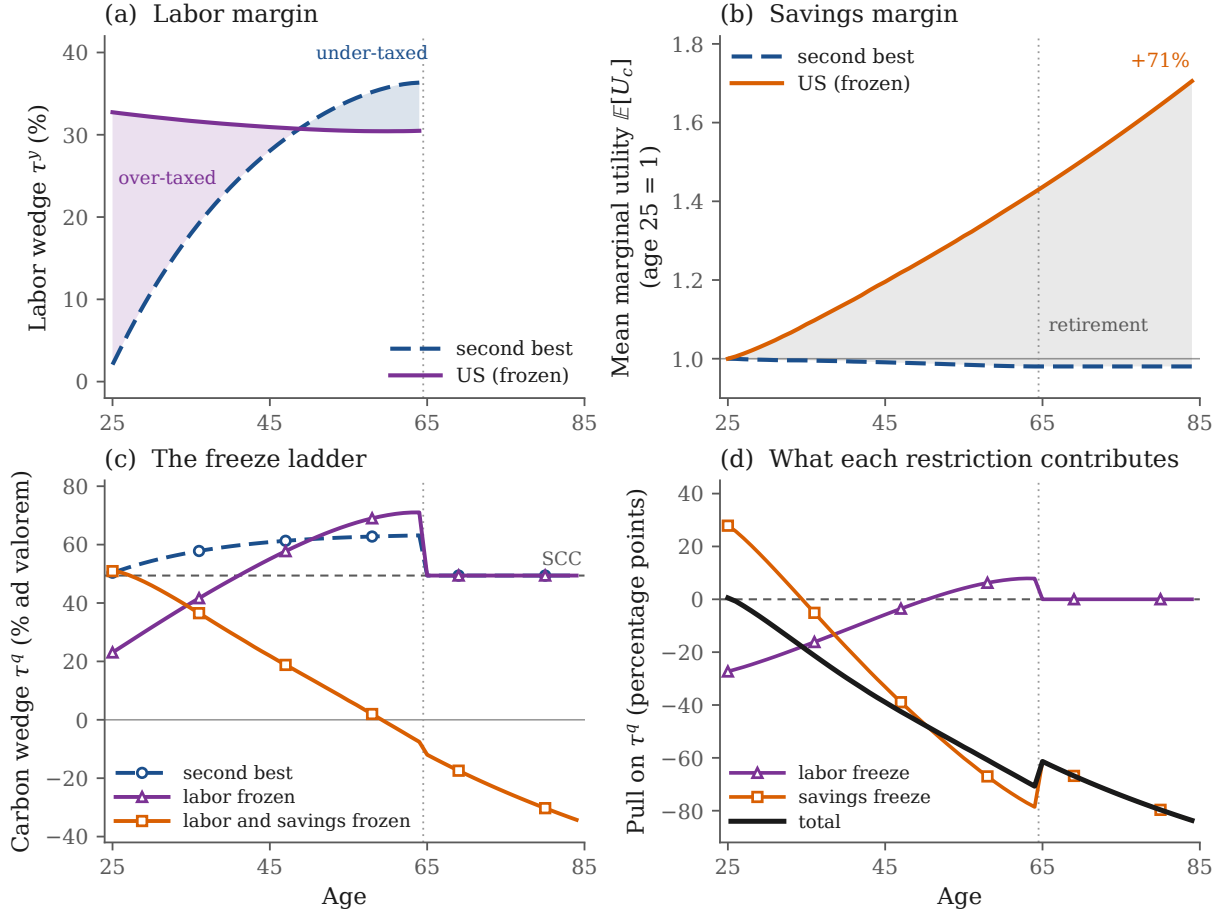


Figure 3: The two frozen schedules and what they do to the carbon wedge. The benchmark $\tau^k = 0.18$ throughout, and a *pull* is the difference between consecutive paths in (c). Blue is the unrestricted second best, purple the labor restriction, orange the savings restriction, black their total; the second best is dashed and anything frozen is solid. (a) The frozen labor schedule at its age- t anchor against the second-best wedge at the same age; shading takes the color of whichever curve is on top, purple where the frozen schedule overdistorts labor and blue where it underdistorts. (b) The savings margin as the accumulated distortion: mean marginal utility under each regime, indexed to one at age 25, with the grey band the gap between them. Levels of U_c are not comparable across allocations; (11) restricts the growth rate, which is what the panel plots. The shading is neutral because a frozen savings wedge constrains a ratio across dates, so its multiplier changes sign (Lemma 5.3) and this gap's sign does not map into the sign of the pull. Both retirement segments are analytical: the second best holds $\mathbb{E}[U_c]$ constant once risk resolves, and the frozen path compounds at $1/(1 - \bar{\tau}^s)$. (c) The carbon wedge with nothing frozen, with the labor schedule frozen, and with both margins frozen; the retirement segments of the first two paths are analytical (Appendix C.12). (d) The two pulls, which sum to the total at every age.

Table 3: The carbon price against a given income tax, across the capital-tax grid. The labor schedule is frozen at the Heathcote et al. (2017) function in every row of the grid, and the savings margin is frozen in every row of the grid including the first: $\tau^k = 0$ sets $\bar{\tau}^s = 0$, which still restricts the Euler equation. The memo row is the unrestricted second best, the reference the frozen rows move from. The labor restriction’s pull alone is isolated by the second rung of Figure 3, not by this table.

τ^k	$\bar{\tau}^s$ (%/yr)	δ at 25	crossing age	δ at 64	δ at 84	working-life mean τ^q
0	0	−0.27	—	−0.02	−0.10	0.36
0.025	0.13	−0.26	—	−0.13	−0.26	0.29
0.05	0.25	−0.22	—	−0.21	−0.41	0.27
0.18	0.90	+0.01	28	−0.62	−1.27	0.23
0.36	1.80	+0.29	36	−1.31	−2.98	0.22
<i>memo</i> : second best	free	+0.01	—	+0.08	0	0.59

Note: $\text{SCC} = e = 0.494$, and $\delta_t \equiv (\tau_t^q - \text{SCC})/(1 + \tau_t^q)$ is the deviation of (4), evaluated at the date- t cross-sectional mean carbon price over simulated histories. Since δ is convex in τ^q this is not the mean of δ across histories; it is reported this way so that it inverts exactly into the τ^q levels of Section 6.2. “Crossing” is the age at which the deviation changes sign, and a dash means it keeps one sign over the whole life cycle. In the memo row the same deviation statistic is the premium of Table 2, computed from the same solve; its retirement value is exactly zero (Appendix C.12), and *free* marks the margins the second best leaves unfrozen. $\tau^k = 0.18$ is the benchmark of Table 1 and $\tau^k = 0.36$ the upper bound of Appendix D.1.

does not depend on where the benchmark falls, and every qualitative feature above survives at every rate on it. What the benchmark fixes is the size of the rotation and the age at which the price crosses the social cost of carbon, both of which rise with τ^k .

6.5 What carbon pricing is worth

The welfare accounting inverts between the two worlds, and the inversion is the section’s main result (Figure 5).

6.5.1 When the income tax is optimal

Table 4 reports consumption-equivalent gains over having no carbon price, under the optimal nonlinear income and savings mechanism throughout. Four findings, in order of size.

The value of a carbon price is overwhelmingly Pigouvian. At the EPA social cost of carbon, having any carbon price is worth 0.31% of consumption, and a flat rate at the social cost of carbon alone captures 98% of the full second-best value while cutting emissions by 13.5%. For scale, Farhi and Werning (2013) put the entire insurance value of their optimal income-tax apparatus over laissez-faire at 1.56% of consumption: the carbon instrument is worth about a

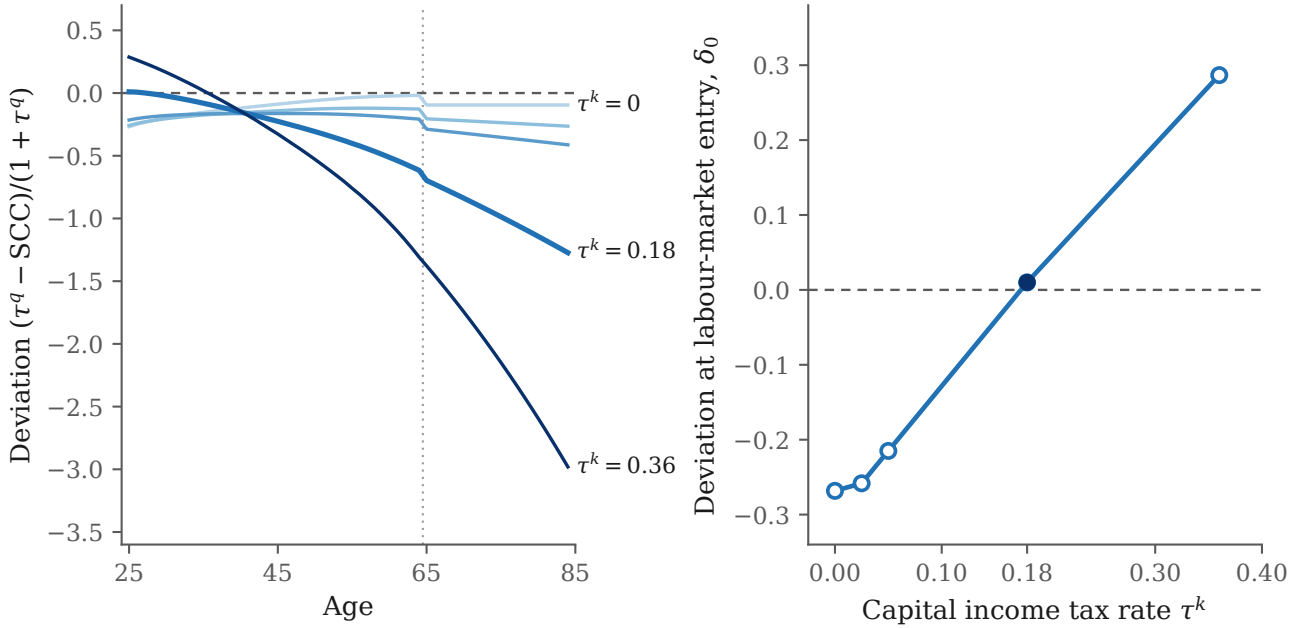


Figure 4: Left: the deviation $\delta_t = (\tau_t^q - \text{SCC}) / (1 + \tau_t^q)$ over the life cycle across the capital-tax grid, with the labor schedule frozen at the US function throughout. Shading runs light to dark in τ^k over the grid of Table 3; the benchmark $\tau^k = 0.18$ is drawn heavier. The dashed horizontal line is Pigouvian pricing and the dotted vertical line is the retirement date. Right: the same deviation at labor-market entry, plotted against τ^k , with the benchmark rate marked by a filled circle.

fifth of that at this social cost of carbon.

The optimal flat rate exceeds the social cost of carbon by about 12%, $\tau^{q} = 0.555$ against $e = 0.494$, or \$214 per ton against \$190, the flat reflection of the Corlett-Hague premium. But the welfare gain from that uplift over pricing at the social cost of carbon is 0.002%: the objective is very flat in the rate, so “price at the social cost of carbon” is a near-optimal rule of thumb even with nonseparable preferences.*

Sophistication buys almost nothing beyond the level. Making the rate age-dependent adds about 0.001 percentage points over the optimal flat rate, and full history dependence about 0.001 more, both at the numerical resolution. The premium varies over life and across histories by a few points of an equilibrium budget share near 6% with a substitution elasticity of 0.4, so the deadweight loss from mistracking it is second-order squared.

The pure Corlett-Hague motive is tiny, and necessity halves the stakes. With the externality off, the entire commodity-wedge apparatus of the second best is worth 0.004% of consumption, and its optimal flat rate is about 4%, materially positive as in West and Williams (2007)

Table 4: Welfare value of carbon pricing under progressively simpler rules, income tax optimal (consumption-equivalent gain over $\tau^q = 0$, % of consumption)

	benchmark ($e = 0.494$)	Stone-Geary (necessity)	$\gamma_c = \gamma_q$ (placebo)	no externality ($e = 0$)
second best (nonlinear τ^q)	0.313	—	0.252	0.0041
age-dependent linear $\tau^q(t)$	0.312	—	—	0.0032
flat τ^{q*} (optimized)	0.311	0.152	—	0.0019
optimal flat rate τ^{q*}	0.56	0.56	—	0.038
flat Pigouvian $\tau^q = e$	0.309	0.150	0.253	—
emissions vs $\tau^q = 0$: second best	−15.0%	—	−13.5%	−1.8%
flat Pigouvian	−13.5%	−5.7%	−13.5%	—

Note: welfare entries are in % of consumption; the optimal-flat-rate row is an ad-valorem rate, and the final two rows report emissions changes relative to $\tau^q = 0$. The rungs are separate solves whose simulated budgets agree only to about 0.003 points of consumption, the ladder’s numerical resolution: differences below it measure that alignment as much as any policy. Sampling error is far smaller. With common random numbers the seed-to-seed standard deviation is about 0.0003 points on each entry and below 0.0001 on any difference between rungs, so sub-resolution differences are stable across simulation panels; they are just not resolved. The no-externality column sits at the resolution throughout. The optimized flat rate abates slightly more than the Pigouvian one where the two differ: 6.3% under Stone-Geary and 1.4% in the no-externality column, where no Pigouvian rate exists. Dashes mark three cases. *Not computed*: the second best requires homothetic preferences, so it and the age-dependent rung are blank under the non-homothetic Stone-Geary calibration. *Degenerate*: under the $\gamma_c = \gamma_q$ placebo the second best already coincides with the flat Pigouvian tax, so the intermediate rungs add nothing. *Not applicable*: with no externality ($e = 0$) there is no Pigouvian benchmark.

but negligible in welfare and barely identified, since the whole column sits at the numerical resolution. Under the Stone-Geary calibration, in which energy’s budget share falls with income, the value of the flat carbon tax falls to 0.15% and its abatement to 6%, because the subsistence part of energy does not respond to price, while the optimal flat rate is essentially unchanged. This is the calibration in which a flat carbon tax is regressive, yet the mechanism absorbs the incidence through the income side, which is why the flat instrument loses so little welfare despite its incidence. As a check on the whole apparatus, switching the leisure complementarity off ($\gamma_c = \gamma_q$) makes the second best and the flat Pigouvian tax coincide in welfare at the numerical resolution and exactly in emissions: dynamic Atkinson-Stiglitz, and the exact knife-edge of Proposition 4.3, reproduced through two structurally different solvers.

6.5.2 When the income tax is the one the US levies

Every one of those conclusions reverses. Table 5 runs the same rate grid with the two income-side wedges frozen, and Figure 5 plots the two rate curves side by side. Both panels come from the same solver family at the same grid with the same simulated histories, so the only difference between them is the freeze.

A flat carbon tax at the social cost of carbon is welfare-negative. It costs 0.115% of consumption relative to having no carbon price at all, and 0.155% at the $\tau^k = 0.36$ upper bound. This is not an artifact of ignoring the damages the tax avoids: the planner’s budget prices energy at the social price $1 + e$ in every regime, including the one with no carbon tax, so the comparison is damage-inclusive by construction. With the income tax stuck at US levels, legislating the social cost of carbon as a flat rate overshoots the constrained optimum badly enough that the economy would rather have no carbon price.

The optimal flat rate is a quarter of the social cost of carbon, and it is worth almost nothing. It is 0.12, against 0.56 when the income tax is optimal, and it buys 0.02% of consumption over charging nothing. Welfare gains run out entirely at 0.254, near \$98 per ton, interpolated between solves on the same geometry at 0.24 and 0.32. The social cost of carbon is therefore four times the best flat rate and twice the highest flat rate that does no harm at all. A single rate must average a profile that runs from above the social cost of carbon at entry to a deep subsidy in retirement, weighted by substitution responses, and the twenty retirement years drag that average far down. The below-damages prescription of the Ramsey tradition reappears here, and more starkly than in the linear-instrument literature, because a frozen income tax cannot absorb any of the burden.

Ambition falls by much less than the flat rate suggests. Within the same frozen world the nonlinear wedge abates 8.0% of discounted emissions against the 4.3% the best flat rate delivers, and its effective carbon price, the average rate a ton of emissions actually faces,¹¹ is 0.28 rather than 0.12: \$107 per ton against \$47. Emissions concentrate in mid-career, where the wedge is still well above zero, and only 9% of discounted emissions fall in the retirement years the subsidy reaches. What a miset income tax mostly destroys is the flat instrument, not the case for pricing carbon.

The near-sufficiency of flat carbon taxation is overturned. When the income tax is optimal, the entire nonlinear fine structure of the wedge is worth 0.002 percentage points over the best flat rate. When it is frozen at the US schedules, the nonlinear wedge is worth 1.38 percentage points, more than five hundred times as much and four times the entire second-best value of carbon pricing. Under a miset income tax the commodity margin’s history dependence performs income-tax repair, and no anonymous rate can imitate it.

That last number is best read as a statement about commodity taxation rather than about carbon. Energy is the only margin the constrained planner still controls freely, so it absorbs the whole repair job; with more consumption goods the same logic would spread across every

¹¹The discounted-emissions-weighted mean of the age-varying wedge, $\sum_t \beta^t E_t \tau_t^q / \sum_t \beta^t E_t$, with E_t the cohort’s date- t emissions under the policy being evaluated and $\beta = 1/R$. Equivalently, the flat rate that would collect the same amount per ton over the same emissions path, which is what makes a life-cycle profile comparable to a single rate. It exceeds the unweighted mean of τ_t^q because emissions concentrate at ages where the wedge is high, and each regime carries its own emission weights.

Table 5: Welfare of a flat carbon tax when the income-side wedges are frozen at the US schedules (consumption-equivalent gain over $\tau^q = 0$ of the same regime, % of consumption). The “income tax optimal” column is the same rate grid solved with the income and savings wedges free, at the same grid resolution and on the same simulated histories, and is reported for comparison.

flat rate τ^q	income tax optimal	income tax given
0 (no carbon price)	0	0
0.10	+0.119	+0.021
0.20	+0.204	+0.014
0.35	+0.280	−0.035
Pigouvian, $e = 0.494$	+0.311	−0.115
0.65	+0.310	−0.226
best flat rate τ^{q*}	0.56	0.12
its welfare gain	+0.315	+0.021
nonlinear τ^q	see Table 4	+1.401

Note: entries are consumption-equivalent gains in % of consumption; the best-flat-rate row is an ad-valorem rate. The right-hand column is the benchmark $\tau^k = 0.18$; at the $\tau^k = 0.36$ upper bound the same rows read +0.011 at the best flat rate 0.087, −0.155 at the Pigouvian rate, and +1.649 for the nonlinear wedge, so nothing in the reading turns on the capital-tax measure. The nonlinear entry is left to Table 4 in the left-hand column because the second best is solved on a finer grid than this rate sweep and the two levels are not directly comparable; the within-column comparison there gives 0.313 against 0.311 for the best flat rate.

commodity wedge and each would move less.

Figure 6 puts both subsections on one scale, in dollars per ton. The second-best mechanism charges \$218 on an emissions-weighted average. Freezing the labor schedule takes off \$49 and freezing the savings margin a further \$62, leaving the \$107 that the nonlinear third-best wedge collects. Restricting the remaining instrument to a flat rate takes off \$60 more and lands at the \$47 of Table 5. That last restriction costs under \$5 when the income tax is set optimally alongside the price, since the best flat rate is then within a few dollars of the \$218 the nonlinear mechanism collects. The flat-rate restriction is nearly free against an optimal income tax and expensive against the one in force. This is the near-sufficiency result and its reversal priced per ton rather than measured as a share of a welfare gain, and the two metrics do not compress the contrast equally: \$5 against \$60 is a factor of twelve where the welfare shares differ by three orders of magnitude.

6.6 Reading and limitations

The practical hierarchy for carbon policy depends on a condition that is easy to leave implicit. If the income tax is set optimally alongside the carbon price, the hierarchy is clean: get the carbon

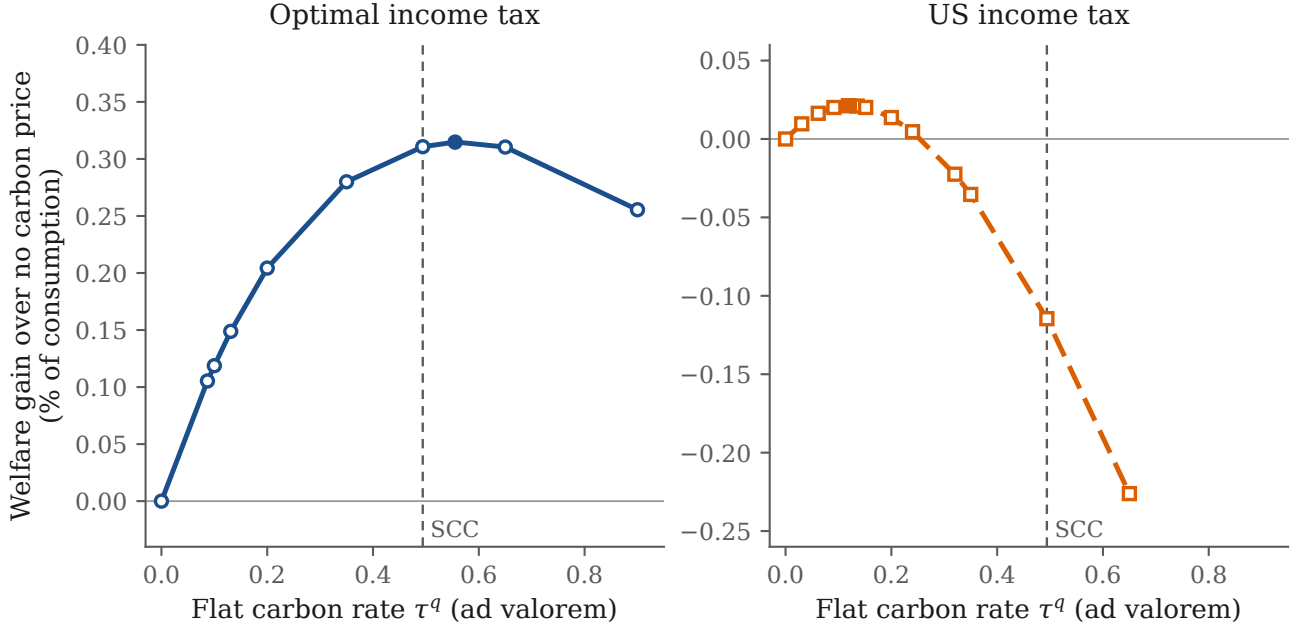


Figure 5: Consumption-equivalent welfare gain from a flat carbon tax, against the rate, when the income tax is set optimally alongside it (left) and when it is frozen at the US schedules (right). Both panels use the same solver family and the same simulated histories; only the freeze differs. The filled marker is the welfare-maximizing rate and the dashed vertical line is the social cost of carbon. Note the different vertical scales.

price to the social cost of carbon, worth 0.31% of consumption; nudge it up by the premium for leisure complementarity, worth a further 0.002%; and spend the sophistication budget on the income tax, where Farhi and Werning (2013)’s own accounting shows the returns to age- and history-dependence are two orders of magnitude larger. The redistributive value embedded in the carbon tax is small, which is the separation principle in welfare terms, and it agrees with Douenne et al. (2026)’s near-Pigouvian conclusion, reached here from the Mirrleesian rather than the Ramsey side.

If the income tax is the one actually in force, that hierarchy does not hold, and the direction of the failure is informative. The below-damages prescription is not a property of distortionary taxation as such: with the income tax optimal, the carbon price is exactly the social cost of carbon at every history under weak separability. It is a property of income-tax misspecification. Set the income tax right and the correction vanishes; set it wrong and the correction is large, larger than the Ramsey literature’s 8–24%, and a flat tax at the social cost of carbon becomes worse than no carbon tax. The two results are one claim: the case for pricing carbon at the social cost of carbon is a case for fixing the income tax, and the two reforms are not separable in practice even though the wedges separate in theory.

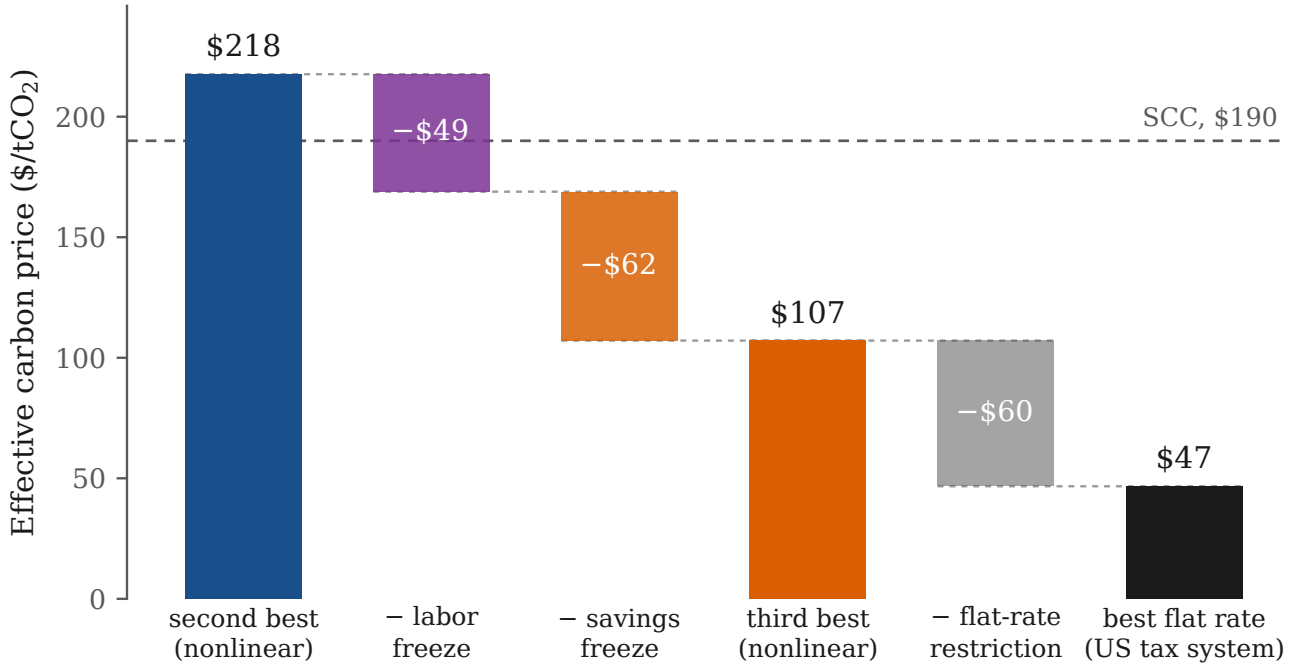


Figure 6: The effective carbon price, one restriction at a time. The three level bars are discounted-emissions-weighted means of the solved wedge, $\sum_t \beta^t E_t \tau_t^q / \sum_t \beta^t E_t$; the three step bars are differences between them; the last bar is the optimal flat rate of Table 5. Here the weights E_t are the benchmark emission path, held common across regimes so that the levels are a reweighting of one object rather than three differently weighted ones. Using each regime’s own path instead, as in Section 6.5.2, moves the second-best level by under a dollar. Each step nets pulls that change sign over the life cycle, so a step is not a statement about any single age; Figure 3(d) shows the age profiles behind the first two. The final step is not an averaging operation either: a flat rate solves a different problem rather than summarizing the nonlinear wedge, and the flat rate matching the nonlinear wedge’s abatement is \$94 rather than \$47.

This does not rehabilitate weakening the carbon price on its own. A government that lowers the carbon price while leaving the income tax misset moves along the right-hand panel of Figure 5 and recovers at most 0.02% of consumption; one that fixes the income tax moves to the left-hand panel and gains 0.31% from the carbon price alone, on top of the income tax’s own insurance value. The ordering of the two reforms matters, and it runs from the income tax to the carbon price rather than the other way around.

Six limitations bound the numbers. The split of Figure 6 between the two margins is the least secure of them, in two respects that the total is not exposed to. It is order-dependent, since the labor restriction is imposed first and the savings restriction second; only the \$111

total is invariant to that choice. And it depends on the life-cycle block. Farhi and Werning (2013) give newborns identical ex ante skills and a driftless random walk thereafter, so the Mirrleesian labor wedge starts near zero and mean earnings fall rather than rise with age. Both features widen the labor gap at young ages, which discounting and the mid-career concentration of emissions weight most heavily. Golosov et al. (2016), calibrating initial dispersion to the data, report a labor wedge at average earnings running from about 25% early in life to about 65% late, against the 2% to 36% here. Those are the two endpoints they report; the path between them is interpolated linearly in what follows. Against a schedule of that shape the US function would underdistort labor over most of the working life. Re-running the labor freeze in exactly that configuration prices the concession: with the frozen schedule shifted so that the frozen-minus-desired gap reproduces, age by age, the gap between the US function and their wedge path (overdistortion of 8 points at 25, a crossing at 32, underdistortion of 35 points by 64), the labor rung flips from $-\$49$ to $+\$54$, and the carbon wedge runs from 43% at 25 to 140% at 64, against 23% to 71% under the US freeze. The $\$49$ is therefore not a point estimate, and not even a sign. Which way frozen labor pushes the carbon price is a statement about where the tax code sits against the desired schedule, overdistortion pulling the price down and underdistortion pushing it up, as Proposition 5.1 prices it. The externality is a constant marginal damage, so the pollution-stock dynamics of Section 4.3, which move the *level* of the tax rather than the wedge structure measured here, are a separate exercise. The second best requires homothetic preferences, so the necessity calibration is read off the exact linear-tax reduction rather than the full nonlinear mechanism. The log consumption index is required by the balanced-growth homogeneity the solver relies on. The energy good carries a single carbon intensity, so within-energy heterogeneity is outside the model. And the third-best exercise freezes the savings wedge through retirement as well as through the working life, which is the right reading of a capital income tax that does not stop at 65 but is nonetheless a modeling choice doing real work: it is what keeps the deviation growing after T_R and what pulls the optimal flat rate down.

None of these bears on the qualitative results, though the results rest on different footings. The premium's proportionality to the labor wedge is structural, the identity $\varkappa l^\alpha \tau^y / (1 - \tau^y)$, as is the near-sufficiency of a flat Pigouvian price under an optimal income tax, which follows from the objective's flatness in the rate. The rotation itself is structural too: Lemma 5.3 forces the savings multiplier to reverse along every history whatever the draw, and Appendix B.7 signs the direction of that reversal at a degenerate draw; the solved profiles reproduce both. What is calibration-contingent is the premium's size, the rotation's magnitude, and the age at which the crossing falls, which the degenerate-draw argument does not date under a random walk. They rest on energy being a leisure complement ($\varkappa > 0$), the empirical input from West and Williams (2007), and on the capital income tax rate. The premium scales roughly linearly in $\varkappa$, so $\varkappa = 0.45$ about doubles the mid-career premium and $\varkappa = 0$ collapses it to the Atkinson-Stiglitz benchmark; the welfare stakes in the second-best world stay small across this range. That upper bound is generous: applying West and Williams (2007)'s nonseparability

component to gasoline alone, with no tempering for the composite, would put κ nearer 0.33.

7 Conclusion

A pollution externality placed inside a dynamic Mirrlees economy leaves the corrective and redistributive tasks separate. When preferences are weakly separable, the carbon wedge is at the social cost of carbon and the labor and savings wedges keep their standard forms, so the below-Pigouvian result that the Ramsey tradition attributes to fiscal distortions does not survive an optimal nonlinear income tax. In a setting where the planner is free to allocate labor and consumption with private information on skill as its only constraint, only nonseparability in preferences justifies setting the carbon price away from the social cost of carbon. More specifically, the dirty good is overdistorted when it is a labor substitute and underdistorted when it is a labor complement. In that case, the carbon price becomes nonlinear and history dependent. In addition, the social cost of carbon rises above the Pigouvian price when working more raises the household's valuation of a cleaner environment, and falls below it when working more lowers that valuation.

However, solving a model calibrated to US data where energy acts as a stand-in for the dirty good shows that the welfare cost of using a flat instrument rather than an optimal nonlinear instrument is small. Indeed, a flat instrument set at the social cost of carbon already captures 98% of the welfare gain from the optimal nonlinear instrument. This assumes that distortions are optimized jointly. When labor and savings distortions are fixed at the levels implied by the US tax system, the best flat rate falls to a quarter of the social cost of carbon, from \$190/tCO₂ to \$47/tCO₂. Here, the best flat tax only captures 1.5% of the welfare gain of a fully optimal nonlinear tax. Worse, a flat tax at the social cost of carbon reduces welfare, as does any flat tax above \$98/tCO₂.

Both exercises answer the paper's opening question through different lenses. If we believe that income taxes are already set optimally, then a flat carbon tax slightly above the social cost of carbon is a good approximation of the optimum. If we believe they are not and we do not have access to nonlinear instruments, then the best flat carbon tax is \$47/tCO₂, which just so happens to be in the ballpark of where many jurisdictions price carbon, consistent with a government doing its best with the instruments it has in a suboptimal tax system. If nonlinear instruments were available, the wedge would price carbon at an effective \$107/tCO₂, still below the social cost of carbon, but over twice the best flat rate. Instrument linearity therefore dampens climate ambition when the income tax system cannot be optimized. The solution: either get the carbon tax closer to a nonlinear form or get the income tax system closer to optimality such that a more ambitious flat carbon tax suffices.

This paper leaves open a number of questions. First, it characterizes wedges and not tax implementation. If we believe we are in a world where income distortions are optimal, then

the question of implementing a nonlinear carbon wedge is likely moot. However, if we believe that income distortions are significantly misset, this question becomes more relevant. Second, the level of the social cost of carbon is exogenous in the quantitative exercise, the reason being that the model considers a single generation living in a single country and is therefore not fit to realistically endogenize externalities. Introducing overlapping generations to the model would allow for a more realistic treatment of carbon dynamics. Third, outside the environmental context, the optimal commodity wedge being a time-varying object raises the question of age-dependent commodity taxation. Since age-based pricing of goods and services is already common practice, their optimality is an interesting question for future research.

References

- Albanesi, Stefania and Christopher Sleet (2006). “Dynamic Optimal Taxation with Private Information”. In: *The Review of Economic Studies* 73.1, pp. 1–30. ISSN: 00346527, 1467937X.
- Atkinson, A.B. and J.E. Stiglitz (July 1976). “The Design of Tax Structure: Direct versus Indirect Taxation”. In: *Journal of Public Economics* 6.1, pp. 55–75. ISSN: 0047-2727. DOI: [10.1016/0047-2727\(76\)90041-4](https://doi.org/10.1016/0047-2727(76)90041-4).
- Barrage, Lint (2020). “Optimal Dynamic Carbon Taxes in a Climate–Economy Model with Distortionary Fiscal Policy”. In: *The Review of Economic Studies* 87.1, pp. 1–39. ISSN: 0034-6527. DOI: [10.1093/restud/rdz055](https://doi.org/10.1093/restud/rdz055).
- Bastani, Spencer, Soren Blomquist, and Luca Micheletto (Nov. 2013). “The Welfare Gains of Age-related Optimal Income Taxation”. In: *International Economic Review* 54.4, pp. 1219–1249. ISSN: 0020-6598. DOI: [10.1111/iere.12034](https://doi.org/10.1111/iere.12034). URL: <https://doi.org/10.1111/iere.12034> (visited on 07/22/2026).
- Bénabou, Roland (Mar. 2002). “Tax and Education Policy in a Heterogeneous-Agent Economy: What Levels of Redistribution Maximize Growth and Efficiency?” In: *Econometrica* 70.2, pp. 481–517. ISSN: 0012-9682. DOI: [10.1111/1468-0262.00293](https://doi.org/10.1111/1468-0262.00293). (Visited on 07/14/2023).
- Blackorby, Charles, Russell Davidson, and William Schworm (Dec. 1991). “The validity of piecemeal second-best policy”. In: *Journal of Public Economics* 46.3, pp. 267–290. ISSN: 0047-2727. DOI: [10.1016/0047-2727\(91\)90008-P](https://doi.org/10.1016/0047-2727(91)90008-P). URL: <https://www.sciencedirect.com/science/article/pii/004727279190008P>.
- Boadway, Robin and Richard Harris (Oct. 1977). “A characterization of piecemeal second best policy”. In: *Journal of Public Economics* 8.2, pp. 169–190. ISSN: 0047-2727. DOI: [10.1016/0047-2727\(77\)90017-2](https://doi.org/10.1016/0047-2727(77)90017-2). URL: <https://www.sciencedirect.com/science/article/pii/0047272777900172>.
- Bovenberg, A. Lans and Ruud A. de Mooij (1994). “Environmental Levies and Distortionary Taxation”. In: *The American Economic Review* 84.4. JSTOR: 2118046, pp. 1085–1089. ISSN: 00028282. (Visited on 03/14/2023).
- Bovenberg, A. Lans and Sjak A. Smulders (1996). “Transitional Impacts of Environmental Policy in an Endogenous Growth Model”. In: *International Economic Review* 37.4, pp. 861–893. ISSN: 00206598, 14682354. DOI: [10.2307/2527315](https://doi.org/10.2307/2527315). URL: <http://www.jstor.org/stable/2527315> (visited on 07/21/2026).
- Burnham, Paul and Dorian Carloni (Feb. 2022). *CBO’s Model for Estimating the Effect That Federal Taxes Have on Capital Income From New Investment*. Working Paper. 2022-01. Washington, DC: Congressional Budget Office. URL: <https://www.cbo.gov/publication/57429>.
- Chamley, Christophe (1986). “Optimal Taxation of Capital Income in General Equilibrium with Infinite Lives”. In: *Econometrica* 54.3, pp. 607–622. ISSN: 00129682, 14680262. DOI: [10.2307/1911310](https://doi.org/10.2307/1911310). URL: <http://www.jstor.org/stable/1911310> (visited on 07/21/2026).

- Chancel, Lucas (Nov. 2022). “Global carbon inequality over 1990–2019”. In: *Nature Sustainability* 5.11, pp. 931–938. ISSN: 2398-9629. DOI: [10.1038/s41893-022-00955-z](https://doi.org/10.1038/s41893-022-00955-z). URL: <https://doi.org/10.1038/s41893-022-00955-z>.
- Christiansen, Vidar (July 1984). “Which commodity taxes should supplement the income tax?”. In: *Journal of Public Economics* 24.2, pp. 195–220. ISSN: 0047-2727. DOI: [10.1016/0047-2727\(84\)90024-0](https://www.sciencedirect.com/science/article/pii/00472727(84)90024-0). URL: [https://www.sciencedirect.com/science/article/pii/00472727\(84\)90024-0](https://www.sciencedirect.com/science/article/pii/00472727(84)90024-0).
- Congressional Budget Office (Dec. 2014). *Taxing Capital Income: Effective Marginal Tax Rates Under 2014 Law and Selected Policy Options*. Report Publication 49817. Written by Paul Burnham, CBO Tax Analysis Division. Washington, DC: Congressional Budget Office. URL: <https://www.cbo.gov/publication/49817>.
- Corlett, W. J. and D. C. Hague (Jan. 1953). “Complementarity and the Excess Burden of Taxation”. In: *The Review of Economic Studies* 21.1, pp. 21–30. ISSN: 0034-6527. DOI: [10.2307/2296257](https://doi.org/10.2307/2296257). URL: <https://doi.org/10.2307/2296257> (visited on 07/23/2026).
- Cremer, Helmuth, Firouz Gahvari, and Norbert Ladoux (Dec. 1998). “Externalities and Optimal Taxation”. In: *Journal of Public Economics* 70.3, pp. 343–364. ISSN: 0047-2727. DOI: [10.1016/S0047-2727\(98\)00039-5](https://www.sciencedirect.com/science/article/pii/S0047-2727(98)00039-5).
- (2001). “Second-Best Pollution Taxes and the Structure of Preferences”. In: *Southern Economic Journal* 68.2, pp. 258–280. ISSN: 00384038. DOI: [10.2307/1061594](https://doi.org/10.2307/1061594).
- (Jan. 2010). “Environmental tax design with endogenous earning abilities (with applications to France)”. In: *Journal of Environmental Economics and Management* 59.1, pp. 82–93. ISSN: 0095-0696. DOI: [10.1016/j.jeem.2009.04.009](https://www.sciencedirect.com/science/article/pii/S0095069609000850). URL: <https://www.sciencedirect.com/science/article/pii/S0095069609000850>.
- Dennig, Francis et al. (Dec. 2015). “Inequality, Climate Impacts on the Future Poor, and Carbon Prices”. In: *Proceedings of the National Academy of Sciences* 112.52, pp. 15827–15832. DOI: [10.1073/pnas.1513967112](https://doi.org/10.1073/pnas.1513967112). (Visited on 08/31/2023).
- Diamond, P.A. (Nov. 1975). “A many-person Ramsey tax rule”. In: *Journal of Public Economics* 4.4, pp. 335–342. ISSN: 0047-2727. DOI: [10.1016/0047-2727\(75\)90009-2](https://www.sciencedirect.com/science/article/pii/004727277590009-2). URL: <https://www.sciencedirect.com/science/article/pii/004727277590009-2>.
- Dietsch, Peter (Dec. 2024). “A fairer and more effective carbon tax”. In: *Nature Sustainability* 7.12, pp. 1584–1591. ISSN: 2398-9629. DOI: [10.1038/s41893-024-01429-0](https://doi.org/10.1038/s41893-024-01429-0). URL: <https://doi.org/10.1038/s41893-024-01429-0>.
- Douenne, Thomas, Albert Jan Hummel, and Marcelo Pedroni (Jan. 2026). “Optimal Fiscal Policy in a Climate-Economy Model with Heterogeneous Households*”. In: *The Economic Journal*, ueag006. ISSN: 0013-0133. DOI: [10.1093/ej/ueag006](https://doi.org/10.1093/ej/ueag006). URL: <https://doi.org/10.1093/ej/ueag006> (visited on 04/19/2026).
- Douenne, Thomas et al. (2025). *Optimal Climate Policy with Incomplete Markets*. Tech. rep. Issue: tecipa-807. University of Toronto, Department of Economics. URL: <https://ideas.repec.org/p/tor/tecipa/tecipa-807.html>.

- Environment and Climate Change Canada (2016). *Technical update to Environment and Climate Change Canada's social cost of greenhouse gas estimates*. Tech. rep. Environment and Climate Change Canada.
- Erosa, Andrés and Martin Gervais (Aug. 2002). “Optimal Taxation in Life-Cycle Economies”. In: *Journal of Economic Theory* 105.2, pp. 338–369. ISSN: 0022-0531. DOI: [10.1006/jeth.2001.2877](https://doi.org/10.1006/jeth.2001.2877). URL: <https://www.sciencedirect.com/science/article/pii/S0022053101928777>.
- Espey, James A. and Molly Espey (2004). “Turning on the Lights: A Meta-Analysis of Residential Electricity Demand Elasticities”. In: *Journal of Agricultural and Applied Economics* 36.1, pp. 65–81.
- Farhi, Emmanuel and Iván Werning (2013). “Insurance and Taxation over the Life Cycle”. In: *The Review of Economic Studies* 80.2 (283), pp. 596–635. ISSN: 0034-6527. DOI: [10.1111/1467-937x.00256](https://doi.org/10.1111/1467-937x.00256).
- Fernandes, Ana and Christopher Phelan (Apr. 2000). “A Recursive Formulation for Repeated Agency with History Dependence”. In: *Journal of Economic Theory* 91.2, pp. 223–247. ISSN: 0022-0531. DOI: [10.1006/jeth.1999.2619](https://doi.org/10.1006/jeth.1999.2619).
- Golosov, Mikhail, Narayana Kocherlakota, and Aleh Tsyvinski (2003). “Optimal Indirect and Capital Taxation”. In: *Review of Economic Studies* 70.3, pp. 569–587. ISSN: 0034-6527. DOI: [10.1111/1467-937x.00256](https://doi.org/10.1111/1467-937x.00256).
- Golosov, Mikhail, Maxim Troshkin, and Aleh Tsyvinski (2016). “Redistribution and Social Insurance”. In: *American Economic Review* 106.2, pp. 359–86. DOI: [10.1257/aer.20111550](https://doi.org/10.1257/aer.20111550).
- Golosov, Mikhail, Aleh Tsyvinski, and Nicolas Werquin (2014a). “A Variational Approach to the Analysis of Tax Systems”. In: *National Bureau of Economic Research Working Paper Series* No. 20780. DOI: [10.3386/w20780](https://doi.org/10.3386/w20780). URL: <http://www.nber.org/papers/w20780>.
- Golosov, Mikhail et al. (Jan. 2014b). “Optimal Taxes on Fossil Fuel in General Equilibrium”. In: *Econometrica* 82.1, pp. 41–88. ISSN: 0012-9682. DOI: [10.3982/ECTA10217](https://doi.org/10.3982/ECTA10217).
- Goulder, Lawrence H. et al. (July 2019). “Impacts of a Carbon Tax across US Household Income Groups: What Are the Equity-Efficiency Trade-Offs?” In: *Journal of Public Economics* 175, pp. 44–64. ISSN: 0047-2727. DOI: [10.1016/j.jpubeco.2019.04.002](https://doi.org/10.1016/j.jpubeco.2019.04.002).
- Guesnerie, Roger (July 1979). “General statements on second best pareto optimality”. In: *Journal of Mathematical Economics* 6.2, pp. 169–194. ISSN: 0304-4068. DOI: [10.1016/0304-4068\(79\)90005-3](https://doi.org/10.1016/0304-4068(79)90005-3). URL: <https://www.sciencedirect.com/science/article/pii/0304406879900053>.
- Guner, Nezih, Remzi Kaygusuz, and Gustavo Ventura (Oct. 2014). “Income taxation of U.S. households: Facts and parametric estimates”. In: *Review of Economic Dynamics* 17.4, pp. 559–581. ISSN: 1094-2025. DOI: [10.1016/j.red.2014.01.003](https://doi.org/10.1016/j.red.2014.01.003). URL: <https://www.sciencedirect.com/science/article/pii/S1094202514000040>.
- Hänsel, Martin C. et al. (Oct. 2022). “Optimal carbon taxation and horizontal equity: A welfare-theoretic approach with application to German household data”. In: *Journal of Environmental Economics and Management* 116, p. 102730. ISSN: 0095-0696. DOI: [10.1016/j.jenvman.2022.102730](https://doi.org/10.1016/j.jenvman.2022.102730).

- 016/j.jeem.2022.102730. URL: <https://www.sciencedirect.com/science/article/pii/S0095069622000833>.
- Hassett, Kevin A., Aparna Mathur, and Gilbert E. Metcalf (Apr. 2009). “The Incidence of a U.S. Carbon Tax: A Lifetime and Regional Analysis”. In: *The Energy Journal* 30.2, pp. 155–178. ISSN: 0195-6574. DOI: [10.5547/ISSN0195-6574-EJ-Vol30-No2-8](https://doi.org/10.5547/ISSN0195-6574-EJ-Vol30-No2-8). URL: <https://doi.org/10.5547/ISSN0195-6574-EJ-Vol30-No2-8> (visited on 07/22/2026).
- Heathcote, Jonathan, Kjetil Storesletten, and Giovanni L. Violante (Nov. 2017). “Optimal Tax Progressivity: An Analytical Framework”. In: *The Quarterly Journal of Economics* 132.4, pp. 1693–1754. ISSN: 0033-5533. DOI: [10.1093/qje/qjx018](https://doi.org/10.1093/qje/qjx018). URL: <https://doi.org/10.1093/qje/qjx018> (visited on 07/28/2026).
- (Sept. 2020). “Optimal progressivity with age-dependent taxation”. In: *Journal of Public Economics* 189, p. 104074. ISSN: 0047-2727. DOI: [10.1016/j.jpubeco.2019.104074](https://doi.org/10.1016/j.jpubeco.2019.104074). URL: <https://www.sciencedirect.com/science/article/pii/S0047272719301355>.
- Heathcote, Jonathan and Hitoshi Tsujiyama (Nov. 2021). “Optimal Income Taxation: Mirrlees Meets Ramsey”. In: *Journal of Political Economy* 129.11, pp. 3141–3184. ISSN: 0022-3808. DOI: [10.1086/715851](https://doi.org/10.1086/715851).
- Hellwig, Christian (June 2021). “Static and Dynamic Mirrleesian Taxation with Non-separable Preferences: A Unified Approach”. Working paper. Toulouse School of Economics.
- Interagency Working Group on Social Cost of Greenhouse Gases (Feb. 2021). *Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide Interim Estimates under Executive Order 13990*. Tech. rep. Interagency Working Group on Social Cost of Greenhouse Gases.
- Jacobs, Bas and Robin Boadway (Sept. 2014). “Optimal Linear Commodity Taxation under Optimal Non-Linear Income Taxation”. In: *Journal of Public Economics* 117, pp. 201–210. ISSN: 0047-2727. DOI: [10.1016/j.jpubeco.2014.04.012](https://doi.org/10.1016/j.jpubeco.2014.04.012).
- Jacobs, Bas and Ruud A. de Mooij (May 2015). “Pigou Meets Mirrlees: On the Irrelevance of Tax Distortions for the Second-Best Pigouvian Tax”. In: *Journal of Environmental Economics and Management* 71, pp. 90–108. ISSN: 0095-0696. DOI: [10.1016/j.jeem.2015.01.003](https://doi.org/10.1016/j.jeem.2015.01.003).
- Jacobs, Bas and Frederick van der Ploeg (May 2019). “Redistribution and pollution taxes with non-linear Engel curves”. In: *Journal of Environmental Economics and Management* 95, pp. 198–226. ISSN: 0095-0696. DOI: [10.1016/j.jeem.2019.01.008](https://doi.org/10.1016/j.jeem.2019.01.008). URL: <https://www.sciencedirect.com/science/article/pii/S0095069617308549>.
- Jewitt, Ian (Oct. 1981). “Preference structure and piecemeal second best policy”. In: *Journal of Public Economics* 16.2, pp. 215–231. ISSN: 0047-2727. DOI: [10.1016/0047-2727\(81\)90025-6](https://doi.org/10.1016/0047-2727(81)90025-6). URL: <https://www.sciencedirect.com/science/article/pii/0047272781900256>.
- Judd, Kenneth L. (Oct. 1985). “Redistributive taxation in a simple perfect foresight model”. In: *Journal of Public Economics* 28.1, pp. 59–83. ISSN: 0047-2727. DOI: [10.1016/0047-2727\(85\)90020-9](https://doi.org/10.1016/0047-2727(85)90020-9). URL: <https://www.sciencedirect.com/science/article/pii/0047272785900209>.

- Kapička, Marek (2013). “Efficient Allocations in Dynamic Private Information Economies with Persistent Shocks: A First-Order Approach”. In: *The Review of Economic Studies* 80.3, pp. 1027–1054. ISSN: 0034-6527. DOI: [10.1093/restud/rds045](https://doi.org/10.1093/restud/rds045).
- Kaplow, Louis (May 2012). “Optimal Control of Externalities in the Presence of Income Taxation”. In: *International Economic Review* 53.2, pp. 487–509. ISSN: 0020-6598. DOI: [10.1111/j.1468-2354.2012.00689.x](https://doi.org/10.1111/j.1468-2354.2012.00689.x). URL: <https://doi.org/10.1111/j.1468-2354.2012.00689.x> (visited on 04/20/2026).
- Klenert, David et al. (Nov. 2018). “Environmental Taxation, Inequality and Engel’s Law: The Double Dividend of Redistribution”. In: *Environmental and Resource Economics* 71.3, pp. 605–624. ISSN: 1573-1502. DOI: [10.1007/s10640-016-0070-y](https://doi.org/10.1007/s10640-016-0070-y).
- Kocherlakota, Narayana R. (2005). “Zero Expected Wealth Taxes: A Mirrlees Approach to Dynamic Optimal Taxation”. In: *Econometrica* 73.5, pp. 1587–1621. ISSN: 00129682, 14680262.
- Kornek, Ulrike et al. (May 2021). “The Social Cost of Carbon and Inequality: When Local Redistribution Shapes Global Carbon Prices”. In: *Journal of Environmental Economics and Management* 107, p. 102450. ISSN: 0095-0696. DOI: [10.1016/j.jeem.2021.102450](https://doi.org/10.1016/j.jeem.2021.102450).
- Labandeira, Xavier, José M. Labeaga, and Xiral López-Otero (2017). “A meta-analysis on the price elasticity of energy demand”. In: *Energy Policy* 102, pp. 549–568.
- Mendoza, Enrique G., Assaf Razin, and Linda L. Tesar (Dec. 1994). “Effective tax rates in macroeconomics: Cross-country estimates of tax rates on factor incomes and consumption”. In: *Journal of Monetary Economics* 34.3, pp. 297–323. ISSN: 0304-3932. DOI: [10.1016/0304-3932\(94\)90021-3](https://doi.org/10.1016/0304-3932(94)90021-3). URL: <https://www.sciencedirect.com/science/article/pii/0304393294900213>.
- Mirrlees, J. A. (1971). “An Exploration in the Theory of Optimum Income Taxation”. In: *The Review of Economic Studies* 38.2. JSTOR: 2296779, pp. 175–208. ISSN: 00346527, 1467937X. DOI: [10.2307/2296779](https://doi.org/10.2307/2296779). (Visited on 07/22/2023).
- (Nov. 1976). “Optimal Tax Theory: A Synthesis”. In: *Journal of Public Economics* 6.4, pp. 327–358. ISSN: 0047-2727. DOI: [10.1016/0047-2727\(76\)90047-5](https://doi.org/10.1016/0047-2727(76)90047-5).
- Pavan, Alessandro, Ilya Segal, and Juuso Toikka (Mar. 2014). “Dynamic Mechanism Design: A Myersonian Approach”. In: *Econometrica* 82.2, pp. 601–653. ISSN: 0012-9682. DOI: [10.3982/ECTA10269](https://doi.org/10.3982/ECTA10269). (Visited on 08/13/2023).
- Ploeg, Frederick van der, Armon Rezai, and Miguel Tovar Reaños (July 2025). “Third-Best Carbon Taxation: Trading Off Emission Cuts, Equity, and Efficiency”. In: *Journal of the Association of Environmental and Resource Economists* 12.4, pp. 1023–1058. DOI: [10.1086/732802](https://doi.org/10.1086/732802). URL: <https://doi.org/10.1086/732802> (visited on 07/21/2026).
- Poterba, James M. (1989). “Lifetime Incidence and the Distributional Burden of Excise Taxes”. In: *The American Economic Review* 79.2, pp. 325–330. ISSN: 00028282. URL: <http://www.jstor.org/stable/1827779> (visited on 07/22/2026).

- Saez, Emmanuel (Jan. 2001). “Using Elasticities to Derive Optimal Income Tax Rates”. In: *The Review of Economic Studies* 68.1, pp. 205–229. ISSN: 0034-6527. DOI: [10.1111/1467-937X.00166](https://doi.org/10.1111/1467-937X.00166). (Visited on 07/20/2023).
- (Feb. 2002). “The desirability of commodity taxation under non-linear income taxation and heterogeneous tastes”. In: *Journal of Public Economics* 83.2, pp. 217–230. ISSN: 0047-2727. DOI: [10.1016/S0047-2727\(00\)00159-6](https://doi.org/10.1016/S0047-2727(00)00159-6). URL: <https://www.sciencedirect.com/science/article/pii/S0047272700001596>.
- Sandmo, Agnar (1975). “Optimal Taxation in the Presence of Externalities”. In: *The Swedish Journal of Economics* 77.1. JSTOR: 3439329, pp. 86–98. ISSN: 00397318. DOI: [10.2307/3439329](https://doi.org/10.2307/3439329). (Visited on 02/18/2023).
- Trabandt, Mathias and Harald Uhlig (May 2011). “The Laffer curve revisited”. In: *Journal of Monetary Economics* 58.4, pp. 305–327. ISSN: 0304-3932. DOI: [10.1016/j.jmoneco.2011.07.003](https://doi.org/10.1016/j.jmoneco.2011.07.003). URL: <https://www.sciencedirect.com/science/article/pii/S030439321100064X>.
- U.S. Environmental Protection Agency (Nov. 2023). *Report on the Social Cost of Greenhouse Gases: Estimates Incorporating Recent Scientific Advances*. Report. Washington, DC: U.S. Environmental Protection Agency. URL: <https://www.epa.gov/environmental-economics/scghg>.
- Weinzierl, Matthew (2011). “The Surprising Power of Age-Dependent Taxes”. In: *The Review of Economic Studies* 78.4, pp. 1490–1518. ISSN: 00346527, 1467937X.
- Werning, Iván (2007). “Optimal Fiscal Policy with Redistribution”. In: *The Quarterly Journal of Economics* 122.3. JSTOR: 25098865, pp. 925–967. ISSN: 00335533, 15314650. (Visited on 11/26/2023).
- West, Sarah E. and Roberton C. Williams (Apr. 2007). “Optimal taxation and cross-price effects on labor supply: Estimates of the optimal gas tax”. In: *Journal of Public Economics* 91.3, pp. 593–617. ISSN: 0047-2727. DOI: [10.1016/j.jpubeco.2006.08.007](https://doi.org/10.1016/j.jpubeco.2006.08.007). URL: <https://www.sciencedirect.com/science/article/pii/S0047272706001162>.
- Zhu, Xing et al. (2018). “A meta-analysis on the price elasticity and income elasticity of residential electricity demand”. In: *Journal of Cleaner Production* 201, pp. 169–177.

A Optimal wedge derivations

A.1 The problem and first-order conditions

Because $U(c, q, l, S)$ can be written as $U(c, q, y/\theta, S)$, I can write U_θ as $-U_l l/\theta$. The Hamiltonian of the planner's problem of Section 4.1 is as follows, with $f_{2,t} \equiv \partial f_t / \partial \theta_{t-1}$ the persistence derivative of the transition density (Assumption 3.2), written f_2 where the date is unambiguous:

$$\begin{aligned} \mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, \theta, \mathbf{S}) \right] f \\ & + \psi \left[-U_l(c, q, y/\theta, S) \frac{l}{\theta} + \beta w_2 \right] \\ & - \lambda_1 u(\theta) \bar{\alpha}_t f_t \\ & + \lambda_2 u(\theta) f_2 \\ & + \phi [u - U(c, q, y/\theta, S) - \beta w] \\ & - \mu [S - S(\dots, Q)] \end{aligned}$$

This yields the following first-order conditions, in which $S_q \equiv \partial S / \partial Q$ is the marginal effect of current aggregate emissions on the current pollution stock.

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial c} : & f - \psi U_c \frac{l}{\theta} = \phi U_c \\ \frac{\partial \mathcal{H}}{\partial q} : & f - \psi U_q \frac{l}{\theta} + \left[\mu S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q} \right] f = \phi U_q \\ \frac{\partial \mathcal{H}}{\partial l} : & -\theta f - \phi U_l = \frac{1}{\theta} \psi [U_{ll} l + U_l] \\ \frac{\partial \mathcal{H}}{\partial u} : & \dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\ \frac{\partial \mathcal{H}}{\partial w} : & \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f = \phi \beta \\ \frac{\partial \mathcal{H}}{\partial w_2} : & \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f = -\psi \beta \\ \frac{\partial \mathcal{H}}{\partial S} : & \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \phi U_S \right) d\theta = \mu \end{aligned}$$

Here $S_q \equiv \partial S / \partial Q$ is the marginal effect of current aggregate emissions on the current pollution stock and the density f in the first-order condition for q comes from differentiating the aggregate

$Q = \int q(\theta)f(\theta) d\theta$ which enters both the current accumulation constraint and, as the newest entry of the emission state $\mathbf{S}$, the continuation value through $\partial V_{t+1}/\partial Q \equiv \partial V_{t+1}/\partial Q_t$. The scalar current stock S is not itself an argument of $V_{t+1}(\cdot, \mathbf{S})$; the continuation value sees current emissions only through $\mathbf{S}$, via $\partial V_{t+1}/\partial Q$, which is why no V term appears in the first-order condition for S . Define the full shadow price of current emissions,

$$\nu_t \equiv \mu_t S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q}. \quad (14)$$

The three wedges can be derived from here. The labor wedge is derived in Section A.3, the savings wedge in Section A.4, and the commodity wedge in Section A.5.

A.2 Validity of the recursive formulation

Section 4.1 takes its recursive form from Kapička (2013), whose consumption set is an arbitrary convex and compact set, so his reduction covers the consumption vector (c, q) . The externality adds two ingredients his environment lacks: the pollution stock enters preferences and the planner internalizes its accumulation from aggregate emissions. This subsection shows that neither disturbs the validity of the recursive approach.

An allocation is a collection of measurable functions $c_t(\theta^t)$, $q_t(\theta^t)$, and $y_t(\theta^t)$ for $t = 0, \dots, T$. Aggregate emissions at an allocation are

$$Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta] dF_0(\theta) \quad t = 0, \dots, T$$

and the implied pollution stocks are $S_t = S(Q_{t-s}, \dots, Q_t)$, with pre-period emissions given. As in the main text, $\Phi_j \equiv \partial S_{t+j}/\partial Q_t$, with $\Phi_j = 0$ for $j > s$ by the truncation of the carbon cycle. Multipliers dated beyond T are zero.

Lemma A.1 (Incentive constraints with an aggregate pollution stock).

Fix a deterministic emission sequence $(Q_0, \dots, Q_T)$ and let $S_t = S(Q_{t-s}, \dots, Q_t)$. Define the dated utility

$$\tilde{U}_t(c, q, l) \equiv U(c, q, l, S_t)$$

Given this sequence, (i) an allocation is incentive compatible in the economy with pollution if and only if it is incentive compatible in the no-externality economy, the $\nu_t = 0$ case of Section 4.1 in which the dirty good is dirty in name only, with period utility $\tilde{U}_t$, and (ii) the local incentive constraint is

$$\dot{u}_t(\theta^t) = U_\theta \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) + \beta w_{2,t}(\theta^t)$$

with no term in the pollution stock.

Proof. Step 1: deviations leave the pollution stock unchanged. Incentive compatibility compares one agent's payoffs under truthful reporting and under a reporting strategy σ , holding the allocation and all other reports fixed. Aggregate emissions integrate the allocation over the population and a single agent has measure zero in that integral, so Q_t and with it S_t take the same values under truth-telling and under the deviation. Every payoff entering the incentive constraint is therefore evaluated at the same pollution-stock sequence.

Step 2: the incentive constraints are those of a dated no-externality economy. By Step 1, the incentive constraint of Section 3 reads for every initial type θ and every reporting strategy σ ,

$$\begin{aligned} & \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t}, S_t \right) \middle| \theta \right] \\ & \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t U \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t}, S_t \right) \middle| \theta \right] \end{aligned}$$

with the same fixed S_t on both sides. Substituting the definition of $\tilde{U}_t$ on both sides,

$$\begin{aligned} & \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t \tilde{U}_t \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t} \right) \middle| \theta \right] \\ & \geq \mathbb{E}_0 \left[\sum_{t=0}^T \beta^t \tilde{U}_t \left(c_t(\sigma^t(\theta^t)), q_t(\sigma^t(\theta^t)), \frac{y_t(\sigma^t(\theta^t))}{\theta_t} \right) \middle| \theta \right] \end{aligned}$$

The two displays are the same statement, so an allocation satisfies one if and only if it satisfies the other. The second is the incentive constraint of the no-externality economy with period utility $\tilde{U}_t$, which proves (i).

Step 3: the local constraint. Part (i) places the reporting problem in the environment of Kapička (2013) with period utility $\tilde{U}_t$. His requirements bear on the period utility date by date and $\tilde{U}_t$ inherits them from Assumption 3.1 because S_t enters U as a fixed parameter: $\tilde{U}_t$

is twice continuously differentiable, increasing in both goods, jointly concave in (c, q, l) , and differentiable in the type. The dating of $\tilde{U}_t$ adds nothing beyond the date dependence Section 4.1 already carries in f_t and $\bar{\alpha}_t$ and the horizon is finite, so the one-shot-deviation reduction of Fernandes and Phelan (2000) and the envelope theorem of Kapička (2013) apply date by date:

$$\dot{u}_t(\theta^t) = \frac{\partial}{\partial \theta_t} \tilde{U}_t \left(c_t(\theta^t), q_t(\theta^t), \frac{y_t(\theta^t)}{\theta_t} \right) + \beta w_{2,t}(\theta^t)$$

The derivative passes through the pollution stock, which does not vary with the agent's type:

$$\frac{\partial}{\partial \theta_t} \tilde{U}_t \left(c_t, q_t, \frac{y_t}{\theta_t} \right) = U_\theta \left(c_t, q_t, \frac{y_t}{\theta_t}, S_t \right) + U_S \left(c_t, q_t, \frac{y_t}{\theta_t}, S_t \right) \frac{\partial S_t}{\partial \theta_t} = U_\theta \left(c_t, q_t, \frac{y_t}{\theta_t}, S_t \right)$$

since $\partial S_t / \partial \theta_t = 0$. This proves (ii). $\square$

The lemma is the formal version of the observation in Section 3 that the pollution stock enters both sides of the incentive constraint identically. To the agent, the pollution stock is a date-specific preference parameter, not a state she controls. The planner is different because it moves the pollution stock. The next proposition shows what internalizing it costs.

Proposition A.1 (Recursive formulation with an externality).

Let the Kapička (2013) reduction be valid for the two-good economy without an externality. The relaxed planner problem with pollution then admits the recursive formulation of Section 4.1 with the emission window $\mathbf{S}_- = (Q_{t-s}, \dots, Q_{t-1})$ as the only new state, the accumulation constraint as the only new restriction, and the first-order system of the recursive problem as that of the sequence problem. In that system, (i) given the emission price ν_t , every date- t history solves the no-externality node problem with period utility $\tilde{U}_t$ and the resource cost of the dirty good raised from 1 to $1 + \nu_t$; (ii) the pollution-stock multiplier μ_t is common to all date- t histories and equals the damage integral over the date- t cross-section; and (iii) the emission price collects the discounted pollution-stock multipliers, $\nu_t = \sum_{j=0}^s R^{-j} \mu_{t+j} \Phi_j$.

Proof. Step 1: the aggregates are deterministic and internalizing them is a finite-dimensional addition. Skills are the only shocks and they are idiosyncratic, allocations are measurable functions of skill histories, and agents form a continuum, so the cross-sectional distribution of histories at each date coincides with the ex-ante law of a single agent's history. Aggregate emissions Q_t are therefore numbers known at date 0 and so are the pollution stocks S_t . The relaxed planner problem is then equivalent to a joint choice of the allocation and two vectors $(Q_0, \dots, Q_T)$ and $(S_0, \dots, S_T)$, subject to the consistency constraint $Q_t = \int_0^\infty \mathbb{E}_0[q_t(\theta^t) | \theta] dF_0(\theta)$ and the accumulation constraint $S_t = S(Q_{t-s}, \dots, Q_t)$ at each date. Substituting the two

constraints back into the objective recovers the original problem, so the two problems select the same allocations.

Step 2: Lagrangian. Attach the multiplier ν_t in present value ν_t/R^t to the date- t consistency constraint and μ_t in present value μ_t/R^t to the date- t accumulation constraint, with the sign convention of the Hamiltonian in Appendix A.1. The Lagrangian of the relaxed problem is

$$\begin{aligned} \mathcal{L} = \mathcal{L}^{alloc}(\{c_t, q_t, y_t\}; \{S_t\}) &+ \sum_{t=0}^T \frac{\nu_t}{R^t} \left(\int_0^\infty \mathbb{E}_0[q_t(\theta^t)|\theta] dF_0(\theta) - Q_t \right) \\ &- \sum_{t=0}^T \frac{\mu_t}{R^t} [S_t - S(Q_{t-s}, \dots, Q_t)] \end{aligned}$$

where $\mathcal{L}^{alloc}$ collects every term of the Section 4.1 Lagrangian: the resource cost, the local incentive constraints, and the promise- and threat-keeping constraints, all with period utility $\tilde{U}_t$, as Lemma A.1 requires. The pollution stocks enter $\mathcal{L}^{alloc}$ only as the dates of $\tilde{U}_t$. The aggregates enter only the two constraint blocks.

Step 3: given the aggregates, the allocation problem is the Section 4.1 problem, re-dated and re-priced. Fix $(Q_t, S_t, \nu_t)_{t=0}^T$. Merging the consistency block with the resource cost turns the date- t flow cost of a history into

$$c_t(\theta^t) + (1 + \nu_t) q_t(\theta^t) - y_t(\theta^t)$$

plus the term $-\nu_t Q_t$, a constant with respect to the allocation. The allocation problem is therefore the no-externality problem with two changes: period utility $\tilde{U}_t$ and dirty-good resource cost $1 + \nu_t$. Both stay inside the class the Section 4.1 reduction covers. Dating the utility adds nothing beyond the date dependence already present in f_t and $\bar{\alpha}_t$, and over the finite horizon the recursion is a backward induction from $V_{T+1} \equiv 0$, so no stationarity is used. Re-pricing keeps the period cost linear, hence continuous and convex, and increasing in q for $\nu_t > -1$. Since ν_t will turn out to be the social cost of carbon, this asks only that the carbon price not subsidize the dirty good by more than its unit resource cost. The factorization of the incentive-constraint set into current controls and continuation promises never involves the objective, so the re-pricing leaves it untouched. Given the aggregates, the allocation problem is therefore recursive in the Section 4.1 states $(\hat{w}, \hat{w}_2, \theta_-)$ and its value functions solve the Bellman equation of that section with $\tilde{U}_t$ and the re-priced dirty good. This is (i).

Step 4: first-order conditions for the aggregates. The pollution stock S_t appears in $\mathcal{L}$ in two places: the accumulation block, with derivative $-\mu_t/R^t$, and the date- t utility terms of $\mathcal{L}^{alloc}$,

at every date- t history. Within a history, utility enters two constraints: the local incentive constraint with costate ψ and the definition of u with multiplier ϕ , exactly as in the Hamiltonian of Appendix A.1. Differentiating each with respect to S ,

$$\frac{\partial}{\partial S} \psi \left[-U_l \left(c, q, \frac{y}{\theta}, S \right) \frac{l}{\theta} + \beta w_2 \right] = -\psi U_{Sl} \frac{l}{\theta} \quad \frac{\partial}{\partial S} \phi \left[u - U \left(c, q, \frac{y}{\theta}, S \right) - \beta w \right] = -\phi U_S$$

The pollution stock is common to all date- t histories, so stationarity of $\mathcal{L}$ in S_t sums these contributions across the date- t cross-section,

$$\mu_t = \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \phi U_S \right) d\theta$$

where the integral runs over the date- t population of skill histories. This is the first-order condition for S in Appendix A.1 whose type integral is to be read over that cross-section: at $t = 0$ the cross-section is F_0 and the reading is literal and at later dates the social-cost-of-carbon formula applies it by writing the date- $(t + j)$ damage integrals as population averages, the $\mathbb{E}_t \int(\cdot) f_{t+j} d\theta$ terms of Proposition 4.2. Because there is no aggregate risk, these averages are deterministic: $\mathbb{E}_t$ denotes a cross-section, the same for every date- t history, not a forecast specific to the current node. In particular, μ_t is one number per date, which is (ii).

Current aggregate emissions Q_t appear in the consistency block with derivative $-\nu_t/R^t$ and, in the accumulation blocks of dates t , through $t + s$ since $S_\tau = S(Q_{\tau-s}, \dots, Q_\tau)$ carries Q_t as an argument exactly when $t \leq \tau \leq t + s$. Stationarity of $\mathcal{L}$ in Q_t gives

$$-\frac{\nu_t}{R^t} + \sum_{\tau=t}^{t+s} \frac{\mu_\tau}{R^\tau} \frac{\partial S_\tau}{\partial Q_t} = 0 \quad \Longleftrightarrow \quad \nu_t = \sum_{j=0}^s \frac{1}{R^j} \mu_{t+j} \Phi_j$$

which is (iii).

Step 5: recursive problem of Section 4.1 encodes the same system. Two checks remain: the state and the first-order conditions. From date t on, the aggregates matter only through the dated utilities $\tilde{U}_\tau$ for $\tau \geq t$, hence through $S_\tau = S(Q_{\tau-s}, \dots, Q_\tau)$ for $\tau \geq t$, and each of these depends on past aggregates only through the window $\mathbf{S}_- = (Q_{t-s}, \dots, Q_{t-1})$, the rest being current and future choices. Appending $\mathbf{S}_-$ to the Section 4.1 state is therefore sufficient and the window updates recursively, appending Q_t and dropping Q_{t-s} . The conditions: differentiate the Hamiltonian of Appendix A.1. The condition for q carries the extra term $\left[\mu_t S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q} \right] f$,

so the dirty good's effective resource cost is $1 + \nu_t$ with $\nu_t = \mu_t S_q + \frac{1}{R} \frac{\partial V_{t+1}}{\partial Q_t}$, and the envelope chain of that appendix resolves $\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^s R^{-(j-1)} \mu_{t+j} \Phi_j$, so that

$$\nu_t = \mu_t \Phi_0 + \frac{1}{R} \sum_{j=1}^s \frac{1}{R^{j-1}} \mu_{t+j} \Phi_j = \sum_{j=0}^s \frac{1}{R^j} \mu_{t+j} \Phi_j$$

The condition for S is the population damage integral of Step 4. The conditions for (c, l, u, w, w_2) contain no aggregate term and are those of the Section 4.1 problem with $\tilde{U}_t$, matching Step 3. The two first-order systems coincide condition by condition. $\square$

The argument concerns the relaxed problem, as everywhere in the paper: the envelope condition stands in for global incentive compatibility, and the sufficiency caveat of Section 3 is unchanged since Lemma A.1 maps the externality economy into the class for which that caveat was stated. Section 6 verifies global incentive compatibility numerically at the computed optima.

A.3 Labor wedge

Dividing the first-order condition for c of Appendix A.1 by U_c and using $\gamma_c \equiv U_{cl}/U_c$, with the multipliers $\phi, \psi, \lambda_1, \lambda_2$ and the costate boundary conditions $\psi(0) = \psi(\infty) = 0$ as introduced there,

$$\phi = \frac{f}{U_c} - \frac{\psi}{\theta} \gamma_c. \quad (15)$$

From the first-order condition for l and since $U_{ll}l + U_l = U_l(1 + \epsilon)$ with $\epsilon \equiv U_{ll}l/U_l$,

$$-\theta f - \phi U_l = \frac{\psi}{\theta} U_l(1 + \epsilon) \implies \phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1 + \epsilon)$$

Equating the two expressions for ϕ and collecting the terms in ψ ,

$$\frac{f}{U_c} + \frac{\theta f}{U_l} = -\frac{\psi}{\theta}(1 + \epsilon - \gamma_c)$$

The labor wedge is $1 - \tau^y = -U_l/(\theta U_c)$, so $\theta U_c/U_l = -1/(1 - \tau^y)$, making the left-hand side $\frac{f}{U_c} \left(1 - \frac{1}{1 - \tau^y}\right) = -\frac{f}{U_c} \frac{\tau^y}{1 - \tau^y}$. Substituting and rearranging,

$$\frac{\tau^y}{1 - \tau^y} = (1 + \epsilon - \gamma_c) \frac{\psi U_c}{\theta f} = A_t(\theta) \frac{\psi U_c}{\theta f}, \quad A_t(\theta) \equiv 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) \quad (16)$$

Next, substitute ϕ from (15) into $\dot{\psi} = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi$,

$$\dot{\psi} - \frac{\gamma_c}{\theta} \psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \frac{f}{U_c}$$

This is a linear first-order ordinary differential equation. Multiplying by the integrating factor $\exp \left[- \int^\theta \gamma_c(\tilde{x}) d\tilde{x}/\tilde{x} \right]$ and integrating from θ to ∞ with the boundary condition $\psi(\infty) = 0$,

$$\psi(\theta) = \int_\theta^\infty \exp \left[- \int_\theta^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx. \quad (17)$$

The remaining boundary condition $\psi(0) = 0$ pins down λ_1 ,

$$\lambda_{1,t} = \frac{\int_0^\infty \exp \left[- \int_0^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{1}{U_{c,t}(x)} f_t(x) + \lambda_{2,t} f_{2,t}(x) \right) dx}{\int_0^\infty \exp \left[- \int_0^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \bar{\alpha}_t(x) f_t(x) dx}. \quad (18)$$

Substituting (17) into (16),

$$\frac{\tau^y}{1 - \tau^y} = (1 + \epsilon - \gamma_c) \frac{U_c(\theta)}{\theta f(\theta)} \int_\theta^\infty \exp \left[- \int_\theta^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left(\frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx$$

I split the integral into the $[f/U_c - \lambda_1 \bar{\alpha}_t f]$ part, which yields the intratemporal term $A_t B_t C_t$, and the $\lambda_2 f_2$ part, which yields the intertemporal term. The three factors are collected in (20) below: B_t is the inverse hazard rate $(1 - F_t)/(\theta f_t)$ and C_t the average of the redistributive term over types above θ , so the $1 - F_t(\theta)$ inserted to make C_t an average cancels against B_t and the product is the integral itself. Splitting it this way is what puts the decomposition in the form Golosov et al. (2016) use. For the first part, carry $U_c(\theta)/U_c(x)$ into the integrand. Along the optimal allocation $U_{c,t}$ depends on the type through $c(\theta)$, $q(\theta)$, and $l(\theta) = y(\theta)/\theta$; the pollution stock is common to the date- t cross-section, so $\partial S_t / \partial \theta = 0$ and no term in U_{cS} appears. Hence,

$$\frac{d}{d\theta} \ln U_{c,t} = \frac{U_{cc} \dot{c} + U_{cq} \dot{q} + U_{cl} \dot{l}}{U_c} = -\sigma_t \frac{\dot{c}}{c} - \sigma_{cq,t} \frac{\dot{q}}{q} + \gamma_c \frac{\dot{l}}{l}$$

Since $l = y/\theta$ gives $\dot{l}/l = \dot{y}/y - 1/\theta$, integrating from θ to x and multiplying by the integrating factor gives the following:

$$\frac{U_{c,t}(\theta)}{U_{c,t}(x)} \exp \left[- \int_{\theta}^x \gamma_c(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] = \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x}) \frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_c(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right], \quad (19)$$

Here, the $\gamma_c/\tilde{x}$ term from $\dot{l}/l$ cancels the integrating factor. The cross-curvature $\sigma_{cq,t}$ enters because, under nonseparability, extracting resources from a higher type shifts the marginal utility of the numéraire through the adjustment of both goods. The first part therefore equals $A_t B_t C_t$. For the second part, two preliminaries pin down $\lambda_{2,t}$. The first-order condition for w_2 rearranges to $\partial V_{t+1}/\partial w_2 = -\beta R \psi/f$, and the envelope theorem applied to the period- $(t+1)$ problem gives $\partial V_{t+1}/\partial \hat{w}_2 = -\lambda_{2,t+1}$, the multiplier with which threat keeping enters the Lagrangian. Equating the two expressions and shifting the date back one period,

$$\lambda_{2,t}(\theta^{t-1}) = \beta R \frac{\psi_{t-1}}{f_{t-1}}$$

Next, evaluating (16) at $t-1$ and solving for the costate,

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{\theta_{t-1}}{U_{c,t-1} A_{t-1}}$$

The $\lambda_2 f_2$ block of the wedge is $\lambda_{2,t}$, a constant in the current type, times the same outer structure as the first part,

$$A_t(\theta) \frac{U_{c,t}(\theta)}{\theta f_t(\theta)} \lambda_{2,t} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx$$

Substituting the two preliminaries and collecting the ratios,

$$\beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)} = \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta)$$

Collecting both parts,

$$\frac{\tau^y}{1 - \tau^y} = A_t(\theta) B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} D_t(\theta)$$

$$\begin{aligned}
A_t(\theta) &= 1 + \epsilon_t(\theta) - \gamma_{c,t}(\theta) \\
B_t(\theta) &= \frac{1 - F_t(\theta)}{\theta f_t(\theta)} \\
C_t(\theta) &= \int_{\theta}^{\infty} \exp \left[\int_{\theta}^x \left(\sigma_t(\tilde{x}) \frac{\dot{c}_t(\tilde{x})}{c_t(\tilde{x})} + \sigma_{cq,t}(\tilde{x}) \frac{\dot{q}_t(\tilde{x})}{q_t(\tilde{x})} - \gamma_{c,t}(\tilde{x}) \frac{\dot{y}_t(\tilde{x})}{y_t(\tilde{x})} \right) d\tilde{x} \right] (1 - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
D_t(\theta) &= \frac{A_t(\theta)}{A_{t-1}} \frac{U_{c,t}(\theta)}{U_{c,t-1}} \frac{\theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \gamma_{c,t}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\theta f_t(\theta)}
\end{aligned} \tag{20}$$

This is Golosov et al. (2016)'s labor-wedge decomposition, with the cross-curvature $\sigma_{cq,t}$ in C_t .

A.4 Savings wedge

The savings wedge follows from the same first-order conditions of Appendix A.1. I define the incentive credit χ_t , the share of the cost of delivering utility through the numéraire that the planner recoups because consumption complements labor, relaxing the incentive constraint:

$$\chi_t(\theta^t) \equiv \frac{\gamma_{c,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y}$$

Using $\frac{\tau_t^y}{1 - \tau_t^y} = \frac{\psi U_c}{\theta f} (1 + \epsilon_t - \gamma_{c,t})$, $\phi = f/U_c - \psi \gamma_c/\theta$ from (15) can be rewritten as follows:

$$\frac{\phi}{f} = \frac{1}{U_c} (1 - \gamma_c \psi U_c / (\theta f)) = \frac{1 - \chi_t}{U_{c,t}}. \tag{21}$$

$\chi_t > 0$ when the numéraire is complementary with labor and labor is taxed and zero when preferences are separable in the numéraire against labor or labor is undistorted. The first-order condition for w and the envelope condition $\partial V_t / \partial \hat{w} = \lambda_{1,t}$ one period ahead give

$$\lambda_{1,t+1}(\theta^t) = \beta R \frac{\phi(\theta)}{f(\theta)} = \beta R \frac{1 - \chi_t}{U_{c,t}}. \tag{22}$$

A second expression for $\lambda_{1,t}$ comes from integrating the costate equation over the whole type space. With $\psi(0) = \psi(\infty) = 0$,

$$0 = \int_0^{\infty} \dot{\psi} d\theta = \lambda_{1,t} \int_0^{\infty} \bar{\alpha}_t f d\theta - \lambda_{2,t} \int_0^{\infty} f_2 d\theta - \int_0^{\infty} \phi d\theta$$

The Pareto weights are normalized so that $\int \bar{\alpha}_t f d\theta = 1$, and $\int f_t(\theta|\theta_-) d\theta = 1$ for every θ_- implies $\int f_{2,t} d\theta = 0$. Hence $\lambda_{1,t} = \int_0^\infty \phi d\theta$, which by (21) is

$$\lambda_{1,t} = \mathbb{E}_{t-1} \left[\frac{1 - \chi_t}{U_{c,t}} \right], \quad (23)$$

where $\mathbb{E}_{t-1}$ integrates over θ_t conditional on θ_{t-1} . The multiplier on promise keeping is set before the current draw is known. Evaluating (23) at $t + 1$ and equating with (22),

$$\frac{1 - \chi_t}{U_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}}{U_{c,t+1}} \right].$$

Defining the incentive-adjusted marginal utility $\widehat{U}_{c,t} \equiv U_{c,t}/(1 - \chi_t)$,

$$\frac{1}{\widehat{U}_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1}{\widehat{U}_{c,t+1}} \right]$$

This is the inverse Euler equation in the adjusted marginal utility $\widehat{U}_c$. Delivering a util through the numéraire costs $1/U_{c,t}$ in resources and, when $U_{cl} > 0$, the unit lowers the information rent $-U_l l/\theta$, an incentive gain worth $\chi_t/U_{c,t}$.

A.5 Commodity wedge

As in the previous proofs, starting from the first-order conditions of Appendix A.1,

$$\phi = \frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c}$$

Because c and q share the same ϕ ,

$$\phi = \frac{f}{U_q} - \frac{\psi U_{ql} \frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q}$$

Hence,

$$\frac{f}{U_q} - \frac{\psi U_{ql} \frac{l}{\theta}}{U_q} + \frac{\nu_t f}{U_q} = \frac{f}{U_c} - \frac{\psi U_{cl} \frac{l}{\theta}}{U_c}$$

Multiplying by U_q and U_c ,

$$U_c \left(f - \psi U_{ql} \frac{l}{\theta} + \nu_t f \right) = U_q \left(f - \psi U_{cl} \frac{l}{\theta} \right)$$

Using $\gamma_c \equiv \frac{U_{cl} l}{U_c}$ and $\gamma_q \equiv \frac{U_{ql} l}{U_q}$, or, equivalently, $\gamma_c U_c = U_{cl} l$ and $\gamma_q U_q = U_{ql} l$,

$$U_c \left(f - \frac{\psi}{\theta} \gamma_q U_q + \nu_t f \right) = U_q \left(f - \frac{\psi}{\theta} \gamma_c U_c \right)$$

Then, putting the γ terms (as well as the externality term) on one side,

$$U_c f - U_q f = \frac{\psi}{\theta} U_c U_q (\gamma_q - \gamma_c) - \nu_t U_c f$$

Finally, dividing by $U_q f$,

$$\frac{U_c}{U_q} - 1 = \frac{\psi U_c}{\theta f} (\gamma_q - \gamma_c) - \nu_t \frac{U_c}{U_q}$$

Note that, by definition, $1 + \tau^q = U_q/U_c$, so $\frac{U_c}{U_q} - 1 = -\frac{\tau^q}{1 + \tau^q}$. Multiplying through by -1 ,

$$\frac{\tau^q}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q) + \nu_t \frac{U_c}{U_q}$$

Since $\frac{U_c}{U_q} = \frac{1}{1 + \tau^q}$,

$$\frac{\tau^q}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q) + \frac{\nu_t}{1 + \tau^q}$$

Rearranging,

$$\frac{\tau^q - \nu_t}{1 + \tau^q} = \frac{\psi U_c}{\theta f} (\gamma_c - \gamma_q)$$

The prefactor $\psi U_c/(\theta f)$ is the same object that drives the labor wedge. The first-order conditions for c and l and the costate equation for ψ are untouched by the externality, so the

integrating-factor steps of Appendix A.3 carry over verbatim. Solving (16) for the prefactor and substituting the $ABCD$ decomposition,

$$\frac{\psi U_c}{\theta f} = \frac{1}{A_t} \frac{\tau^y}{1 - \tau^y} = B_t C_t + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{D_t}{A_t}$$

Multiplying by the complementarity gap $\gamma_{c,t} - \gamma_{q,t}$,

$$\frac{\tau_t^q - \nu_t}{1 + \tau_t^q} = (\gamma_{c,t} - \gamma_{q,t}) \left[B_t(\theta) C_t(\theta) + \beta R \frac{\tau_{t-1}^y}{1 - \tau_{t-1}^y} \frac{D_t(\theta)}{A_t(\theta)} \right]$$

with A_t, B_t, C_t, D_t as in Appendix A.3. Alternatively, multiplying $\frac{\psi U_c}{\theta f} = \frac{1}{A_t} \frac{\tau^y}{1 - \tau^y}$ by the gap,

$$\frac{\tau_t^q - \nu_t}{1 + \tau_t^q} = \frac{\gamma_{c,t} - \gamma_{q,t}}{1 + \epsilon_t - \gamma_{c,t}} \frac{\tau_t^y}{1 - \tau_t^y}$$

The commodity wedge inherits every dynamic feature of the labor wedge. Its shape is that of the labor wedge, scaled by the elasticity term. The externality leaves this block untouched and enters only by replacing τ^q with the deviation $\tau^q - \nu_t$ on the left.

A.6 The social cost of carbon

Now consider the shadow price of current emissions ν_t of (14). Its first piece is the date- t pollution-stock multiplier μ_t , the marginal damage of a unit of the current pollution stock. Substituting ϕ from the first-order condition for c of Appendix A.1 into the first-order condition for S , then dividing and multiplying by f ,

$$\mu_t = \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \left[\frac{f}{U_c} - \frac{\psi U_{cl} l}{U_c} \right] U_S \right) d\theta = \int_0^\infty \left(-\frac{\psi}{\theta f} U_{Sl} l - \left[1 - \frac{\psi}{\theta f} U_{cl} l \right] \frac{U_S}{U_c} \right) f d\theta$$

Rewrite $\psi/\theta f$ by noting that

$$\frac{\psi}{\theta f} = \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{U_c} \frac{1}{1 + \epsilon - \gamma_c}$$

Hence, also defining $\xi \equiv \frac{U_{Sl} l}{U_S}$,

$$\begin{aligned}
\mu_t &= \int \left(- \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{1 + \epsilon - \gamma_c} \frac{U_{Sl} l}{U_c} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{1 + \epsilon - \gamma_c} \frac{U_{cl} l}{U_c} \right] \frac{U_S}{U_c} \right) f d\theta \\
&= \int \left(- \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{1}{1 + \epsilon - \gamma_c} \frac{U_{Sl} l}{U_c} \frac{U_S}{U_S} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma_c}{1 + \epsilon - \gamma_c} \right] \frac{U_S}{U_c} \right) f d\theta \\
&= \int \left(- \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\xi}{1 + \epsilon - \gamma_c} \frac{U_S}{U_c} - \left[1 - \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\gamma_c}{1 + \epsilon - \gamma_c} \right] \frac{U_S}{U_c} \right) f d\theta \\
&= - \int \left[1 + \left(\frac{\tau^y}{1 - \tau^y} \right) \frac{\xi - \gamma_c}{1 + \epsilon - \gamma_c} \right] \frac{U_S}{U_c} f d\theta
\end{aligned}$$

The second piece of ν_t prices the echo of current emissions in future pollution stocks. By the envelope theorem applied to the period- $(t + 1)$ problem, Q_t enters continuation costs only through the accumulation constraints of the next s periods: $S_{t+j} = S(Q_{t+j-s}, \dots, Q_{t+j})$ carries Q_t as an argument for $j \leq s$ and drops it afterward. Each envelope step peels off one date,

$$\frac{\partial V_{t+1}}{\partial Q_t} = \mu_{t+1} \Phi_1 + \frac{1}{R} \frac{\partial V_{t+2}}{\partial Q_t} \quad \Phi_j \equiv \frac{\partial S_{t+j}}{\partial Q_t}$$

and iterating until Q_t drops out ($\Phi_j = 0$ for $j > s$ by construction),

$$\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^s \frac{1}{R^{j-1}} \mu_{t+j} \Phi_j$$

Substituting into (14), with $\Phi_0 = S_q$,

$$\nu_t = \mu_t \Phi_0 + \frac{1}{R} \sum_{j=1}^s \frac{1}{R^{j-1}} \mu_{t+j} \Phi_j = \sum_{j=0}^s \frac{1}{R^j} \mu_{t+j} \Phi_j$$

Substituting the damage block above for each μ_{t+j} and writing the date- $(t + j)$ population integrals as expectations over future skill histories,

$$\text{SCC}_t \equiv \nu_t = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \left[1 + \left(\frac{\tau_{t+j}^y}{1 - \tau_{t+j}^y} \right) \frac{\xi_{t+j} - \gamma_{c,t+j}}{1 + \epsilon_{t+j} - \gamma_{c,t+j}} \right] \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta$$

Here, s is the number of periods emissions affect the pollution stock.

B Constrained-optimal commodity wedge

B.1 Validity of the recursive formulation

Part (i): Both constraints restrict the functions (c_t, q_t, y_t) that the planner may choose. No reporting strategy enters either constraint, and neither appears on either side of the incentive inequalities, so incentive compatibility is decided as usual. The one-shot-deviation reduction of Fernandes and Phelan (2000) and the envelope theorem of Kapička (2013) therefore apply.

Neither constraint involves the promised value u , the continuation promise w , or the marginal continuation value w_2 . Both constraints restrict the current bundle and the current pollution stock, but nothing else. The conditions for those three are therefore those of Appendix A.1, such that the costate equation is unchanged in form, its boundary conditions $\psi(0) = \psi(\infty) = 0$ still hold, and the envelope identities for $\hat{w}$ and $\hat{w}_2$ carry over. What changes is ϕ , which the condition for c defines and which both constraints reach.

Part (ii): At every date- t node, the labor constraint involves the current bundle $(c(\theta), q(\theta), y(\theta))$ and the current pollution stock only with no promised value, no continuation object, and no other date's controls. It enters the date- t component problem the way the resource cost does, pointwise with multiplier $\eta(\theta)f_t$, and adds no state.

Part (iii): Fix a history θ^t with $t < T$. The savings constraint reads

$$U_{c,t}(\theta^t) = (1 - \bar{\tau}_t^s(\theta_t)) \beta R \int_0^\infty U_{c,t+1}(\theta^t, \theta') f_{t+1}(\theta'|\theta_t) d\theta'$$

The left side is pinned by the date- t allocation at θ^t while the right side integrates over possible future allocations conditional on the current skill. Isolating the integral, define

$$M(\theta^t) \equiv \frac{U_{c,t}(\theta^t)}{(1 - \bar{\tau}_t^s(\theta_t)) \beta R}$$

This is a function of date- t controls alone, so constraint (7) can be rewritten as

$$M(\theta^t) = \int_0^\infty U_{c,t+1}(\theta^t, \theta') f_{t+1}(\theta'|\theta_t) d\theta'$$

That is, the constraint requires marginal utility at date $t+1$ to average to $M(\theta^t)$ over the draws that follow θ^t . The draws that follow θ^t are succeeded by the date- $(t+1)$ problem at the state Section 4.1 writes $(w(\theta^t), w_2(\theta^t), \theta_t, \mathbf{S})$. Two histories reaching that same state can compute

different requirements since M is read off the date- t allocation and not off the promises. The requirement can therefore be appended to that state as $\hat{M}$. Two terms enter the Hamiltonian. The first delivers the inherited requirement.

$$\vartheta_t \left[\int_0^\infty U_c \left(c(\theta), q(\theta), \frac{y(\theta)}{\theta}, S \right) f_t(\theta|\theta_-) d\theta - \hat{M} \right]$$

The second is the continuation value's dependence on the requirement passed forward, which Appendix B.2 differentiates by the chain rule through M . Only the first involves $\hat{M}$, so $\partial V_t / \partial \hat{M} = -\vartheta_t$ by the envelope theorem. The date-0 problem is the continuation of no history, so it carries no such term and $\vartheta_0 = 0$. The date- T problem has no successor, so it passes no requirement and (9) reads $\eta_T^s = \vartheta_T U_{c,T}$.

An allocation satisfying (7) at every history hands every component problem a requirement it meets, and component problems that each meet their inherited requirement assemble into an allocation satisfying (7) at every history. Minimizing the same cost over the same constraint set, the two problems have the same first-order conditions, which are the ones in Appendix B.2.

B.2 First-order conditions

The Hamiltonian is that of Appendix A.1 with two added terms and a continuation value that now carries the marginal-utility requirement M as an argument:

$$\begin{aligned} \mathcal{H} = & \left[c + q - \theta l + \frac{1}{R} V_{t+1}(w, w_2, M(c, q, l, S), \theta, \mathbf{S}) \right] f \\ & + \psi \left[-U_l(c, q, y/\theta, S) \frac{l}{\theta} + \beta w_2 \right] \\ & - \lambda_1 u(\theta) \bar{\alpha}_t f_t \\ & + \lambda_2 u(\theta) f_2 \\ & + \phi [u - U(c, q, y/\theta, S) - \beta w] \\ & - \mu [S - S(\dots, Q)] \\ & + \eta \left[-\frac{U_l(c, q, y/\theta, S)}{\theta U_c(c, q, y/\theta, S)} - (1 - \bar{\tau}_t^y(\theta)) \right] f \\ & + \vartheta_t \left[U_c(c, q, y/\theta, S) - \hat{M} \right] f \end{aligned}$$

Differentiating the continuation term with respect to a control $z \in \{c, q, l, S\}$, by the chain rule through M and the envelope condition at $t + 1$,

$$\frac{1}{R} \frac{\partial V_{t+1}}{\partial \hat{M}} \frac{\partial M}{\partial z} f = -\frac{\vartheta_{t+1}(\theta^t)}{(1 - \bar{\tau}_t^s) \beta R^2} U_{cz} f$$

Adding the $\vartheta_t U_{cz} f$ term, each first-order condition gains $(\eta_t^s/U_{c,t}) U_{cz} f$ with η_t^s as in (9), alongside the labor-constraint term $\eta f \Gamma_z$. The first-order conditions are

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial c} : f - \psi U_{cl} \frac{l}{\theta} + \eta f \Gamma_c + \frac{\eta_t^s}{U_c} U_{cc} f &= \phi U_c \\ \frac{\partial \mathcal{H}}{\partial q} : f - \psi U_{ql} \frac{l}{\theta} + \nu_t f + \eta f \Gamma_q + \frac{\eta_t^s}{U_c} U_{cq} f &= \phi U_q \\ \frac{\partial \mathcal{H}}{\partial l} : -\theta f - \phi U_l + \eta f \Gamma_l + \frac{\eta_t^s}{U_c} U_{cl} f &= \frac{1}{\theta} \psi [U_{ll} l + U_l] \\ \frac{\partial \mathcal{H}}{\partial u} : \dot{\psi} &= \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - \phi \\ \frac{\partial \mathcal{H}}{\partial w} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w} f &= \phi \beta \\ \frac{\partial \mathcal{H}}{\partial w_2} : \frac{1}{R} \frac{\partial V_{t+1}}{\partial w_2} f &= -\psi \beta \\ \frac{\partial \mathcal{H}}{\partial S} : \int_0^\infty \left(-\psi U_{sl} \frac{l}{\theta} - \phi U_S + \eta f \Gamma_S + \frac{\eta_t^s}{U_c} U_{cS} f \right) d\theta &= \mu \end{aligned}$$

with $\nu_t \equiv \mu_t S_q + \frac{1}{R} \partial V_{t+1} / \partial Q$ the shadow price of current emissions. The derivatives of the labor constraint are

$$\begin{aligned} \Gamma_c &= (1 - \bar{\tau}_t^y) \frac{\sigma}{c} - \frac{\gamma_c}{y} \\ \Gamma_q &= (1 - \bar{\tau}_t^y) \frac{\sigma_{cq}}{q} - \frac{\gamma_q (1 + \tau^q)}{y} \\ \Gamma_l &= (1 - \bar{\tau}_t^y) \frac{\epsilon - \gamma_c}{l} \\ \Gamma_S &= -\frac{U_S}{U_c} \left[\frac{\xi}{y} + (1 - \bar{\tau}_t^y) \frac{\xi_c}{c} \right] \end{aligned} \tag{24}$$

All four are $\partial \Gamma / \partial z$ evaluated with $U_l = -\theta U_c (1 - \bar{\tau}_t^y)$, which collapses the quotient rule to $\partial \Gamma / \partial z = -U_{lz} / (\theta U_c) - (1 - \bar{\tau}_t^y) U_{cz} / U_c$. The factor $1 + \tau^q$ in Γ_q comes from the first piece: γ_q is defined against U_q , so $U_{ql} = \gamma_q U_q / l = \gamma_q (1 + \tau^q) U_c / l$. It is what cancels the prefactor

$1/(1 + \tau^q)$ in Appendix B.9 and lets $\gamma_{q,t}/y_t$ pair with $\gamma_{c,t}/y_t$ there. Each savings term is η_t^s times the log-derivative of U_c in the corresponding direction since $U_{cz}/U_c = \partial \ln U_c / \partial z$. Written with elasticities, these are $-\sigma/c$ for the numéraire, $-\sigma_{cq}/q$ for the dirty good, γ_c/l for hours, and $\xi_c(U_S/U_c)/c$ for the pollution stock, with $\xi_c \equiv U_{Sc}c/U_S$ the complementarity of the pollutant with the clean good.

B.3 Statistics

That $\Gamma_c > 0$ signals normality of leisure follows from the standard problem's labor constraint. Give an agent with $-U_l/(\theta U_c) = 1 - \bar{\tau}_t^y$ a unit of the numéraire at fixed hours. The margin she requires moves by $\partial [-U_l/(\theta U_c)] / \partial c = -U_{lc}/(\theta U_c) + U_l U_{cc}/(\theta U_c^2)$. Using $U_l = -\theta U_c (1 - \bar{\tau}_t^y)$, this is $\Gamma_{c,t} = (1 - \bar{\tau}_t^y) \sigma_t/c_t - \gamma_{c,t}/y_t$. The first term means that diminishing marginal utility of consumption raises the margin she requires and the second term means that complementarity between the numéraire and labor lowers it. Normality of leisure is a statement about unearned income at a fixed wage. Her budget has slope $\theta(1 - \bar{\tau}_t^y)$ at the assigned point, whether the schedule is indexed on the type or on earnings, so a perturbation dm of unearned income gives $dc = \theta(1 - \bar{\tau}_t^y) dl + dm$. Differentiating, her first-order condition for hours becomes

$$\Gamma_{c,t} [\theta(1 - \bar{\tau}_t^y) dl + dm] + \Gamma_{l,t} dl = 0 \quad \implies \quad \frac{\partial l}{\partial m} = -\frac{\Gamma_{c,t}}{\theta(1 - \bar{\tau}_t^y) \Delta_t}$$

This uses $\theta(1 - \bar{\tau}_t^y) \Gamma_{c,t} + \Gamma_{l,t} = \theta(1 - \bar{\tau}_t^y) \Delta_t$, which is (24) collected at $y = \theta l$ against (8), with $\Gamma_{l,t}$ and Δ_t . Under an earnings-indexed schedule, both carry the progressivity parameter p_t of (42) and the identity is unchanged. The denominator is positive under $\Delta_t > 0$, so hours fall in unearned income when $\Gamma_{c,t} > 0$. The argument is unaffected by the savings constraint, which does not enter the agent's hours choice at the frozen wedge.

Δ_t has a reading of its own, and it is the one Section 5.1 assumes. Fix the dirty-good assignment and perturb (c, l) along an indifference curve of an agent holding the margin at $-U_l/(\theta U_c) = 1 - \bar{\tau}_t^y$: utility compensation imposes $dc = \theta(1 - \bar{\tau}_t^y) dl$, and differentiating the hours condition along the perturbation,

$$0 = U_c d(1 - \bar{\tau}_t^y) + \left[\theta(1 - \bar{\tau}_t^y)^2 U_{cc} + 2(1 - \bar{\tau}_t^y) U_{cl} + \frac{U_{ll}}{\theta} \right] dl$$

In the statistics of (1) and (2) the bracket equals $-U_c(1 - \bar{\tau}_t^y) \theta \Delta_t$, so

$$s_{yy,t} \equiv \frac{\partial y}{\partial (1 - \bar{\tau}_t^y)} \Big|_u = \theta \frac{dl}{d(1 - \bar{\tau}_t^y)} \Big|_u = \frac{1}{(1 - \bar{\tau}_t^y) \Delta_t} > 0 \quad (25)$$

the compensated response of earnings to the net-of-wedge rate, holding the commodity split fixed: the hours margin's own compensated response, the object the labor constraint prices, not the full two-good demand response. The same bracket, up to a factor of θ , is the agent's second-order condition. Let her face the linear budget $c = \theta(1 - \bar{\tau}_t^y)l + m$ at the frozen net wage with the dirty-good assignment held fixed, and choose l to maximize utility. Then $dU/dl = \theta(1 - \bar{\tau}_t^y)U_c + U_l$, whose vanishing is (6), and

$$\frac{d^2U}{dl^2} = \theta^2(1 - \bar{\tau}_t^y)^2 U_{cc} + 2\theta(1 - \bar{\tau}_t^y)U_{cl} + U_{ll} = -\theta^2(1 - \bar{\tau}_t^y)U_c \Delta_t$$

which is negative exactly when $\Delta_t > 0$. So the assumption says the agent's hours choice at the frozen wedge is an interior maximum rather than a minimum: a restriction on the agent's problem, not on the planner's. Under an earnings-indexed schedule both displays carry $\Delta_t + p_t/y_t$: the schedule's own dependence on earnings enters through $d(1 - \bar{\tau}_t^y)/dl = -p_t(1 - \bar{\tau}_t^y)\theta/y_t$ and shifts each bracket by exactly the term that replaces Δ_t with $\Delta_t + p_t/y_t$, which is the $\Gamma_{l,t}$ of Appendix B.12.

B.4 The labor multiplier

Divide the first-order condition for c by f and use $U_{cl}l = \gamma_c U_c$ and $U_{cc}/U_c = -\sigma/c$.

$$\frac{U_c \phi}{f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta \Gamma_c - \eta_t^s \frac{\sigma}{c} \equiv 1 - \chi_t \quad (26)$$

Here, χ is the incentive credit of Appendix A.4 generalized to the case with frozen wedges and later used in Proposition 5.2. Solving for ϕ twice, once from the condition for c and once from the condition for l using $U_{ll}l + U_l = U_l(1 + \epsilon)$,

$$\begin{aligned} \phi &= \frac{f}{U_c} - \frac{\psi}{\theta} \gamma_c + \eta \frac{\Gamma_c}{U_c} f + \eta_t^s \frac{U_{cc}}{U_c} \frac{f}{U_c} \\ \phi &= -\frac{\theta f}{U_l} - \frac{\psi}{\theta} (1 + \epsilon) + \eta \frac{\Gamma_l}{U_l} f + \eta_t^s \frac{U_{cl}}{U_c} \frac{f}{U_l} \end{aligned}$$

Equating the two and multiplying through by U_l ,

$$\left(\frac{U_l}{\theta U_c} + 1 \right) \theta f = -\frac{\psi U_l}{\theta} (1 + \epsilon - \gamma_c) + \eta U_l \left[\frac{\Gamma_l}{U_l} - \frac{\Gamma_c}{U_c} \right] f + \eta_t^s \left[\frac{U_{cl}}{U_c} - \frac{U_{cc} U_l}{U_c U_c} \right] f$$

Because $\bar{\tau}_t^y = 1 + U_l/(\theta U_c)$, divide by $\theta f(1 - \bar{\tau}_t^y)$ and substitute $U_l = -\theta U_c(1 - \bar{\tau}_t^y)$ term by term. On the left, $U_l/(\theta U_c) = -(1 - \bar{\tau}_t^y)$ leaves $[1 - (1 - \bar{\tau}_t^y)]/(1 - \bar{\tau}_t^y) = \bar{\tau}_t^y/(1 - \bar{\tau}_t^y)$. On the right, the first term becomes $A_t \psi U_c/(\theta f)$ with $A_t = 1 + \epsilon - \gamma_c$, and the labor-wedge-multiplier term becomes $-\eta [\Gamma_l U_c/U_l - \Gamma_c]$ which by (24) equals

$$-\eta \left[-\frac{\epsilon - \gamma_c}{y} - (1 - \bar{\tau}_t^y) \frac{\sigma}{c} + \frac{\gamma_c}{y} \right] = \eta \left[(1 - \bar{\tau}_t^y) \frac{\sigma}{c} + \frac{\epsilon - 2\gamma_c}{y} \right] = \eta \Delta_t$$

Using $U_d/U_c = \gamma_c/l$, $U_{cc}/U_c = -\sigma/c$, and $U_l/U_c = -\theta(1 - \bar{\tau}_t^y)$, the savings-wedge-multiplier term is

$$\frac{\eta_t^s}{\theta(1 - \bar{\tau}_t^y)} \left[\frac{\gamma_c}{l} - \frac{\sigma}{c} \theta(1 - \bar{\tau}_t^y) \right] = -\eta_t^s \left[\frac{\sigma}{c} - \frac{\gamma_c}{(1 - \bar{\tau}_t^y)y} \right] = -\eta_t^s \Lambda_t \quad \text{with } \Lambda_t \text{ as in (8)}$$

Putting everything together,

$$\frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} = A_t \frac{\psi U_{c,t}}{\theta f_t} + \eta_t \Delta_t - \eta_t^s \Lambda_t \quad (27)$$

Next, define the desired labor wedge $\tau_t^{y,*}$ and define $\tau_t^{y,*}/(1 - \tau_t^{y,*}) \equiv A_t \psi U_{c,t}/(\theta f_t)$. This is the same function with the same statistics as the standard labor wedge, so it is what the planner would set if it could. Moving it to the left of (27) and adding $\eta_t^s \Lambda_t$ to both sides,

$$E_t \equiv \frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} - \frac{\tau_t^{y,*}}{1 - \tau_t^{y,*}} = \eta_t \Delta_t - \eta_t^s \Lambda_t$$

$$\eta_t \Delta_t = E_t + \eta_t^s \Lambda_t$$

This is the expression in Lemma 5.2, with E_t being the labor wedge gap. To infer its sign, divide by Δ_t , which is positive by assumption,

$$\eta_t = \frac{E_t + \eta_t^s \Lambda_t}{\Delta_t} \quad (28)$$

η_t carries the sign of $E_t + \eta_t^s \Lambda_t$. It is positive when the frozen schedule holds the labor wedge above the desired wedge adjusted for the savings multiplier, negative where it holds it below, and zero where the two cross.

B.5 The intertemporal condition

The chain of Appendix A.4 is unchanged in form because neither constraint enters the conditions for u , w and w_2 or the boundary conditions. What the constraints reach is the value of ϕ/f , which (26) evaluates at the generalized incentive credit. The first-order condition for w and the envelope condition give

$$\lambda_{1,t+1} = \beta R \frac{\phi}{f} = \beta R \frac{1 - \chi_t}{U_{c,t}}$$

Integrating the costate equation over $(0, \infty)$ with $\psi(0) = \psi(\infty) = 0$,

$$\lambda_{1,t} = \int_0^\infty \phi d\theta = \mathbb{E}_{t-1} \left[\frac{1 - \chi_t}{U_{c,t}} \right]$$

Evaluating the second at $t + 1$ and equating with the first,

$$\frac{1 - \chi_t}{U_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}}{U_{c,t+1}} \right] \quad (29)$$

This is the generalized inverse Euler equation of Proposition 5.2. Multiplying (29) by $U_{c,t}$ and substituting the frozen Euler equation,

$$1 - \chi_t = (1 - \bar{\tau}_t^s) \mathbb{E}_t[U_{c,t+1}] \mathbb{E}_t \left[\frac{1 - \chi_{t+1}}{U_{c,t+1}} \right]$$

Because $\mathbb{E}[X] \mathbb{E}[Y] = \mathbb{E}[XY] - \text{Cov}(X, Y)$ with $X = U_{c,t+1}$ and $Y = (1 - \chi_{t+1})/U_{c,t+1}$,

$$1 - \chi_t = (1 - \bar{\tau}_t^s) \left\{ \mathbb{E}_t[1 - \chi_{t+1}] - \text{Cov}_t \left(U_{c,t+1}, \frac{1 - \chi_{t+1}}{U_{c,t+1}} \right) \right\} \quad (30)$$

This is the general form that the covariance display below specializes.

Both (29) and (30) are statements in χ_t , which by (26) carries the two multipliers together. The law of motion is to be an equation in the savings multiplier alone, so split the delivery cost into the part free of η_t^s and the part that carries it:

$$\mathcal{G}_t \equiv 1 - \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} + \eta_t \Gamma_{c,t} \quad \text{so that} \quad 1 - \chi_t = \mathcal{G}_t - \eta_t^s \frac{\sigma_t}{c_t} \quad (31)$$

$\mathcal{G}_t$ is what a util delivered through the numéraire costs once the incentive credit and the frozen labor schedule are counted when the savings constraint is inactive. Substituting (31) collects η^s on one side and everything else on the other. I maintain $\mathcal{G}_t > 0$ at every date. By (31), it equals one when the numéraire is separable from labor and the labor wedge is not frozen, but nothing signs it in general, and both the definition below and the factoring further down need it. Define the desired savings wedge as

$$1 - \tau_t^{s,*} \equiv \frac{\mathcal{G}_t}{\mathbb{E}_t[\mathcal{G}_{t+1}]}$$

This is the intertemporal margin the mechanism delivers when only the labor wedge schedule is frozen. When the date- $(t+1)$ draw is degenerate, the covariance vanishes and the conditional expectation drops. Writing (31) on both sides,

$$\mathcal{G}_t - \eta_t^s \frac{\sigma_t}{c_t} = (1 - \bar{\tau}_t^s) \left[\mathcal{G}_{t+1} - \eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} \right]$$

Collecting the η^s terms on the left and factoring the positive $\mathcal{G}_{t+1}$ out of the right,

$$\eta_t^s \frac{\sigma_t}{c_t} - (1 - \bar{\tau}_t^s) \eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} = \mathcal{G}_{t+1} \left[\frac{\mathcal{G}_t}{\mathcal{G}_{t+1}} - (1 - \bar{\tau}_t^s) \right] = \mathcal{G}_{t+1} (\bar{\tau}_t^s - \tau_t^{s,*}) \quad (32)$$

This is the law of motion of Proposition 5.2 and, with $\mathcal{G}_{t+1} > 0$, its right-hand side carries the sign of the wedge gap $\bar{\tau}_t^s - \tau_t^{s,*}$. Away from a degenerate draw, the same steps applied to (30) carry the date- $(t+1)$ terms as conditional expectations and leave the covariance standing.

$$\eta_t^s \frac{\sigma_t}{c_t} - (1 - \bar{\tau}_t^s) \mathbb{E}_t \left[\eta_{t+1}^s \frac{\sigma_{t+1}}{c_{t+1}} \right] = \mathbb{E}_t[\mathcal{G}_{t+1}] (\bar{\tau}_t^s - \tau_t^{s,*}) + (1 - \bar{\tau}_t^s) \text{Cov}_t \left(U_{c,t+1}, \frac{1 - \chi_{t+1}}{U_{c,t+1}} \right) \quad (33)$$

At a constant $\chi_{t+1} \equiv \bar{\chi}$, the covariance is $(1 - \bar{\chi}) \text{Cov}_t(U_{c,t+1}, 1/U_{c,t+1})$. Its second factor is negative whenever $U_{c,t+1}$ is non-degenerate, since $\mathbb{E}_t[U_{c,t+1} \cdot 1/U_{c,t+1}] = 1$ while $\mathbb{E}_t[U_{c,t+1}] \mathbb{E}_t[1/U_{c,t+1}] > 1$ by Jensen's inequality and $U_{c,t+1} > 0$ by Assumption 3.1. The product is negative because $1 - \bar{\chi} > 0$: by (26) it equals $U_{c,t+1} \phi_{t+1}/f_{t+1}$, and $\phi > 0$ because delivering utility costs resources. It stays negative unless χ_{t+1} varies enough with $U_{c,t+1}$ to reverse it and it is negative outright when $1 - \chi_{t+1}$ is independent of $U_{c,t+1}$ with positive mean. Risk therefore adds a negative drift to the wedge gap, offsetting a positive error and deepening a negative one.

When the labor schedule is left free, $\eta_t \equiv 0$ and $\mathcal{G}_t = 1 - \gamma_{c,t}\psi U_{c,t}/(\theta f_t)$. The law of motion keeps its form, with the labor-constraint term $+\eta_t\Gamma_{c,t}$ absent from the savings-free delivery cost and from the desired savings wedge. Setting $\bar{\tau}_t^s = \tau_t^{s,*}$ at every date solves (32) with $\eta^s \equiv 0$.

From T_R on, the same term is absent for a different reason. The labor constraint is not imposed there, so $\eta_t \equiv 0$ and $\psi \equiv 0$ by Corollary 5.3 and $\eta_t\Gamma_{c,t}$ is absent from χ_t and $\mathcal{G}_t$ rather than evaluated at $y_t = 0$, where the $1/y_t$ that $\Gamma_{c,t}$ carries is undefined. The retiree's delivery cost is $\mathcal{G}_t = 1$ and the desired savings wedge is zero, so the law of motion reads $\eta_t^s\sigma_t/c_t - (1 - \bar{\tau}_t^s)\eta_{t+1}^s\sigma_{t+1}/c_{t+1} = \bar{\tau}_t^s$. That is, any nonzero frozen wedge keeps injecting error through retirement.

B.6 The sign of the savings multiplier

The date-0 problem is the continuation of no history, so $\vartheta_0 = 0$. Likewise, the date- T problem has no successor and passes no requirement forward, so there is no ϑ_{T+1} term. Equation (9) carries the two multipliers in a single bracket, $\vartheta_t - \vartheta_{t+1}(\theta^t)/\kappa_t$ with $\kappa_t \equiv (1 - \bar{\tau}_t^s)\beta R^2 > 0$ because $\bar{\tau}_t^s < 1$ and $\beta, R > 0$. At $t = 0$, the leading term vanishes, leaving $\eta_0^s = -\vartheta_1(\theta^0)U_{c,0}/\kappa_0$. At $t = T$, the trailing term vanishes, leaving $\eta_T^s = \vartheta_T U_{c,T}$. Solving (9) for ϑ_{t+1} gives:

$$\vartheta_{t+1}(\theta^t) = \kappa_t \left[\vartheta_t - \frac{\eta_t^s}{U_{c,t}} \right] \quad (34)$$

Fix a history. Write $P_t \equiv \prod_{j=0}^{t-1} \kappa_j$, with $P_0 = 1$ and $P_t > 0$ for all $t > 0$, and define $x_t \equiv \vartheta_t/P_t$. Dividing (34) by $P_{t+1} = P_t\kappa_t$ leaves the lagged term with no coefficient.

$$x_{t+1} = x_t - \frac{\eta_t^s}{P_t U_{c,t}} \quad (35)$$

η_t^s is then the increment the date- t constraint adds to the normalized multiplier. Summing (35) from $t = 0$ to $T - 1$ telescopes the left side to $x_T - x_0$. $\vartheta_0 = 0$ gives $x_0 = 0$. The terminal condition $\eta_T^s = \vartheta_T U_{c,T}$ gives $x_T = \eta_T^s/(P_T U_{c,T})$. Collecting both endpoints on one side,

$$\sum_{t=0}^T \frac{\eta_t^s}{P_t U_{c,t}} = 0 \quad (36)$$

The weights $1/(P_t U_{c,t})$ are strictly positive so, if $\eta_t^s \geq 0$ at every date along the history, every term of (36) is non-negative and a sum of non-negative terms equals zero only if each term equals zero. So $\eta_t^s = 0$ at every date along the history, and (34) run forward from $\vartheta_0 = 0$ then gives $\vartheta_t = 0$ at every date along it. The case $\eta_t^s \leq 0$ at every date is the same argument with

the inequalities reversed. Nothing compares two nodes at the same date and the history was arbitrary, so a schedule leaving η^s one-signed at every node leaves it zero at every node. By contraposition, if $\eta^s \neq 0$ along the history, then (36) means terms of both signs must appear.

B.7 The direction of the reversal

At a degenerate draw, the law of motion pins where the multiplier can change sign and what sign it ends with. Let $z_t \equiv \eta_t^s \sigma_t / c_t$. I assume $\sigma_t > 0$ throughout, which makes z_t carry the sign of η_t^{s12} , and read (32) as a backward recursion.

$$z_t = (1 - \bar{\tau}_t^s) z_{t+1} + \mathcal{G}_{t+1} (\bar{\tau}_t^s - \tau_t^{s,*}) \quad (37)$$

Write $a_t \equiv 1 - \bar{\tau}_t^s$, positive because $\bar{\tau}_t^s < 1$, and $g_t \equiv \mathcal{G}_{t+1} (\bar{\tau}_t^s - \tau_t^{s,*})$, which carries the sign of the wedge gap where $\mathcal{G}_{t+1} > 0$. Iterating (37) down from the last date,

$$z_t = A_{t,T} z_T + \sum_{s=t}^{T-1} A_{t,s} g_s, \quad A_{t,s} \equiv \prod_{j=t}^{s-1} a_j > 0 \quad (38)$$

Every $A_{t,s}$ is a product of positive terms, so each z_t is a positive multiple of z_T plus a positively weighted sum of the gaps from t onward. Equation (36) then fixes the terminal value. In these variables, it reads $\sum_t \omega_t z_t = 0$ with $\omega_t \equiv c_t / (\sigma_t P_t U_{c,t}) > 0$. Substituting (38) and collecting,

$$z_T = - \frac{\sum_{s < T} W_s g_s}{\sum_t \omega_t A_{t,T}}, \quad W_s \equiv \sum_{t \leq s} \omega_t A_{t,s} > 0 \quad (39)$$

Every weight is strictly positive, so the multiplier ends the life cycle negative when the wedge gaps are positive in weighted average. This holds whatever the pattern of underdistortion or overdistortion over the life cycle. Whenever that weighted sum is nonzero, $z_T \neq 0$, so η_t^s is nonzero at some date and Lemma 5.3 gives it both signs along the history such that a crossing occurs.

Where a crossing can sit is settled date by date by the sign of the local gap. Where $g_t > 0$, $z_t \leq 0$ forces $z_{t+1} < 0$, so the multiplier cannot rise through zero and only a downward crossing is available. Where $g_t < 0$, $z_t \geq 0$ forces $z_{t+1} > 0$, so only an upward crossing is available. Where the gap is zero, $z_t = (1 - \bar{\tau}_t^s) z_{t+1}$ and the two dates share a sign, so no crossing occurs. Each case rules a direction out rather than forcing one in. Whether the multiplier crosses at a

¹²Assumption 3.1 delivers only $\sigma_t \geq 0$. The strict version, which the calibration of Section 6 satisfies, is what the argument here needs.

given date depends on its level there. Retirement is where the gap's sign is known in advance since $\mathcal{G}_{t+1} = 1$ and $\tau_t^{s,*} = 0$ leave $g_t = \bar{\tau}_t^s$, which is positive at any positive capital income tax, meaning retirement can host only a downward crossing.

Away from the degenerate draw case, the reversal result still survives because it never reads the law of motion away from retirement. There, the draw is degenerate whatever the skill process and $\mathcal{G}_{t+1} = 1$ with $\tau_t^{s,*} = 0$, so $\eta_t^s = 0$ at every date would reduce (32) to $0 = \bar{\tau}_t^s$, which any nonzero capital income tax rules out. Lemma 5.3, free of both the draw and the first-order system, then gives both signs along every history. However, the results on the terminal sign of (39), the date-by-date restriction on where a crossing can sit, as well as the count and direction that follow from them do not apply.

All three read (32), whereas risk replaces this with (33), which relates the realized z_t to the conditional mean $\mathbb{E}_t[z_{t+1}]$ rather than to z_{t+1} itself. The date-by-date restriction then weakens: the most $z_t \leq 0$ can deliver is $\mathbb{E}_t[z_{t+1}] < 0$, a statement about the average successor and not about any particular one. The terminal sign fares worse, since (38) unrolls the recursion from one date to the next and there is no such chain to unroll here.

B.8 The net shadow price

Substituting (28) into the net shadow price $\zeta_t \equiv \eta_t (1 - \bar{\tau}_t^y) - \eta_t^s$ and collecting terms in η_t^s ,

$$\zeta_t = (1 - \bar{\tau}_t^y) \frac{E_t + \eta_t^s \Lambda_t}{\Delta_t} - \eta_t^s = \frac{(1 - \bar{\tau}_t^y) E_t}{\Delta_t} + \eta_t^s \left[\frac{(1 - \bar{\tau}_t^y) \Lambda_t}{\Delta_t} - 1 \right]$$

By (8), $(1 - \bar{\tau}_t^y) \Lambda_t = \Gamma_{c,t}$, so the bracket is $(\Gamma_{c,t} - \Delta_t) / \Delta_t$. The two statistics carry the term $(1 - \bar{\tau}_t^y) \sigma_t / c_t$ identically, so it cancels in their difference and

$$\Gamma_{c,t} - \Delta_t = -\frac{\gamma_{c,t}}{y_t} - \frac{\epsilon_t - 2\gamma_{c,t}}{y_t} = \frac{\gamma_{c,t} - \epsilon_t}{y_t} = -\frac{A_t - 1}{y_t}$$

The last equality uses $A_t \equiv 1 + \epsilon_t - \gamma_{c,t}$. Substituting the bracket gives (10). The coefficient on E_t is positive because $\Delta_t > 0$ and $1 - \bar{\tau}_t^y = -U_l / (\theta U_c) > 0$ by (6) and Assumption 3.1. The coefficient on η_t^s , taken with its sign, is $-(A_t - 1) / (y_t \Delta_t)$, which is negative wherever $A_t > 1$. This proves the identity and the coefficient signs. Appendix B.12 records what an earnings-indexed schedule does to them.

Consequence (i). Fix a history and suppose $\eta^s \neq 0$ along it. Lemma 5.3, whose walk runs along a fixed history, then delivers dates with $\eta_t^s > 0$ and dates with $\eta_t^s < 0$ along that one. Nothing in that argument restricts where they fall, though at a degenerate draw (37) puts the positive dates first when the savings schedule overdistorts at every date. Over the working life, the

coefficient is negative under $A_t > 1$. From the retirement year onward, the decomposition has no content, but Corollary 5.3 gives $\zeta_t = -\eta_t^s$, a coefficient of -1 . The coefficient is therefore negative at every date of the life cycle, so the contribution carries the opposite sign to the multiplier throughout and a multiplier taking both signs delivers a contribution of both signs.

The two regimes are not patched together at T_R . Clearing y_t out of the working-life coefficient,

$$-\frac{A_t - 1}{y_t \Delta_t} = -\frac{\epsilon_t - \gamma_{c,t}}{y_t (1 - \bar{\tau}_t^y) \sigma_t / c_t + \epsilon_t - 2\gamma_{c,t}} \quad (40)$$

$\gamma_{c,t} = U_{clt}/U_{c,t}$ vanishes with labor supply, as does y_t itself. That is, wherever ϵ_t stays bounded away from zero the coefficient tends to -1 . Retirement's -1 is the limit of the working-life coefficient rather than a second case, which is why $A_t > 1$ and no other threshold is what carries one sign across the seam.

Consequence (ii). E_t is not a within-date object since $\tau_t^{y,*}$ carries the costate, which the persistence term of Proposition 4.1 ties to date $t - 1$. What the labor side lacks is not a link across dates, but the pair of endpoint conditions that gives Lemma 5.3 its force, and the identity carries no cross-date term in E_t . With the schedule free to move date by date, any sign profile for the labor contribution is therefore consistent with the identity.

Consequence (iii). Read (10) at a schedule that overdistorts labor at every date. The labor term $(1 - \bar{\tau}_t^y) E_t / \Delta_t$ is then positive at every date whatever the sign of η_t itself, which (28) leaves ambiguous wherever $\eta_t^s \Lambda_t < 0$. With normal leisure, this is wherever $\eta_t^s < 0$. Write $\zeta_t = L_t + \mathcal{T}_t$, with $L_t > 0$ the labor term and $\mathcal{T}_t$ the savings term, which by Appendix B.7 rises through zero exactly once when the savings schedule overdistorts throughout and the date- $(t + 1)$ draw is degenerate. Under weak separability, the carbon-price deviation is $-\zeta_t \mathcal{I}_t$ by Corollary 5.1, carrying the sign opposite to ζ_t where the dirty good is normal, so raising the labor level shifts the deviation down at every age.

What that does to the crossing takes more care than adding a constant to a rising profile. Raising L_t raises ζ_t pointwise, so the set of ages carrying a positive deviation, $\{t : \mathcal{T}_t < -L_t\}$, can only shrink. Where that set is an initial segment of the life cycle, shrinking it is the crossing coming forward. A large enough labor level empties it and leaves the deviation negative throughout. That is what the identity supports and it is weaker than a crossing date falling smoothly in L , which would need $\mathcal{T}_t$ increasing. Appendix B.7 counts the crossings of $\mathcal{T}_t$ and rules out downward ones. It does not make the profile monotone between them.

This is a reading of the identity at fixed multipliers rather than a comparative static in the schedule. Freezing labor moves η_t into $\mathcal{G}_t$, hence into $\tau_t^{s,*}$ and the law of motion, so $\mathcal{T}_t$ responds too. What the identity delivers is that the two channels enter additively at the solved allocation, not that one can be moved with the other held still.

B.9 Commodity wedge

Divide the first-order condition for q by f , using $U_{ql}l = \gamma_q U_q$ and $U_{cq}/U_c = -\sigma_{cq}/q$.

$$1 - \gamma_q \frac{\psi U_q}{\theta f} + \nu_t + \eta \Gamma_q - \eta_t^s \frac{\sigma_{cq}}{q} = \frac{\phi U_q}{f}$$

Dividing by U_q , substituting $\phi/f = (1 - \chi_t)/U_c$ from (26), and multiplying by U_c ,

$$\frac{U_c}{U_q} \left(1 + \nu_t + \eta \Gamma_q - \eta_t^s \frac{\sigma_{cq}}{q} \right) - \gamma_q \frac{\psi U_c}{\theta f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta \Gamma_c - \eta_t^s \frac{\sigma}{c}$$

With $1 + \tau^q = U_q/U_c$, multiply through by $1 + \tau^q$ and rearrange terms.

$$\frac{\tau_t^q - \nu_t}{1 + \tau_t^q} = (\gamma_c - \gamma_q) \frac{\psi U_c}{\theta f} - \eta \Gamma_c + \eta_t^s \frac{\sigma}{c} + \frac{1}{1 + \tau^q} \left[\eta \Gamma_q - \eta_t^s \frac{\sigma_{cq}}{q} \right]$$

Substituting Γ_c and Γ_q from (24),

$$\begin{aligned} \frac{\tau_t^q - \nu_t}{1 + \tau_t^q} &= (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t} - \eta_t (1 - \bar{\tau}_t^y) \frac{\sigma_t}{c_t} + \eta_t \frac{\gamma_{c,t}}{y_t} + \eta_t^s \frac{\sigma_t}{c_t} \\ &\quad + \eta_t (1 - \bar{\tau}_t^y) \frac{\sigma_{cq,t}}{q_t (1 + \tau_t^q)} - \eta_t \frac{\gamma_{q,t}}{y_t} - \eta_t^s \frac{\sigma_{cq,t}}{q_t (1 + \tau_t^q)} \\ &= (\gamma_{c,t} - \gamma_{q,t}) \left(\frac{\psi U_{c,t}}{\theta f_t} + \frac{\eta_t}{y_t} \right) - [\eta_t (1 - \bar{\tau}_t^y) - \eta_t^s] \left[\frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t (1 + \tau_t^q)} \right] \end{aligned}$$

The first bracket is the net shadow price ζ_t and the second bracket is the income-effect gap. It is immediate that $\mathcal{I}_t > 0$ if and only if the dirty good is normal. Let q^d be the dirty-good demand from the household's split of expenditure x at the prevailing consumer price, and let $\mathcal{D} \equiv U_{qq} - 2(1 + \tau^q)U_{cq} + (1 + \tau^q)^2 U_{cc}$ be the curvature of utility along that budget line, negative because the split sets $U_q = (1 + \tau^q)U_c$ and Assumption 3.1 signs $\mathcal{D} = -(2U_c U_q U_{cq} - U_c^2 U_{qq} - U_q^2 U_{cc})/U_c^2$. This is the $\mathcal{D}$ of Section 5.2 at $\bar{p} = 1 + \tau^q$, the frozen price having no counterpart in this problem. Differentiating the split's tangency along the budget line,

$$\frac{\partial q^d}{\partial x} = \frac{(1 + \tau^q)U_{cc} - U_{cq}}{\mathcal{D}} = \frac{U_c}{\mathcal{D}} \left[-(1 + \tau^q) \frac{\sigma}{c} + \frac{\sigma_{cq}}{q} \right] = \frac{U_c (1 + \tau^q)}{|\mathcal{D}|} \mathcal{I}_t$$

From T_R on, the conditions for c and q are redone rather than continued. The labor constraint is not imposed there, so $\eta_t \equiv 0$, and $\psi \equiv 0$ by Appendix C.12. The conditions for c and q lose their costate and labor-constraint terms, leaving $\phi U_c/f = 1 - \eta_t^s \sigma/c$ and $1 + \nu_t - \eta_t^s \sigma_{cq}/q = \phi U_q/f$. Dividing the second by U_q , multiplying by U_c , and equating the two,

$$\frac{1 + \nu_t}{1 + \tau_t^q} = 1 - \eta_t^s \left[\frac{\sigma_t}{c_t} - \frac{\sigma_{cq,t}}{q_t (1 + \tau_t^q)} \right] = 1 - \eta_t^s \mathcal{I}_t \quad \implies \quad \frac{\tau_t^q - \nu_t}{1 + \tau_t^q} = \eta_t^s \mathcal{I}_t$$

This is Corollary 5.3. No separability assumption enters and nothing is read at $y_t = 0$. Both $\gamma_{c,t}$ and $\gamma_{q,t}$ carry the factor l_t , so they disappear at $l_t = 0$ and the Corlett-Hague term is absent for that reason as well as through ψ .

Appendix B.10 identifies $\nu_t = \text{SCC}_t$ in both, completing Proposition 5.1 and Corollary 5.3.

B.10 The constrained social cost of carbon

Take the first-order condition for S , divide the integrand by f , and substitute $\phi/f = (1 - \chi_t)/U_c$ together with $U_{Sl} = \xi U_S$, Γ_S from (24) and $U_{cS}/U_c = \xi_c (U_S/U_c)/c$. The integrand becomes

$$-\frac{U_S}{U_c} \left\{ \xi \frac{\psi U_c}{\theta f} + (1 - \chi_t) + \eta \left[\frac{\xi}{y} + (1 - \bar{\tau}_t^y) \frac{\xi_c}{c} \right] - \eta_t^s \frac{\xi_c}{c} \right\}$$

Expanding $1 - \chi_t = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta \left[(1 - \bar{\tau}_t^y) \frac{\sigma}{c} - \frac{\gamma_c}{y} \right] - \eta_t^s \frac{\sigma}{c}$ and collecting,

$$\begin{aligned} \{\cdot\} &= 1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta \frac{\xi - \gamma_c}{y} + [\eta (1 - \bar{\tau}_t^y) - \eta_t^s] \frac{\sigma + \xi_c}{c} \\ &= 1 + (\xi - \gamma_c) \left(\frac{\psi U_c}{\theta f} + \frac{\eta}{y} \right) + \zeta_t \frac{\sigma + \xi_c}{c} \end{aligned}$$

The date- t pollution-stock multiplier is

$$\mu_t = - \int_0^\infty \left[1 + (\xi - \gamma_c) \left(\frac{\psi U_c}{\theta f} + \frac{\eta}{y} \right) + \zeta_t \frac{\sigma + \xi_c}{c} \right] \frac{U_S}{U_c} f d\theta \quad (41)$$

Iterating the emission-price recursion $\nu_t = \mu_t S_q + \frac{1}{R} \partial V_{t+1} / \partial Q$ forward as in Appendix A.6 with $\Phi_j \equiv \partial S_{t+j} / \partial Q_t$ and $\Phi_j = 0$ for $j > s$,

$$\nu_t = \text{SCC}_t = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \left[1 + (\xi_{t+j} - \gamma_{c,t+j}) \left(\frac{\psi_{t+j} U_{c,t+j}}{\theta f_{t+j}} + \frac{\eta_{t+j}}{y_{t+j}} \right) + \zeta_{t+j} \frac{\sigma_{t+j} + \xi_{c,t+j}}{c_{t+j}} \right] \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta$$

The sum runs over dates so, read at a working-age history, it reaches nodes with $t+j \geq T_R$, at which the labor constraint is not imposed. With $\psi = 0$ and no Γ_S term, the braces of (41) collapse to $1 - \eta_{t+j}^s (\sigma_{t+j} + \xi_{c,t+j}) / c_{t+j} = 1 + \zeta_{t+j} (\sigma_{t+j} + \xi_{c,t+j}) / c_{t+j}$, which is the displayed bracket with the screening term deleted. Only the savings channel prices damages from T_R on, which is Corollary 5.3 read inside the damage stream rather than at the commodity margin.

Setting $\eta = 0$ gives the weight under a frozen savings schedule alone, setting $\eta^s = 0$ gives the weight under a frozen labor schedule alone, and setting both to zero leaves the formula in Appendix A.6. Under $U = V(h(c, q, S), l)$, which gives $\gamma_c = \gamma_q$ and $\xi = \gamma_c$ together, $1 + \zeta_{t+j} (\sigma_{t+j} + \xi_{c,t+j}) / c_{t+j}$, so weak separability is not enough to restore a Pigouvian social cost of carbon once a margin is frozen.

B.11 The optimal flat pair

Let the two schedules be flat at levels $\bar{\tau}_t^y$ and $\bar{\tau}_t^s$ and treat each as a choice variable. Write $\mathcal{L}$ for the date-0 Lagrangian of the problem of Section 5.1: the resource cost with every constraint appended, whose period block is the Hamiltonian of Appendix B.2. The rate $\bar{\tau}_t^y$ enters it only through the date- t labor constraints, one per node, so its derivative collects those terms and nothing else:

$$\frac{\partial \mathcal{L}}{\partial \bar{\tau}_t^y} = \int_0^\infty \eta_t(\theta) f_t(\theta) d\theta = \mathbb{E}_0[\eta_t] = 0$$

The rate $\bar{\tau}_t^s$ enters only through the requirement $M(\theta)$ passed to the successor, so

$$\frac{\partial \mathcal{L}}{\partial \bar{\tau}_t^s} = \int_0^\infty \frac{1}{R} \frac{\partial V_{t+1}}{\partial \hat{M}} \frac{\partial M(\theta)}{\partial \bar{\tau}_t^s} f_t(\theta) d\theta = - \int_0^\infty \frac{\vartheta_{t+1}(\theta) U_{c,t}(\theta)}{(1 - \bar{\tau}_t^s)^2 \beta R^2} f_t(\theta) d\theta = 0$$

which, with $(1 - \bar{\tau}_t^s)^2 \beta R^2 > 0$ constant across the date- t cross-section, gives $\mathbb{E}_0[\vartheta_{t+1} U_{c,t}] = 0$. Taking the cross-sectional expectation of (9) then gives $\mathbb{E}_0[\eta_t^s] = \mathbb{E}_0[\vartheta_t U_{c,t}]$, which equals zero at $t = 0$ because $\vartheta_0 = 0$.

For the closed form, solve (27) for η_t and impose $\mathbb{E}_0[\eta_t] = 0$:

$$\mathbb{E}_0 \left[\frac{1}{\Delta_t} \left(\frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} - A_t \frac{\psi U_{c,t}}{\theta f_t} + \eta_t^s \Lambda_t \right) \right] = 0$$

The weight is $1/\Delta_t$, which Appendix B.3 reads as the compensated earnings response $s_{yy,t} = 1/((1 - \bar{\tau}_t^y) \Delta_t)$ of (25): the hours margin's own compensated response, the object the labor constraint prices, not the full two-good demand response. With $\bar{\tau}_t^y$ constant across the cross-section, the zero-average condition rearranges to

$$\frac{\bar{\tau}_t^y}{1 - \bar{\tau}_t^y} = \frac{\mathbb{E}_0 \left[s_{yy,t} \left(A_t \frac{\psi U_{c,t}}{\theta f_t} - \eta_t^s \Lambda_t \right) \right]}{\mathbb{E}_0[s_{yy,t}]}$$

which is Proposition 5.5. At $\eta_t^s = 0$ the savings term drops and the rate averages the desired wedges alone.

B.12 An earnings-indexed schedule

Section 5.1 leaves the frozen labor schedule arbitrary. The one this paper freezes is the two-parameter tax and transfer function of Bénabou (2002) and Heathcote et al. (2017),

$$T_t(y) = y - \lambda_t^T y^{1-p_t} \implies 1 - \bar{\tau}_t^y(y) = \lambda_t^T (1 - p_t) y^{-p_t} \quad (42)$$

where p_t indexes progressivity and λ_t^T the level.

Indexing the schedule on earnings rather than on type changes one statistic and nothing else, which is what lets Section 5.1 be read at (42) throughout. Three things have to be shown: that $\Gamma_{c,t}$, $\Gamma_{q,t}$ and $\Gamma_{S,t}$ survive intact, since $\bar{\tau}_t^y$ now depends on the type through assigned earnings $y_t(\theta^t)$; that $\Gamma_{l,t}$ and hence Δ_t each gain a term in p_t ; and that the resulting substitution $\epsilon_t \mapsto \epsilon_t + p_t$ stays inside Δ_t everywhere downstream of (8) but one.

With the frozen labor wedge following (42), $1 - \bar{\tau}_t^y(y) = \lambda_t^T (1 - p_t) y^{-p_t}$ with $y = \theta l$. Differentiating,

$$\frac{\partial \bar{\tau}_t^y}{\partial y} = \lambda_t^T (1 - p_t) p_t y^{-p_t-1} = \frac{p_t (1 - \bar{\tau}_t^y)}{y}$$

The constraint is $\Gamma = -U_l/(\theta U_c) - (1 - \bar{\tau}_t^y(y))$. Its derivatives with respect to c , q , and S are unchanged, because $\bar{\tau}_t^y$ depends on none of them. Its derivative with respect to l gains one term through $y = \theta l$:

$$\Gamma_l = (1 - \bar{\tau}_t^y) \frac{\epsilon - \gamma_c}{l} + \frac{\partial \bar{\tau}_t^y}{\partial y} \theta = (1 - \bar{\tau}_t^y) \frac{\epsilon - \gamma_c}{l} + \frac{p_t (1 - \bar{\tau}_t^y)}{l} = (1 - \bar{\tau}_t^y) \frac{\epsilon - \gamma_c + p_t}{l}$$

using $\theta/y = 1/l$. Only Δ_t changes, since Γ_l enters the derivation of Appendix B.4 solely through $-\eta [\Gamma_l U_c/U_l - \Gamma_c]$:

$$-\eta \left[-\frac{\epsilon - \gamma_c + p_t}{y} - (1 - \bar{\tau}_t^y) \frac{\sigma}{c} + \frac{\gamma_c}{y} \right] = \eta \left[(1 - \bar{\tau}_t^y) \frac{\sigma}{c} + \frac{\epsilon + p_t - 2\gamma_c}{y} \right]$$

so $\Delta_t \mapsto \Delta_t + p_t/y_t$, that is

$$\Gamma_{l,t} = (1 - \bar{\tau}_t^y) \frac{\epsilon_t - \gamma_{c,t} + p_t}{l_t} \quad \text{and} \quad \Delta_t = (1 - \bar{\tau}_t^y) \frac{\sigma_t}{c_t} + \frac{\epsilon_t + p_t - 2\gamma_{c,t}}{y_t}$$

The same p_t/y_t reaches the compensated-perturbation bracket of Appendix B.3, so $s_{yy,t}$ carries the shifted Δ_t as well.

Every other object downstream of (8) is untouched but one. The exception is the cancellation of Appendix B.8, where $\Gamma_{c,t}$ is unchanged while Δ_t is not, so the substitution survives the difference:

$$\Gamma_{c,t} - \Delta_t = -\frac{A_t - 1}{y_t} - \frac{p_t}{y_t} = -\frac{A_t - 1 + p_t}{y_t}$$

Equation (10) therefore holds with $A_t - 1 + p_t$ in the numerator of the coefficient on η_t^s and $\Delta_t + p_t/y_t$ in both denominators, so that the coefficient reads $-(A_t - 1 + p_t)/(y_t \Delta_t + p_t)$: p_t appears outside Δ_t as well as inside it, not instead of it, and the condition signing that coefficient weakens from $A_t > 1$ to $A_t - 1 + p_t > 0$. Because $p_t > 0$ for a progressive schedule, Δ_t rises: the schedule's convexity in earnings makes the agent's objective more concave in hours, so an interior labor choice is easier to sustain than under a schedule flat in earnings, and the interior-hours requirement $\Delta_t > 0$ is weaker than under a type-indexed schedule.

Freezing a tax function rather than a type-indexed wedge schedule is the more faithful exercise and it strengthens the reading of Corollary 5.1. A tax code conditions on earnings, so a planner constrained by (42) cannot make the type-by-type adjustment Kaplow (2012) relies on even in principle, whereas a planner constrained by a type-indexed schedule is denied it only by assumption.

C Derivations: a given carbon price

This appendix derives the results of Section 5.2 in one pass. The commodity wedge is held to a schedule $\bar{p}_t(\theta) \equiv 1 + \bar{\tau}_t^q(\theta)$: the given-schedule results read every condition at an arbitrary exogenous schedule, and the optimal-anonymous-price results at a flat schedule, $\bar{p}_t(\theta) = 1 + \bar{\tau}_t$, whose level is chosen optimally. The shadow emission price ν_t of (14) is carried throughout; the no-externality benchmark sets $\nu_t = 0$. Multiplier conventions follow Appendix A.3, and $\eta^q(\theta)f_t(\theta|\theta_-)$ is the multiplier on the wedge constraint. Two shorthands recur: the ad-valorem deviation and its desired counterpart,

$$\varpi_t(\theta) \equiv \frac{\bar{\tau}_t^q(\theta) - \nu_t}{1 + \bar{\tau}_t^q(\theta)} \quad \varpi_t^*(\theta^t) \equiv (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t}$$

which at a flat schedule are the ϖ_t and ϖ_t^* of Section 5.2.

C.1 The problem and first-order conditions

The Hamiltonian is that of Appendix A.1 plus the constraint term $\eta^q [U_q/U_c - \bar{p}] f$. Write $\Gamma^q \equiv U_q/U_c - \bar{p}$ and Γ_z^q for its partial derivatives; each follows from the quotient rule,

$$\Gamma_z^q = \frac{U_{qz}}{U_c} - \frac{U_q}{U_c} \frac{U_{cz}}{U_c} = \frac{U_{qz}}{U_c} - \bar{p} \frac{U_{cz}}{U_c}$$

evaluated at the constrained allocation, where $U_q/U_c = \bar{p}$. For the numéraire, with $U_{qc}/U_c = -\sigma_{cq}/q$ and $U_{cc}/U_c = -\sigma/c$,

$$\Gamma_c^q = -\frac{\sigma_{cq}}{q} + \bar{p} \frac{\sigma}{c} = \bar{p} \left[\frac{\sigma}{c} - \frac{\sigma_{cq}}{\bar{p}q} \right] = \bar{p} \mathcal{I}_t, \quad (43)$$

the income-effect gap (3) evaluated at $1 + \tau^q = \bar{p}$. For the dirty good, define $\sigma_{q,t} \equiv -U_{qq} q_t / U_q$, the own-curvature of the dirty good's marginal utility, so that $U_{qq}/U_c = (U_{qq}q/U_q)(U_q/U_c)/q = -\bar{p} \sigma_q/q$ and

$$\Gamma_q^q = -\bar{p} \frac{\sigma_q}{q} + \bar{p} \frac{\sigma_{cq}}{q} = -\bar{p} \frac{\sigma_q - \sigma_{cq}}{q} \equiv -\bar{p} \mathcal{J}_t, \quad (44)$$

with $\mathcal{J}_t \equiv (\sigma_{q,t} - \sigma_{cq,t})/q_t$ the mirror image of $\mathcal{I}_t$. For labor, with $U_{ql}/U_c = (U_{ql}l/U_q)(U_q/U_c)/l = \bar{p} \gamma_q/l$ and $U_{cl}/U_c = \gamma_c/l$,

$$\Gamma_l^q = \bar{p} \frac{\gamma_q}{l} - \bar{p} \frac{\gamma_c}{l} = -\bar{p} \frac{\gamma_c - \gamma_q}{l}. \quad (45)$$

For the pollution stock, with $\xi_c \equiv U_{Sc}c/U_S$ and $\xi_q \equiv U_{Sq}q/U_S$ the complementarities of the pollutant with each good, $U_{qS}/U_c = \xi_q(U_S/U_c)/q$ and $U_{cS}/U_c = \xi_c(U_S/U_c)/c$, so

$$\Gamma_S^q = \frac{U_S}{U_c} \left[\frac{\xi_q}{q} - \bar{p} \frac{\xi_c}{c} \right]. \quad (46)$$

The first-order conditions are those of Appendix A.1, each augmented by $\eta^q f \Gamma_z^q$:

$$\begin{aligned} \frac{\partial \mathcal{H}}{\partial c} &: f - \psi U_{cl} \frac{l}{\theta} + \eta^q f \Gamma_c^q = \phi U_c \\ \frac{\partial \mathcal{H}}{\partial q} &: f - \psi U_{ql} \frac{l}{\theta} + \nu_t f + \eta^q f \Gamma_q^q = \phi U_q \\ \frac{\partial \mathcal{H}}{\partial l} &: -\theta f - \phi U_l + \eta^q f \Gamma_l^q = \frac{1}{\theta} \psi [U_{ll}l + U_l] \\ \frac{\partial \mathcal{H}}{\partial S} &: \int_0^\infty \left(-\psi U_{Sl} \frac{l}{\theta} - \phi U_S + \eta^q f \Gamma_S^q \right) d\theta = \mu \end{aligned}$$

with the conditions for u , w , and w_2 unchanged, so the costate equation, the boundary conditions $\psi(0) = \psi(\infty) = 0$, and the promise-keeping chain carry over. In the no-externality economy $\nu_t = 0$ and the pollution-stock condition is absent. When the level of a flat schedule is itself chosen, one more derivative is needed; it is taken in Appendix C.9.

C.2 Implementation and first-order approach

Fix a common price $\bar{p} = 1 + \bar{\tau}$ and consider the agent's split of expenditure x ,

$$\tilde{U}(x, l; \bar{\tau}) = \max_{c, q \geq 0} U(c, q, l) \quad \text{s.t.} \quad c + \bar{p}q = x$$

The first-order condition of the split is $U_q = \bar{p} U_c$. If we substitute c using the budget constraint, $U(x - \bar{p}q, q, l)$ and therefore $\tilde{U}_q = -U_c \bar{p} + U_q$ and $\tilde{U}_{qq} = U_{cc} \bar{p}^2 - 2U_{cq} \bar{p} + U_{qq}$. Assumption 3.1 implies that $\mathcal{D} < 0$ and that an interior split exists and is unique. A bundle (c, q) therefore satisfies the wedge constraint $U_q/U_c = \bar{p}$ if and only if it is the split of $x = c + \bar{p}q$.

For the envelope identities, differentiate $\tilde{U} = U(c^d, q^d, l)$, writing $c^d(x, l; \bar{p})$ and $q^d(x, l; \bar{p})$ for the maximizers of the split problem and $\mathcal{D} \equiv \tilde{U}_{qq} < 0$ for the curvature of utility along the

budget line, the statistic of Section 5.2 that carries the identities below. Differentiating the budget over x gives $\frac{\partial c^d}{\partial x} + \bar{p} \frac{\partial q^d}{\partial x} = 1$. By the first-order condition,

$$\tilde{U}_x = U_c \frac{\partial c^d}{\partial x} + U_q \frac{\partial q^d}{\partial x} = U_c \left(\frac{\partial c^d}{\partial x} + \bar{p} \frac{\partial q^d}{\partial x} \right) = U_c$$

Differentiating the budget in l at fixed x gives $\frac{\partial c^d}{\partial l} + \bar{p} \frac{\partial q^d}{\partial l} = 0$.

$$\tilde{U}_l = U_l + U_c \left(\frac{\partial c^d}{\partial l} + \bar{p} \frac{\partial q^d}{\partial l} \right) = U_l$$

A deviator who reports $\hat{\theta}$ receives the bundle assigned to $\hat{\theta}$, meaning the pair $(x(\hat{\theta}), y(\hat{\theta}))$. They split $x(\hat{\theta})$ at the same common price, so the deviation space is that of a one-good economy in (x, y) with period utility $\tilde{U}(x, y/\theta; \bar{\tau})$. The type enters $\tilde{U}$ only through $l = y/\theta$, so the envelope condition is $\dot{u} = -\tilde{U}_l l/\theta + \beta w_2 = -U_l l/\theta + \beta w_2$ and the first-order approach applies as in the one-good economy of Appendix A.3.

C.3 Demand identities

Differentiating the tangency $U_q = \bar{p}U_c$ together with the budget gives the demand derivatives. In x at fixed $l, \bar{p}$, the budget gives $\partial_x c^d = 1 - \bar{p} \partial_x q^d$. Substituting into $U_{qc} \partial_x c^d + U_{qq} \partial_x q^d = \bar{p}(U_{cc} \partial_x c^d + U_{cq} \partial_x q^d)$ and collecting,

$$\mathcal{D} \partial_x q^d = -(U_{cq} - \bar{p}U_{cc}) \implies \partial_x q^d = \frac{U_{cq} - \bar{p}U_{cc}}{|\mathcal{D}|} = \frac{\bar{p}U_c \mathcal{I}_t}{|\mathcal{D}|}, \quad (47)$$

where the last step uses $U_{cq} - \bar{p}U_{cc} = U_c(\sigma \bar{p}/c - \sigma_{cq}/q) = \bar{p}U_c \mathcal{I}_t$ from the definitions of the statistics. Since $\partial_x q^d$ is the income expansion of dirty-good demand, $\mathcal{I}_t > 0$ exactly when the dirty good is normal. The budget then gives $\partial_x c^d = 1 - m_q$ with $m_q \equiv \bar{p} \partial_x q^d$, which satisfies $U_q \mathcal{J}_t = \bar{p}U_{cq} - U_{qq}$ and hence $\partial_x c^d = U_q \mathcal{J}_t / |\mathcal{D}|$ is positive when c is normal. In l at fixed $x, \bar{p}$, the budget gives $\partial_l c^d = -\bar{p} \partial_l q^d$ and the differentiated tangency yields $\mathcal{D} \partial_l q^d = \bar{p}U_{cl} - U_{ql}$, with $U_{cl} = \gamma_c U_c / l$ and $U_{ql} = \gamma_q \bar{p} U_c / l$,

$$\partial_l q^d = \frac{\bar{p}U_c(\gamma_c - \gamma_q)}{l \mathcal{D}} = \frac{\bar{p}U_c(\gamma_q - \gamma_c)}{l |\mathcal{D}|}. \quad (48)$$

Compensated own-price response: along an indifference curve $dc = -\bar{p}dq$ and differentiating the tangency in $\bar{p}$ at constant utility gives $\mathcal{D} d\bar{p} = U_c d\bar{p}$. Hence,

$$s_{qq} \equiv \frac{\partial q^d}{\partial \bar{p}} \Big|_u = \frac{U_c}{\mathcal{D}} = -\frac{U_c}{|\mathcal{D}|} < 0 \quad \frac{\partial q^d}{\partial \bar{p}} = s_{qq} - q^d \partial_x q^d \quad (\text{Slutsky})$$

Three identities tie the constraint statistics to the demand objects. Combining (48) with s_{qq} ,

$$l \partial_l q^d = -\bar{p} |s_{qq}| (\gamma_c - \gamma_q). \quad (49)$$

Adding $\bar{p} \mathcal{I}_t = (U_{cq} - \bar{p} U_{cc})/U_c$ and $\mathcal{J}_t = (\bar{p} U_{cq} - U_{qq})/U_q = (\bar{p} U_{cq} - U_{qq})/(\bar{p} U_c)$,

$$\Delta_t^q \equiv \bar{p} \mathcal{I}_t + \mathcal{J}_t = \frac{\bar{p} U_{cq} - \bar{p}^2 U_{cc} + \bar{p} U_{cq} - U_{qq}}{\bar{p} U_c} = \frac{-\mathcal{D}}{\bar{p} U_c} = \frac{1}{\bar{p} |s_{qq}|} > 0. \quad (50)$$

Combining (47) and (50), the marginal budget share is the numéraire-side curvature share,

$$m_q = \bar{p} \partial_x q^d = \frac{\bar{p}^2 U_c \mathcal{I}_t}{|\mathcal{D}|} = \frac{\bar{p} \mathcal{I}_t}{\Delta_t^q}. \quad (51)$$

C.4 The multiplier

Dividing the first-order condition for c by $U_c f$, using $U_c l = \gamma_c U_c$ and (43),

$$\frac{U_c \phi}{f} = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta^q \bar{p} \mathcal{I}_t. \quad (52)$$

Dividing the first-order condition for q by $U_q f$, multiplying by U_c , and substituting $U_c/U_q = 1/\bar{p}$ and (44),

$$\frac{U_c \phi}{f} = \frac{1 + \nu_t}{\bar{p}} - \gamma_q \frac{\psi U_c}{\theta f} + \eta^q \frac{\Gamma_q^q}{\bar{p}} = \frac{1 + \nu_t}{\bar{p}} - \gamma_q \frac{\psi U_c}{\theta f} - \eta^q \mathcal{J}_t. \quad (53)$$

Equating (52) and (53) and collecting,

$$1 - \frac{1 + \nu_t}{\bar{p}} = (\gamma_c - \gamma_q) \frac{\psi U_c}{\theta f} - \eta^q (\bar{p} \mathcal{I}_t + \mathcal{J}_t)$$

Since $1 - (1 + \nu_t)/\bar{p} = (\bar{\tau}_t^q - \nu_t)/(1 + \bar{\tau}_t^q) = \varpi_t$ with Δ_t^q from (50),

$$\eta_t^q \Delta_t^q = \varpi_t^* - \varpi_t \quad \Longleftrightarrow \quad \eta_t^q = \bar{p}_t |s_{qq,t}| (\varpi_t^* - \varpi_t). \quad (54)$$

This is Lemma 5.6 with $\text{SCC}_t \equiv \nu_t$; a type-varying schedule changes nothing in the display. At $\eta_t^q = 0$, (54) is the primitive corrective-wedge condition of Appendix A.5, so the multiplier vanishes where the frozen deviation equals the desired one.

C.5 Labor wedge

From the first-order condition for l , using $U_{ll}l + U_l = U_l(1 + \epsilon)$,

$$\phi = -\frac{\theta f}{U_l} - \frac{\psi}{\theta}(1 + \epsilon) + \frac{\eta^q f \Gamma_l^q}{U_l}$$

With $U_l = -(1 - \tau_t^y)\theta U_c$, where τ_t^y is the endogenous labor wedge, and (45),

$$\frac{\Gamma_l^q}{U_l} = \frac{-\bar{p}(\gamma_c - \gamma_q)/l}{-(1 - \tau_t^y)\theta U_c} = \frac{\bar{p}(\gamma_c - \gamma_q)}{(1 - \tau_t^y)y U_c}$$

Multiplying by U_c/f ,

$$\frac{U_c \phi}{f} = \frac{1}{1 - \tau_t^y} - (1 + \epsilon) \frac{\psi U_c}{\theta f} + \eta^q \frac{\bar{p}(\gamma_c - \gamma_q)}{(1 - \tau_t^y)y} \quad (55)$$

Equating (52) and (55) and collecting, with $1/(1 - \tau_t^y) - 1 = \tau_t^y/(1 - \tau_t^y)$ and $A_t = 1 + \epsilon - \gamma_c$,

$$\frac{\tau_t^y}{1 - \tau_t^y} = A_t \frac{\psi U_c}{\theta f} + \eta_t^q \bar{p}_t \left[\mathcal{I}_t - \frac{\gamma_{c,t} - \gamma_{q,t}}{(1 - \tau_t^y)y_t} \right] \quad (56)$$

This is the primitive labor wedge of Proposition 5.3. The bracket is the response of the implied commodity wedge $\ln(U_q/U_c)$ to a utility-compensated unit of earnings. Along $(dc, dl) = (1 - \tau_t^y, 1/\theta)$, we have $d\ln(U_q/U_c) = (1 - \tau_t^y)\mathcal{I}_t + (\gamma_q - \gamma_c)/y$. Dividing by $1 - \tau_t^y$ gives the bracket. Under weak separability the bracket is $\mathcal{I}_t$. Using (54) with $\varpi^* = 0$ and (51),

$$\eta_t^q \bar{p}_t \mathcal{I}_t = -\frac{\bar{p}_t \mathcal{I}_t}{\Delta_t^q} \varpi_t = -m_{q,t} \varpi_t = m_{q,t} \frac{\nu_t - \bar{\tau}_t^q}{1 + \bar{\tau}_t^q}$$

which is Corollary 5.4.

C.6 Savings wedge

Define $\chi_t^q \equiv \gamma_{c,t} \frac{\psi U_{c,t}}{\theta f_t} - \eta_t^q \bar{p}_t \mathcal{I}_t$, so that (52) reads

$$\frac{\phi}{f} = \frac{1 - \chi_t^q}{U_{c,t}}.$$

The promise-keeping chain of Appendix A.4 is untouched, since the wedge constraint enters none of its steps. The first-order condition for w and the envelope condition $\partial V_{t+1}/\partial \hat{w} = \lambda_{1,t+1}$ give

$$\lambda_{1,t+1}(\theta^t) = \beta R \frac{\phi(\theta)}{f(\theta)} = \beta R \frac{1 - \chi_t^q}{U_{c,t}}.$$

Integrating the costate equation over the type space with $\psi(0) = \psi(\infty) = 0$, $\int \bar{\alpha}_t f d\theta = 1$, and $\int f_2 d\theta = 0$ gives the second expression

$$\lambda_{1,t} = \int_0^\infty \phi d\theta = \mathbb{E}_{t-1} \left[\frac{1 - \chi_t^q}{U_{c,t}} \right].$$

Evaluating this at $t + 1$ and equating with the previous display one period apart,

$$\frac{1 - \chi_t^q}{U_{c,t}} = \frac{1}{\beta R} \mathbb{E}_t \left[\frac{1 - \chi_{t+1}^q}{U_{c,t+1}} \right]$$

Substituting the multiplier (54) and the curvature-share identity (51) assembles the adjustment into composite form,

$$\chi_t^q = \gamma_c \frac{\psi U_c}{\theta f} - m_q (\varpi_t^* - \varpi_t) = [\gamma_c - m_q(\gamma_c - \gamma_q)] \frac{\psi U_c}{\theta f} + m_q \varpi_t = \tilde{\gamma}_t \frac{\psi U_{c,t}}{\theta f_t} + m_{q,t} \varpi_t$$

with $\tilde{\gamma} \equiv (1 - m_q)\gamma_c + m_q\gamma_q$ as in Proposition C.1, and $m_q\varpi = (\bar{\tau}^q - \nu) \partial_x q^d$ is the commodity revenue net of marginal damages returned by a marginal dollar of expenditure. It remains to pass from the inverse Euler equation to the savings wedge. The savings wedge τ_t^s is defined by the agent's intertemporal condition $U_{c,t} = \beta R (1 - \tau_t^s) \mathbb{E}_t[U_{c,t+1}]$, so

$$1 - \tau_t^s = \frac{U_{c,t}}{\beta R \mathbb{E}_t[U_{c,t+1}]}.$$

The generalized inverse Euler equation gives $\beta R = U_{c,t} \mathbb{E}_t[(1 - \chi_{t+1}^q)/U_{c,t+1}] / (1 - \chi_t^q)$. Substituting,

$$1 - \tau_t^s = \frac{1 - \chi_t^q}{\mathbb{E}_t[(1 - \chi_{t+1}^q)/U_{c,t+1}] \mathbb{E}_t[U_{c,t+1}]}, \quad (57)$$

the exact wedge. To approximate it, write $g_{t+1} \equiv \ln U_{c,t+1} = \bar{g} + \hat{g}$ with $\mathbb{E}_t \hat{g} = 0$, and treat the fluctuation $\hat{g}$ and the adjustment χ_{t+1}^q as small. Expanding $e^{\pm g_{t+1}}$ to second order in $\hat{g}$ and keeping first order in χ^q ,

$$\begin{aligned}\mathbb{E}_t[U_{c,t+1}] &= e^{\bar{g}} \left(1 + \frac{1}{2} \text{Var}_t \hat{g} \right), \\ \mathbb{E}_t \left[\frac{1 - \chi_{t+1}^q}{U_{c,t+1}} \right] &= e^{-\bar{g}} \left(1 - \mathbb{E}_t \chi_{t+1}^q + \text{Cov}_t(\chi_{t+1}^q, \hat{g}) + \frac{1}{2} \text{Var}_t \hat{g} \right),\end{aligned}$$

where $\mathbb{E}_t \hat{g} = 0$ turns the linear term of the second expansion into the covariance. Their product is $1 - \mathbb{E}_t \chi_{t+1}^q + \text{Cov}_t(\chi_{t+1}^q, \hat{g}) + \text{Var}_t \hat{g}$ to this order, so inverting (57),

$$1 - \tau_t^s = (1 - \chi_t^q) \left[1 + \mathbb{E}_t \chi_{t+1}^q - \text{Cov}_t(\chi_{t+1}^q, \hat{g}) - \text{Var}_t \hat{g} \right] \approx 1 - \chi_t^q + \mathbb{E}_t \chi_{t+1}^q - \text{Cov}_t(\chi_{t+1}^q, \hat{g}) - \text{Var}_t \hat{g}.$$

Since $\text{Cov}_t(\chi_{t+1}^q, \hat{g}) = \text{Cov}_t(\chi_{t+1}^q, \ln U_{c,t+1})$,

$$\tau_t^s \approx \text{Var}_t(\ln U_{c,t+1}) + (\chi_t^q - \mathbb{E}_t \chi_{t+1}^q) + \text{Cov}_t(\chi_{t+1}^q, \ln U_{c,t+1}),$$

the second-order form of Proposition 5.3. The baseline expansion of Appendix A.4 is the special case $\chi^q \mapsto \chi$.

C.7 Composite statistics and the comprehensive wedge

Proposition C.1 (Composite statistics).

Define

$$\tilde{\gamma} \equiv (1 - m_q) \gamma_c + m_q \gamma_q \quad \tilde{\epsilon} \equiv \epsilon - \frac{\bar{p}^2 U_c^2 (\gamma_c - \gamma_q)^2}{l |U_l| |\mathcal{D}|} \leq \epsilon \quad \tilde{A} \equiv 1 + \tilde{\epsilon} - \tilde{\gamma}$$

Then $\tilde{\gamma} = \tilde{U}_{xl} l / \tilde{U}_x$ and $\tilde{\epsilon} = \tilde{U}_{ll} l / \tilde{U}_l$ for the indirect utility of Lemma 5.5. Under weak separability $\gamma_c = \gamma_q$, so $\tilde{\gamma} = \gamma_c$, $\tilde{\epsilon} = \epsilon$, and the restricted problem is that of Section 4.

Substituting the multiplier (54) into the labor condition (56) yields the comprehensive form of the wedge. Using $\bar{p}/\Delta^q = \bar{p}^2 |s_{qq}|$ from (50), the correction term expands into four pieces,

$$\eta^q \bar{p} \left[\mathcal{I} - \frac{\gamma_c - \gamma_q}{(1 - \tau^y) y} \right] = \left[m_q - \frac{\bar{p}^2 |s_{qq}| (\gamma_c - \gamma_q)}{(1 - \tau^y) y} \right] (\varpi^* - \varpi)$$

and, expanding with $\varpi^* = (\gamma_c - \gamma_q) \psi U_c / (\theta f)$,

$$\begin{aligned}
m_q \varpi^* &= m_q(\gamma_c - \gamma_q) \frac{\psi U_c}{\theta f} = (\gamma_c - \tilde{\gamma}) \frac{\psi U_c}{\theta f} \\
-\frac{\bar{p}^2 |s_{qq}|(\gamma_c - \gamma_q)}{(1 - \tau^y)y} \varpi^* &= -\frac{\bar{p}^2 |s_{qq}|(\gamma_c - \gamma_q)^2}{(1 - \tau^y)y} \frac{\psi U_c}{\theta f} = -(\epsilon - \tilde{\epsilon}) \frac{\psi U_c}{\theta f} \\
-m_q \varpi &= -(\bar{\tau}^q - \nu) \partial_x q^d \\
+\frac{\bar{p}^2 |s_{qq}|(\gamma_c - \gamma_q)}{(1 - \tau^y)y} \varpi &= -\frac{\bar{\tau}^q - \nu}{1 - \tau^y} \cdot \frac{\partial_l q^d}{\theta}
\end{aligned}$$

where the second line uses $l|U_l| = (1 - \tau^y)\theta l U_c = (1 - \tau^y)y U_c$ to match Proposition C.1's Le Chatelier correction, $\epsilon - \tilde{\epsilon} = \bar{p}^2 U_c^2 (\gamma_c - \gamma_q)^2 / (l|U_l||\mathcal{D}|) = \bar{p}^2 |s_{qq}|(\gamma_c - \gamma_q)^2 / ((1 - \tau^y)y)$, and the fourth line uses $\partial_l q^d / \theta = -\bar{p} |s_{qq}|(\gamma_c - \gamma_q) / y$ from (49) and $\varpi \bar{p} = \bar{\tau}^q - \nu$. Moving the two fiscal pieces to the left-hand side of (56) and collecting the $\psi U_c / (\theta f)$ terms on the right,

$$\frac{\tau_t^y}{1 - \tau_t^y} + \frac{\bar{\tau}_t^q - \nu_t}{1 - \tau_t^y} \left[(1 - \tau_t^y) \partial_x q^d + \frac{\partial_l q^d}{\theta} \right] = [A_t + (\gamma_c - \tilde{\gamma}) - (\epsilon - \tilde{\epsilon})] \frac{\psi U_c}{\theta f} = \tilde{A}_t \frac{\psi U_c}{\theta f}$$

which is the comprehensive wedge Ω_t of (13), with $\bar{\tau}^q - \nu$ the screening tax and the bracket the compensated demand response $dq^d/dy|_u$. Along a perturbation dy with $dx = (1 - \tau^y)dy$, utility is unchanged and $dq^d = [(1 - \tau^y)\partial_x q^d + \partial_l q^d/\theta]dy$.

The composite statistics are also the curvatures of the indirect utility of Lemma 5.5, which proves the second claim of Proposition C.1. Since $\tilde{U}_x = U_c(c^d, q^d, l)$ and $\partial_l c^d = -\bar{p} \partial_l q^d$,

$$\tilde{U}_{xl} = U_{cl} + (U_{cq} - \bar{p} U_{cc}) \partial_l q^d$$

Dividing by $\tilde{U}_x = U_c$, multiplying by l , and substituting (47) and (48),

$$\frac{\tilde{U}_{xl} l}{\tilde{U}_x} = \gamma_c + \frac{l(U_{cq} - \bar{p} U_{cc})}{U_c} \cdot \frac{\bar{p} U_c (\gamma_q - \gamma_c)}{l|\mathcal{D}|} = \gamma_c + m_q (\gamma_q - \gamma_c) = \tilde{\gamma}$$

Similarly $\tilde{U}_l = U_l(c^d, q^d, l)$ gives, using $U_{lq} - \bar{p} U_{lc} = (\bar{p} U_c / l)(\gamma_q - \gamma_c)$,

$$\tilde{U}_{ll} = U_{ll} + (U_{lq} - \bar{p} U_{lc}) \partial_l q^d = U_{ll} + \frac{\bar{p}^2 U_c^2 (\gamma_c - \gamma_q)^2}{l^2 |\mathcal{D}|}$$

and dividing by $\tilde{U}_l = U_l < 0$ flips the sign of the positive correction,

$$\frac{\tilde{U}_u l}{\tilde{U}_l} = \epsilon - \frac{\bar{p}^2 U_c^2 (\gamma_c - \gamma_q)^2}{l |U_l| |\mathcal{D}|} = \tilde{\epsilon} \leq \epsilon$$

Under weak separability, the corrections vanish and the problem is that of Appendix A.3.

C.8 The costate and the comprehensive-wedge decomposition

Proposition C.2 (Comprehensive-wedge decomposition).

At every history,

$$\Omega_t = \tilde{A}_t(\theta) \tilde{B}_t(\theta) \tilde{C}_t(\theta) + \beta R \Omega_{t-1} \tilde{D}_t(\theta)$$

where $\tilde{B}_t = B_t$ is the hazard ratio of Proposition 4.1 and $\tilde{C}_t, \tilde{D}_t$ are its C_t and D_t with the integrating kernel built on the composite $\tilde{\gamma}$ and, in $\tilde{C}_t$, the redistributive weight carrying the revenue term $1 - (\bar{\tau}_t^q - \nu_t) \partial_x q^d - \lambda_{1,t} \bar{\alpha}_t U_{c,t}$, the commodity credit net of marginal damages.

Substituting $\phi = f/U_c - \psi \gamma_c / \theta + \eta^q \bar{p} \mathcal{I}_t f / U_c$ from (52) into the costate equation and eliminating the multiplier with $\eta^q \bar{p} \mathcal{I}_t = m_q(\varpi^* - \varpi)$ and $m_q \varpi^* f / U_c = m_q(\gamma_c - \gamma_q) \psi / \theta$,

$$\dot{\psi} - \frac{\tilde{\gamma}}{\theta} \psi = \lambda_1 \bar{\alpha}_t f - \lambda_2 f_2 - [1 - m_q \varpi] \frac{f}{U_c}$$

Solving this ODE with the integrating factor $\exp[-\int \tilde{\gamma} d\tilde{x}/\tilde{x}]$ and $\psi(\infty) = 0$,

$$\psi(\theta) = \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \tilde{\gamma}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] \left([1 - m_q(x) \varpi(x)] \frac{f(x)}{U_c(x)} - \lambda_1 \bar{\alpha}_t(x) f(x) + \lambda_2 f_2(x) \right) dx$$

Here, $\psi(0) = 0$ pins $\lambda_{1,t}$ as in (18). To carry $U_c(\theta)$ into the integrand, multiply the identity (19) by $\exp[\int_{\theta}^x (\gamma_c - \tilde{\gamma}) d\tilde{x}/\tilde{x}]$, with $\gamma_c - \tilde{\gamma} = m_q(\gamma_c - \gamma_q)$:

$$\frac{U_c(\theta)}{U_c(x)} \exp \left[- \int_{\theta}^x \tilde{\gamma}(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] = \exp \left[\int_{\theta}^x \left(\sigma \frac{\dot{c}}{c} + \sigma_{cq} \frac{\dot{q}}{q} - \gamma_c \frac{\dot{y}}{y} + m_q (\gamma_c - \gamma_q) \frac{1}{\tilde{x}} \right) d\tilde{x} \right] \equiv \tilde{K}(\theta, x)$$

By Appendix C.7, $\Omega_t = \tilde{A}_t \psi U_c / (\theta f)$, so multiplying the costate solution by $\tilde{A}_t U_c / (\theta f)$ and splitting the integral as in Appendix A.3,

$$\Omega_t = \tilde{A}_t(\theta) \tilde{B}_t(\theta) \tilde{C}_t(\theta) + \beta R \Omega_{t-1} \tilde{D}_t(\theta)$$

$$\begin{aligned}
\tilde{B}_t(\theta) &= \frac{1 - F_t(\theta)}{\theta f_t(\theta)} \\
\tilde{C}_t(\theta) &= \int_{\theta}^{\infty} \tilde{K}(\theta, x) (1 - m_q(x) \varpi(x) - \lambda_{1,t} \bar{\alpha}_t(x) U_{c,t}(x)) \frac{f_t(x) dx}{1 - F_t(\theta)} \\
\tilde{D}_t(\theta) &= \frac{\tilde{A}_t(\theta) U_{c,t}(\theta) \theta_{t-1} \int_{\theta}^{\infty} \exp \left[- \int_{\theta}^x \tilde{\gamma}_t(\tilde{x}) \frac{d\tilde{x}}{\tilde{x}} \right] f_{2,t}(x) dx}{\tilde{A}_{t-1} U_{c,t-1} \theta f_t(\theta)}
\end{aligned}$$

The persistence block uses $\lambda_{2,t} = \beta R \psi_{t-1} / f_{t-1}$ and, from $\Omega_{t-1} = \tilde{A}_{t-1} \psi_{t-1} U_{c,t-1} / (\theta_{t-1} f_{t-1})$,

$$\frac{\psi_{t-1}}{f_{t-1}} = \frac{\theta_{t-1} \Omega_{t-1}}{\tilde{A}_{t-1} U_{c,t-1}}$$

Hence, the recursing object is the comprehensive wedge, which is Proposition C.2. The revenue term $-m_q \varpi$ inside the $\tilde{C}_t$ weight is the only trace of the second good in the redistributive valuation.

C.9 The optimal uniform price

Let the schedule be flat, $\bar{\tau}_t^q(\theta) = \bar{\tau}_t$, with the level chosen by the planner. The level enters the date-0 Lagrangian only through the date- t wedge constraints, each contributing $\eta^q f \partial[U_q/U_c - (1 + \bar{\tau}_t)] / \partial \bar{\tau}_t = -\eta^q f$. The R^{-t} discounting and the transition densities convert the sum over date- t nodes into the unconditional population expectation, as in Step 4 of Proposition A.1. By the envelope theorem,

$$\frac{dW_0}{d\bar{\tau}_t} = -\frac{1}{R^t} \mathbb{E}_0[\eta_t^q], \quad W_0 \equiv -\mathcal{L}_0$$

Hence, the optimal level sets $\mathbb{E}_0[\eta_t^q] = 0$. In the reduction bookkeeping, the same condition emerges from Roy's identity offsetting the incentive value of compensation. The constraint formulation nets the two automatically and only the substitution distortion against the screening benefit survives. Substituting (54) with $\bar{p}_t$ and ϖ_t common across the date- t cross-section,

$$\mathbb{E}_0[\eta_t^q] = \bar{p}_t \{ \mathbb{E}_0[|s_{qq,t}| \varpi_t^*] - \varpi_t \mathbb{E}_0[|s_{qq,t}|] \} = 0 \quad \implies \quad \varpi_t = \frac{\mathbb{E}_0 \left[|s_{qq,t}| (\gamma_{c,t} - \gamma_{q,t}) \frac{\psi U_{c,t}}{\theta f_t} \right]}{\mathbb{E}_0[|s_{qq,t}|]}$$

Multiplying this by $\bar{p}_t \mathbb{E}_0[|s_{qq}|]$ and replacing $\bar{p} |s_{qq}| (\gamma_c - \gamma_q) \psi U_c / (\theta f) = -(\psi U_c / (\theta f)) l \partial_l q^d$ and $\bar{p} \varpi_t = \bar{\tau}_t - \nu_t$,

$$(\bar{\tau}_t - \nu_t) \mathbb{E}_0[|s_{qq,t}|] = -\mathbb{E}_0\left[\frac{\psi U_{c,t}}{\theta f_t} l_t \partial_l q_t^d\right] \iff (\bar{\tau}_t - \nu_t) \mathbb{E}_0[s_{qq,t}] = \mathbb{E}_0\left[\frac{\psi U_{c,t}}{\theta f_t} l_t \partial_l q_t^d\right]$$

At node-specific prices the same condition without the integral gives $\varpi(\theta^t) = \varpi^*(\theta^t)$, the unconstrained wedge of Appendix A.5. The uniform tax is its $|s_{qq}|$ -weighted average. Under weak separability, every node wedge is zero, so $\bar{\tau}_t = \nu_t$; in retirement, $\psi \equiv 0$, so $\psi U_c/(\theta f) \equiv 0$ and again $\bar{\tau}_t = \nu_t$ (both cases of Corollary 5.5).

C.10 The cost of uniformity

By the envelope theorem applied node by node, the date- t resource value of moving a single node's price is $-\eta^q(\theta)f(\theta)$ per unit of $\bar{\tau}^q(\theta)$. By (54),

$$-\eta^q f = \bar{p} |s_{qq}| (\varpi - \varpi^*) f$$

This value zeroes out at $\varpi = \varpi^*$, reproducing the node-specific optimum. Per unit of the screening component ϖ the cost is $\bar{p}^2/(1+\nu)$ times this, since $\varpi = 1 - (1+\nu)/\bar{p}$ gives $d\bar{\tau}^q = \bar{p}^2 d\varpi/(1+\nu)$. The loss from setting ϖ rather than ϖ^* at a node is then, to second order

$$\int_{\varpi^*}^{\varpi} \frac{\bar{p}^3}{1+\nu} |s_{qq}| (v - \varpi^*) dv = \frac{1}{2} \frac{\bar{p}^3}{1+\nu} |s_{qq}| (\varpi - \varpi^*)^2$$

Aggregating over the date- t cross-section, with $\bar{p}$ common because the price is uniform and ν common because the emission price is an aggregate, and minimizing the resulting quadratic over the common ϖ yields the weighted mean $\varpi = \mathbb{E}_t^{|s|}[\varpi_t^*]$ of Appendix C.9 and, at that minimizer,

$$\text{Loss}_t \approx \frac{1}{2} \frac{(1 + \bar{\tau}_t)^3}{1 + \nu_t} \mathbb{E}_0[|s_{qq,t}|] \text{Var}_t^{|s|}(\varpi_t^*)$$

where $\mathbb{E}_t^{|s|}$ and $\text{Var}_t^{|s|}$ are the $|s_{qq,t}|$ -weighted population mean and variance over date- t histories, which is Proposition 5.4.

C.11 The externality and the social cost of carbon

The conditions above already carry the shadow emission price: every deviation is measured net of ν_t , so all of Section 5.2's results hold with $\bar{\tau}_t^q - \nu_t$ the screening tax and every statistic

evaluated at the externality allocation, and the optimal flat level of Appendix C.9 delivers the additive split of Proposition 5.4. What remains is the pollution-stock multiplier. Substitute the three pieces of the first-order condition for S ,

$$-\psi U_{sl} \frac{l}{\theta} = -\frac{\psi U_c}{\theta f} \xi \frac{U_S}{U_c} f \quad -\phi U_S = -(1 - \chi_t^q) \frac{U_S}{U_c} f \quad \eta^q f \Gamma_S^q = \eta^q \frac{U_S}{U_c} \left[\frac{\xi_q}{q} - \bar{p} \frac{\xi_c}{c} \right] f$$

the second from (52) and the third from (46), to get

$$\mu_t = - \int_0^\infty \left[\frac{\psi U_c}{\theta f} \xi + 1 - \chi_t^q - \eta^q \left(\frac{\xi_q}{q} - \bar{p} \frac{\xi_c}{c} \right) \right] \frac{U_S}{U_c} f d\theta$$

Expanding $1 - \chi_t^q = 1 - \gamma_c \frac{\psi U_c}{\theta f} + \eta^q \bar{p} \mathcal{I}_t$ with $\bar{p} \mathcal{I}_t = \bar{p} \sigma / c - \sigma_{cq} / q$ and collecting,

$$\mu_t = - \int_0^\infty \left[1 + (\xi - \gamma_c) \frac{\psi U_c}{\theta f} + \eta^q \left(\bar{p} \frac{\sigma + \xi_c}{c} - \frac{\sigma_{cq} + \xi_q}{q} \right) \right] \frac{U_S}{U_c} f d\theta$$

The shadow emission price aggregates these date- $(t+j)$ damages exactly as in Appendix A.6. Current emissions enter the continuation value only through the accumulation constraints of the next s periods, so each envelope step peels off one date,

$$\frac{\partial V_{t+1}}{\partial Q_t} = \sum_{j=1}^s \frac{1}{R^{j-1}} \mu_{t+j} \Phi_j, \quad \Phi_j \equiv \frac{\partial S_{t+j}}{\partial Q_t},$$

and, with $\Phi_0 = S_q$, (14) gives $\nu_t = \sum_{j=0}^s R^{-j} \mu_{t+j} \Phi_j$. Substituting this expression for each μ_{t+j} and writing the date- $(t+j)$ population integrals as expectations over future skill histories,

$$\text{SCC}_t \equiv \nu_t = - \sum_{j=0}^s \frac{1}{R^j} \mathbb{E}_t \int \left[1 + (\xi_{t+j} - \gamma_{c,t+j}) \frac{\psi_{t+j} U_{c,t+j}}{\theta f_{t+j}} + \eta_{t+j}^q \left(\bar{p}_{t+j} \frac{\sigma_{t+j} + \xi_{c,t+j}}{c_{t+j}} - \frac{\sigma_{cq,t+j} + \xi_{q,t+j}}{q_{t+j}} \right) \right] \frac{U_{S,t+j}}{U_{c,t+j}} \Phi_j f_{t+j} d\theta$$

whose bracket is the damage weight of Proposition 5.3 at each future node. The new piece is the response of the damage valuation to a budget-neutral clean-for-dirty swap: along $(dc, dq) = (\bar{p}, -1)$, which holds expenditure at consumer prices fixed, $d \ln(U_S/U_c) = \bar{p}(\sigma + \xi_c)/c - (\sigma_{cq} + \xi_q)/q$, using $\partial_c \ln(U_S/U_c) = (\sigma + \xi_c)/c$ and $\partial_q \ln(U_S/U_c) = (\sigma_{cq} + \xi_q)/q$. At $\eta^q \equiv 0$ the weight collapses to the fiscal factor of Appendix A.6; at the optimal flat level the multiplier averages to zero but is not zero node by node, so the weight carries the term even when the level is chosen optimally, which is how the corrective part of Proposition 5.4's additive split is repriced relative to Proposition 4.2. The corrective component of the price is nonetheless uniform across types and histories, because damages depend only on aggregate emissions, so the emissions margin carries no information about types and cannot screen; the uniformity restriction binds only the Corlett-Hague screening component and never the corrective one.

C.12 Retirement

This subsection proves Corollary 5.5 and the retirement claims of Section 5. Retirement dates are $t \geq T_R$. Output is zero and the period utility is $U(c, q, 0, S_t)$, so the current type enters neither preferences nor technology.

Lemma C.1 (Retirement kills the costate).

At every retirement date an optimal allocation is measurable with respect to the retirement-entry history θ^{T_R-1} , the marginal continuation values vanish, $w_{2,t} \equiv 0$ for $t \geq T_R - 1$, the retirement component problem is deterministic node by node, and the costate on the local incentive constraint is $\psi_t \equiv 0$ for $t \geq T_R$. At the junction, the envelope condition of the last working date loses its future-rent term, $\dot{u}_{T_R-1} = U_\theta$.

Proof. Step 1: payoff-irrelevant reports cannot be separated. At a retirement date the report moves the assignment but neither preferences nor an output requirement, so the utility an agent obtains from reporting $\hat{\theta}$,

$$U(c(\hat{\theta}), q(\hat{\theta}), 0, S_t) + \beta w(\hat{\theta})$$

does not depend on the true type. Incentive compatibility then forces it to be constant in $\hat{\theta}$ on the support. Were it higher at some report, every type would prefer that report and truth-telling would fail for a positive mass of types. Hence, u_t is constant across the date- t draws given θ^{t-1} at every retirement date. The assignment can then be taken flat as well: every draw faces the same delivery problem, minimizing $c + q + \frac{1}{R}V_{t+1}$ over the bundles and continuation promises that deliver the common utility, so an optimal assignment common to all draws exists, and without loss the allocation from T_R on is measurable with respect to θ^{T_R-1} .

Step 2: the marginal continuation value dies at the junction. For $t \geq T_R - 1$, the marginal continuation value integrates a constant against the derivative of a density. By Step 1, $u_{t+1}(\theta^t, \theta')$ is constant in θ' , equal to some $\bar{u}_{t+1}(\theta^t)$, so

$$w_{2,t}(\theta^t) = \int_0^\infty u_{t+1}(\theta^t, \theta') \partial_{\theta_t} f_{t+1}(\theta' | \theta_t) d\theta' = \bar{u}_{t+1}(\theta^t) \frac{\partial}{\partial \theta_t} \int_0^\infty f_{t+1}(\theta' | \theta_t) d\theta' = 0$$

since the density integrates to one at every θ_t . At the junction, $w_2 = 0$ enters the date- $(T_R - 1)$ problem as a feasibility restriction rather than a first-order condition: the retirement block can deliver no other value, so the choice of w_2 disappears, the last working date's costate is left free, and its envelope condition reads $\dot{u} = U_\theta$.

Step 3: the retirement block is deterministic and carries no costate. With the allocation measurable in θ^{T_R-1} and $w_2 \equiv 0$, the envelope condition at retirement dates reads $\dot{u} =$

$U_\theta + \beta w_2 = 0$, which the flat profile satisfies identically. The type integral drops from the component problem, which becomes deterministic node by node,

$$V_t(\hat{w}, \mathbf{S}_-) = \min_{c,q,w,S} c + q + \frac{1}{R} V_{t+1}(w, \mathbf{S}) \quad \text{s.t.} \quad \hat{w} = U(c, q, 0, S) + \beta w$$

with the emission state and its aggregate constraint carried as in Section 4.1. There is no envelope constraint and no threat-keeping constraint, hence no costate and no threat multiplier: $\psi_t \equiv 0$ and $\lambda_{2,t} = 0$ for $t \geq T_R$. The working-age first-order system confirms the consistency of this reading: at $\lambda_2 = 0$, $\bar{\alpha}_t = 1$, and a flat allocation, the condition for the numéraire gives $\phi = f/U_c$, promise keeping gives $\lambda_{1,t} = \int \phi d\theta = 1/U_{c,t}$, and the costate equation integrates to

$$\dot{\psi} = \lambda_1 f - \phi = \frac{f}{U_{c,t}} - \frac{f}{U_{c,t}} = 0$$

which, with $\psi(0) = 0$, gives $\psi \equiv 0$.

Step 4: the relaxed problem alone would not deliver this. With persistent skills, retirement draws remain informative about working-life types and the relaxed problem, which keeps only the envelope condition, would tilt retirement utility against those draws to relax the last working date's incentive constraint through w_2 . Step 1 rules the tilt out: nothing disciplines a payoff-irrelevant report, so no such tilt is globally incentive compatible. The measurability restriction of Step 1 is therefore a restriction to the implementable class, not an approximation, and it is how the life-cycle block of Section 6, following Farhi and Werning (2013), treats retirement.

Step 5: the formulas do not need the costate. In retirement, the formulas of interest are the commodity wedge, the savings wedge, and the social cost of carbon. In each of them, ψ enters multiplied by one of γ_c , γ_q , or ξ . Each of these statistics is proportional to l through its defining cross-partial, so each vanishes at $l = 0$. The retirement conclusions, a zero screening tax, the standard inverse Euler equation and Pigouvian carbon pricing therefore hold whatever the costate. $\square$

Corollary 5.5 follows: $\psi \equiv 0$ gives $\psi U_c/(\theta f) \equiv 0$ at every retirement node, so the optimal uniform level of Appendix C.9 is $\bar{\tau}_t = \nu_t$, the purely corrective price. The retirement statements of Section 5.1 follow by the same steps. With no hours choice to decentralize, the frozen labor-wedge constraint is not imposed in retirement, so $\eta_t \equiv 0$ there — the labor multiplier of Section 5.1, not the commodity multiplier η_t^q of this appendix — while the frozen savings wedge keeps operating across retirement dates, which is why its distortions survive where the labor-frozen ones die.

D The quantitative exercise

D.1 Calibrating the given schedules to the US tax system

The schedules the exercise needs are not hypothetical. For the labor margin, take the earnings-indexed schedule (42) of Appendix B.12. Heathcote et al. (2017) estimate $p = 0.181$ for US households aged 25 to 60 by regressing log post-government on log pre-government earnings in the PSID, with an R^2 of 0.91. Heathcote and Tsujiyama (2021) pair it with $\lambda^T = 0.840$. Progressivity is flat in age. Re-estimating with both parameters conditioned on age, Heathcote et al. (2020) find no significant age dependence in p , so an age-invariant progressivity is a feature of the US system rather than an approximation error. Their null covers the slope alone; the level λ_t^T is left free to vary with age.

For the savings margin there is no counterpart to (42) with the same standing, since the fitted tax functions of Heathcote et al. (2017) and Guner et al. (2014) are functions of earnings and imply nothing about the intertemporal margin. The practical route is a linear capital income tax. With private saving at gross return R taxed at rate τ_t^k on the interest component, the household's Euler equation and the definition of the savings wedge give

$$1 - \bar{\tau}_t^s = \frac{1 + (R - 1)(1 - \tau_t^k)}{R} \quad (58)$$

so a single number fixes the whole schedule. With $\beta R = 1$ as in Section 6, (58) collapses to $\bar{\tau}_t^s = (1 - \beta)\tau_t^k$: at $\beta = 0.95$ the frozen wedge is one twentieth of the capital income tax rate, because the tax falls on the interest component alone. A wedge of under one percent of the gross return is therefore a tax of tens of percent on the margin it actually distorts, and I report the schedule in tax units throughout to keep that visible.

The rate (58) asks for is a marginal one on the household's own intertemporal margin, since it comes from the Euler equation on a marginal unit of saving. The natural US anchor is the Congressional Budget Office (2014) effective marginal tax rate on capital income, which it puts at 18 percent, and I take that as the benchmark. Two qualifications are worth stating, because the headline figure is not quite the object the mapping wants. It is a capital-stock-weighted composite of a 29 percent rate on business capital and a rate near zero on owner-occupied housing, which is close to half the stock; and it measures the total wedge between the return an investment must earn and the return the saver receives, so it includes a corporate layer that a household facing an exogenous gross return R never sees. The personal layer alone is what the mapping freezes, and the same study prices it: an average marginal rate of 27 percent on interest income, applied to the share of marginal saving that is not assigned to tax-deferred or Roth accounts, where the marginal rate is zero.¹³ The benchmark is also computed under 2014

¹³The composite rate is CBO's Table 2 and the sectoral weights its Table A-1; the marginal rate on interest

law, before the 2017 reform cut rates on business capital, so as a description of the current code it errs high rather than low.

The aggregate effective rates computed by the method of Mendoza et al. (1994), which put τ^k near 0.36 for the US (Trabandt and Uhlig 2011), are averages: capital tax revenue divided by capital income in the national accounts. They coincide with the marginal rate in the linear-tax representative-agent settings they were built for, which is precisely the identification a Mirrleesian model drops. They do reach state, local and property taxes that the federal marginal measure leaves out, so I carry 0.36 as an upper bound in the sensitivity analysis of Section 6.4 and use 0.18 everywhere else.

At $\tau^k = 0.18$ the frozen wedge is $\bar{\tau}^s = 0.009$, nine tenths of a percent of the gross return per year. Section 6 computes the unconstrained-optimal savings wedge, the one the unconstrained mechanism would set, as falling from 0.0056 to 0.0002 over the working life; in the same tax units that is a capital income tax of about 11% at labor-market entry declining to nearly zero by retirement. The US system therefore overdistsorts the intertemporal margin at every age, and by more as the cohort ages, at any τ^k above 0.11, which every measure of the US rate clears.

The exercise inherits one sensitivity worth stating, and the deviations survive it. How far the given schedules sit from their desired counterparts, and therefore how large the net shadow price ζ_t of Section 5.1 is, depends on the Pareto weights $\alpha(\theta)$ of Section 3. Under a utilitarian objective Heathcote and Tsujiyama (2021) find a welfare gain of about 2% of consumption from replacing the estimated US schedule with the fully optimal one, so the schedule is far from optimal; under the weights that rationalize observed US progressivity within their parametric class the gain falls to about 0.05%, and one might expect the deviations of Corollary 5.1 to vanish with it. Re-solving the joint exercise at $\tau^k = 0.18$ under the family $\alpha(\theta) \propto \theta^{-\varrho}$ on the initial skill shows that they do not.¹⁴ Across the whole family the working-life mean of the deviation stays in $[-0.278, -0.249]$, against a utilitarian value of -0.264 ; the retirement mean stays in $[-1.038, -0.999]$, against -1.019 ; and the age at which η_t^s changes sign falls between 39 and 40 at every ϱ . Lemma 5.3 forces that reversal along every history without dating it, so the date is a property of the solve rather than of the lemma. A six-fold tilt in the weights therefore moves the deviation by six percent, and it saturates as the weights tilt toward the unlucky. The labor multiplier η_t moves the way its tightness reading predicts, its working-life

income and the account shares are its Tables A-4 and A-3. The personal-layer figure is my own arithmetic from those two tables, not a number CBO publishes: at their fully-taxable shares it lands in the high teens. That it agrees with the 18 percent headline is a coincidence of construction rather than a corroboration, since the headline nets a corporate layer against a housing subsidy and this does neither.

¹⁴Primitive weights enter the planner problem only through the birth-period objective, so each ϱ re-solves the birth block against the same continuation surfaces; $\varrho = 0$ reproduces the production run, and $\varrho < 0$ weights the fortunate draws. The sweep runs $\varrho \in [-4, 1]$; at $\varrho = -4$ the weight ratio across the 1–99 percent range of the initial draw is six.

mean shrinking by a quarter as ϱ falls, but it saturates well short of zero. No power weighting of the initial skill, however steep, rationalizes the frozen pair.

The weights have so little room because almost none of this economy’s inequality is theirs to reweight. Agents here are ex ante identical, so the primitive weights attach only to the information in the first draw; every later difference across agents is risk, and a welfarist planner prices risk with the agents’ own expected utility whatever α . A calibration with initial dispersion in the data’s range would hand the weights more of the labor gap, the margin Section 6.6 shows changing sign outright under the Golosov et al. (2016) wedge path. It would hand them none of the savings margin: the retiree’s desired savings wedge is zero under every admissible weighting while the frozen wedge stays $(1 - \beta)\tau^k > 0$, so the savings deviation is not the weights’ to remove. I take the utilitarian case throughout, matching the planner of Section 3; the sweep bounds what a reader with different distributional tastes should carry, and what moves is the working-life level, not the sign change of η_t^s . The carbon price’s own crossing is not invariant: it moves later as ϱ rises, from age 26 at $\varrho = -0.5$ to age 27 at the utilitarian weights and age 29 at $\varrho = 1$, and it leaves the life cycle at the young end once the weights tilt far enough toward the fortunate, with no crossing at any $\varrho \leq -1$. This is Lemma 5.4 again, since that crossing is where the savings term offsets the labor term rather than where either vanishes: tilting toward the fortunate lowers the early-life deviation until it is already negative at labor-market entry.

D.2 Solvers and numerical implementation

The three solvers share the architecture of the Farhi and Werning (2013) replication they are built on. Each period’s problem is solved by backward induction on a grid over the two multipliers dual to promise keeping and threat keeping. At every grid point the within-period problem is a two-point boundary value problem in the type dimension, integrated by an adaptive Dormand-Prince 5(4) scheme at relative tolerance 10^{-3} and absolute tolerance 10^{-6} and closed by shooting on the value promised at the bottom of the type support, with Brent’s method over expanding brackets. Values at the storage nodes are read from the dense-output interpolant of the same trajectory the shooting residual integrates, so the stored allocation and the residual that certifies it are the same object; the transversality condition is met to a median of 8×10^{-15} .

The third-best solvers add the marginal-utility requirement $\hat{M}$ of Lemma 5.1 as a third state and replace the shooting pass with direct collocation. The node conditions, the trapezoid collocation rows, the boundary rows and the row that delivers the inherited $\hat{M}$ are stacked and solved by Newton with an analytic sparse Jacobian, warm-started from the neighbouring date.

Grids differ across the stack because the state space does. The second best and the linear commodity tax run on 100×45 dual grids with 20 type nodes. The third best runs on 16×8 dual grids with 40 type nodes and an adaptive promise grid of 11 to 27 points: the third state has to be carried at every dual node, and the dual grid pays for it. This is the coarsening Section 6.1 refers to when it reads the incentive-compatibility audit against a control at the

same grid. Shooting residuals are driven to 10^{-11} , pointwise roots to between 10^{-10} and 10^{-12} , and the Newton stacks to residuals of order 10^{-11} .

Welfare is measured in consumption-equivalent units, $\exp(\Delta u/\mathcal{A}) - 1$, with Δu the difference in mean simulated date-0 lifetime utility between the two allocations compared and $\mathcal{A} = (1 - \beta^{60})/(1 - \beta)$ the present value of a unit consumption stream, and it is computed from simulated lifetime utilities rather than from the grid value functions: at each simulated history the realized bundle is evaluated in the direct utility function, retirement included. Simulation draws one random-number stream per agent, keyed by the run seed, so the shock panel is identical across regimes and across thread counts and every comparison is a common-random-numbers comparison. What bounds the reported differences is not sampling error. For the second-best ladder at 300,000 histories the binding figure is the spread in the present-value resource residual across runs, about 0.003 percentage points; the seed-to-seed sampling error on gains is an order of magnitude tighter.

One consequence for the quantitative exercise. The multiplier η_t is now determined at the fixed point between the assigned earnings $y_t(\theta^t)$ and the marginal rate the schedule attaches to them, since $\bar{\tau}_t^y$ appears on both sides of (27) through y . Under (42) the left-hand side of (27) is strictly increasing in y whenever $p_t \in (0, 1)$, since $\bar{\tau}_t^y/(1 - \bar{\tau}_t^y) = y^{p_t}/(\lambda_t^T(1 - p_t)) - 1$. The right-hand side moves with y too, through A_t , Δ_t , Λ_t and the costate, so any fixed point at which it is non-increasing in y is locally unique. Global uniqueness is not established, and the solver iterates to a fixed point rather than reading the schedule off the type.

D.3 Validation

Validation runs in three layers, and everything reported here was re-run on the production code.

The first layer checks the preference block. Every analytic derivative the solvers use is checked against central finite differences at a battery of parameter points, together with the structural identities the two-good extension has to satisfy: consistency of the indirect utility function, equality of the household's marginal rate of substitution with the consumer price, the budget and resource identities, the spending identity to under 10^{-9} , uniqueness of the root of the share-consistency residual, and the zero-twist limit in which the commodity wedge collapses to the Pigouvian rate. These pass at between 10^{-5} and 10^{-9} .

The second layer checks the solvers against each other on a common 48×24 dual grid with 20 type nodes. With the twist, damages, subsistence and the commodity tax all switched off, the linear-commodity-tax solver reproduces the Farhi and Werning (2013) replication: policies agree to 4×10^{-3} and the initial resource multiplier to 2.6×10^{-4} , and both converge under refinement to the published value. Re-solving both at relative tolerance 10^{-5} collapses the remaining gaps by more than two orders of magnitude, which identifies them as integration noise rather than

a difference in the problem being solved. With the twist off and no externality the second-best and linear solvers agree to 4.4×10^{-7} on the same multiplier, agree to machine precision at the last working date where the retirement closed form removes all continuation interpolation, and return a commodity wedge that is identically zero. With the twist off and a positive damage the second best sets the commodity wedge equal to marginal damage at every grid point and every period to 2.8×10^{-16} . That is the dynamic Atkinson-Stiglitz agreement of Section 6.1, and it holds to solver precision rather than approximately.

The third layer checks the third-best stack against the solvers it nests. At a zero labor multiplier the constrained node system is zeroed by the stored second-best solution to 5×10^{-13} , and at a zero savings multiplier the joint node system equals the system one stage below it exactly. The retirement chain reproduces the closed-form annuity at a zero frozen savings wedge to 1.8×10^{-15} . The stacked residual vanishes at stored second-best solutions to order 10^{-14} , every row of the joint stacked system vanishes at stored slices of the benchmark store to 8×10^{-10} , and warm re-convergence from a refreshed seed returns the stored solution within one Newton iteration. At the driver level, the unconstrained third best approaches the second best at the second-order rate the type discretization implies, the gap in the initial multiplier falling from 1.2×10^{-2} to 3.8×10^{-3} as the type grid doubles from 20 nodes to 40; freezing the schedules at their desired counterparts returns multipliers indistinguishable from zero, which is the second agreement of Section 6.1; and the US freeze reproduces a recorded cross-solver anchor, two structurally different solvers agreeing on the deviation profile to better than 10^{-2} .

One check is a diagnostic rather than a gate, and it is worth saying why. Evaluating Proposition 5.1 directly at simulated states, with every multiplier reconstructed by interpolation from the stored solution, reproduces the solved deviation only to about 0.2 in absolute value, against a mean deviation of 0.26. The failure is one of conditioning on the read side rather than of the solution. The right-hand side of the proposition is a small residual of two large offsetting terms: at labor-market entry the labor and savings channels are -0.25 and $+0.31$, each several times the deviation they sum to, so errors of a few percent in the interpolated multipliers produce absolute misses of the same order as the deviation itself while leaving the solved allocation untouched. Separating the two sources confirms the diagnosis. The proposition's own algebra, evaluated against the node system at the same states and given the same reconstructed inputs, agrees to a mean of 0.024, an order of magnitude below the input error, and the node conditions the proposition collects are enforced exactly at grid points by the gates above. No number reported in this paper is computed from the identity.

D.4 The incentive-compatibility audit

The first-order approach behind every solver is verified ex post by a one-shot-deviation audit. At simulated states of each solved store, a deviating agent picks the best point on the entire reported-type menu and the audit records the utility gain over truth telling. Gains at the

Table 6: The one-shot-deviation audit across the solved stores. Entries are utility gains in utils at simulated states; “p99 gain” is the 99th percentile over the tightest window, “junction p99” the same statistic at the retirement junction, and the last column counts non-monotone reported-earnings menus among sampled states. The horizontal rule separates the second-best grid from the third-best production grid; the control row is the unconstrained solve at the third-best grid, whose gains are pure menu-interpolation noise.

economy	grid	p99 gain	junction p99	non-monotone menus
second best, benchmark	100×45	6.4×10^{-3}	2.5×10^{-3}	0/140
second best, $\gamma_c = \gamma_q$ placebo	100×45	7.4×10^{-3}	2.5×10^{-3}	0/840
Stone-Geary, flat Pigouvian rung	100×45	6.6×10^{-3}	2.5×10^{-3}	0/840
<i>control</i> : unconstrained, third-best grid	16×8	7.4×10^{-2}	6.0×10^{-4}	0/840
labor frozen (stage 1)	16×8	7.0×10^{-2}	5.8×10^{-4}	0/840
joint freeze, $\tau^k = 0.18$	16×8	4.4×10^{-2}	6.8×10^{-4}	0/840
joint freeze, $\tau^k = 0.36$	16×8	6.2×10^{-2}	1.2×10^{-3}	2/840
joint freeze, flat τ^{q*}	16×8	6.7×10^{-2}	9.1×10^{-4}	0/840

scale of the menu-interpolation noise are the audit’s resolution, not violations, and that scale differs by an order of magnitude across grids: the second best solves on 100×45 dual nodes, the third-best production grid on 16×8 . The third-best stores are therefore read against an unconstrained solve at the identical 16×8 grid, whose deviation gains are pure interpolation noise by construction.

Table 6 reports, for each store, the 99th-percentile deviation gain in the tightest re-simulation window, the set of simulated states whose reported type lies within half a log innovation standard deviation of the truth, $\hat{\sigma}$ being the log-skill innovation dispersion of Section 6.1, the retirement-junction row, where the continuation value is closed form and the audit is sharpest, and the count of non-monotone reported-earnings menus. Two readings carry the section’s claim. The four frozen-wedge stores sit at or below the unconstrained control at the identical grid, so the freeze introduces no deviation gains the grid itself does not already produce. And the junction rows sit well over an order of magnitude below the local windows on the third-best grid, so where the audit is sharp it is clean.